MULTIMODAL FASHION RECOMMENDATION FRAMEWORK USING FINE-GRAINED ATTRIBUTE-AWARE RETRIEVAL

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ABSTRACT

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by Shao-Yu Huang

Accurate garment attribute recognition is essential for fashion recommendation systems, trend analysis, and large-scale retrieval applications. However, most current approaches or methods are based on rough visual features and fail to consider about fine-grained semantic attributes that determine clothing style and compatibility. This research investigates visual-to-textual garment labeling using CLIP and extends its capabilities through fine-tuning on curated subset of the Polyvore dataset, which contains diverse images of clothing items. We further extend the idea by introducing a new pipeline incorporates linear probe over CLIP's embedding structure to capture five attributes: category, subcategory, color, material, and pattern, which are the basic elements of clothing. The process involves freezing both image and text encoders attaching lightweight linear heads for classification. This design preserves CLIP's rich semantic representations while enabling efficient fashion specific learning. While the accuracy performance reach 93% for category and 72% for subcategory, and high semantic similarity scores of 87%, 87%, and 85% for color, material, and pattern. Another contribution is incorporating CLIP-based semantic similarity into the evaluation pipeline, which provides a more nuanced and interpretable assessment of attribute recognition capturing partial correctness and closely aligned with semantics rather than perfect matches. Together, these contributions advance fine-grained fashion understanding and provide a practical and explainable framework for fashion recommendation systems.

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Chapter 1: Introduction

Background and Motivation

E-commerce has significantly transformed and accelerated how people understand and consume fashion, enhancing both trend diffusion and the influence of fast fashion (Cheng et al., 2021). However, existing e-commerce recommendation systems primarily focus on suggesting similar items from users' historical purchases, reflecting the perspectives of sellers and platforms, rather than addressing users' broader styling needs (Han et al., 2017; Hou et al., 2019).

From the user's perspective, the goal is often not to repeatedly purchase similar items, like shirts of the same style, but rather to compose a complete outfit. In addition, when users provide detailed or personalized queries in search interfaces, such as specific styles, intended occasions, or preferred materials, these descriptions often do not align well with the metadata or textual annotations provided by sellers. As a result, users frequently encounter difficulties in retrieving relevant fashion items. Such fine-grained and context-aware requirements remain challenging for current recommendation systems to handle effectively.

Consequently, recommendation systems that rely primarily on global visual similarity or limited metadata tend to retrieve items that are visually related but fail to capture the user's underlying semantic intent or stylistic preference. Prior studies have also highlighted the importance of explainability and semantic-attribute understanding in fashion recommendation, particularly when users focus on specific garment characteristics rather than overall appearance (Hou et al., 2019).

From this point of view, current recommendation systems do not fully satisfy user needs, which motivates our work, moreover, the applicability is not limited to e-commerce scenarios. It can also assist users who wish to follow fashion trends but lack confidence in styling their outfits, or those who are preparing for a special occasion to attend and need guidance. In these cases, a recommendation system that understands attribute-level garments becomes valuable.

Problem Statement

First, a key challenge in fashion recommendation is understanding and representing fine-grained attributes in clothing images, where given a single garment image, the system is expected to capture multiple attributes simultaneously. Although datasets such as DeepFashion (Liu et al., 2016), DeepFashion2 (Ge et al., 2019), Fashionpedia (C. Jia et al., 2020), and Polyvore (Han et al., 2017) provide rich annotations and metadata, prior work has rarely treated fine-grained attribute extraction as a central research problem, particularly in the context of vision-language models.

In many existing approaches, attribute information is only used as supplementary metadata or incorporated into broader tasks such as compatibility prediction, retrieval, segmentation, or captioning. As a result, there is still limited understanding of how to effectively model and utilize detailed garment attributes for downstream recommendation tasks. This issue is especially important because users often overlook or omit detailed clothing descriptions when searching for fashion items, if a system can infer these missing details directly from garment images, it can enhance item representations beyond just surface appearance and support recommendations that are not only visually similar but also semantically aligned with user preferences. Therefore, fashion attribute extraction should be seen not as a standalone classification task but as a crucial step toward a more explainable and user-focused outfit recommendation system. This limitation becomes more critical where users often provide incomplete or ambiguous queries when searching for fashion items. In many cases, users may not explicitly specify detailed properties such as material, pattern, or subcategory, even though these attributes strongly influence their preferences. Consequently, recommendation systems that rely primarily on global visual similarity or sparse metadata may fail to capture users' underlying semantic intent.

Furthermore, existing systems lack the ability to represent higher-level fashion concepts, such as style, in a structured and interpretable manner. While style plays a crucial role in how users perceive and select clothing, it is often weakly modeled or entirely absent in traditional recommendation pipelines. Therefore, there is a need for a unified representation that can bridge low-level visual attributes and high-level semantic concepts, enabling more accurate, interpretable, and user-aligned fashion recommendations.

Thesis Objectives

To address the problem and limitations stated above, this research aims to develop a unified framework that bridges the gap between fine-grained garment understanding and attribute-aware retrieval.

First, this work aims to build a robust fashion attribute prediction model that can accurately capture the five core elements of a garment: category, subcategory, color, material, and pattern. Given a single garment image, the model jointly predicts these attributes in both single-label and multi-label settings: category and subcategory are treated as single-label tasks, while color, material, and pattern are modeled as multi-label attributes to reflect the compositional nature of real-world garments.

Second, this work further extends the attribute space by incorporating style as a higher-level semantic attribute. This allows the system to learn and capture more abstract fashion concepts, thereby improving the system's ability to model user preferences and contextual styling needs.

Third, we design an attribute-aware retrieval mechanism that leverages the extracted attributes to improve recommendation quality. Instead of relying solely on visual similarity, the system evaluates similarity at attribute level, enabling more precise and semantically consistent retrieval results.

Finally, this work proposes a flexible evaluation strategy for fashion attribute prediction. With the reason that fashion labels are often noisy and linguistically inconsistent, we use CLIP-based encoding to measure semantic similarity. This approach provides a more interpretable assessment of prediction quality by capturing semantic alignment rather than strict lexical matching.

Overall, this research aims to create a practical connection between image understanding and fashion suggestions, allowing systems to not only find visually similar items but also better align with users' semantic and stylistic needs.

Chapter 2: Related Work

Recommendations for fashion outfits have attracted significant interest, particularly as online shopping and visual social platforms generate rich image-based data. Early works in fashion recommendation primarily focused on global visual features or collaborative filtering. Still, they often lacked fine-grained semantic understanding of garments such as color, material, pattern, and style, and interpretability in recommendation results.

Fashion Vision, Attributes, and Datasets

Fashion computer vision research has long focused on recognizing clothing items, their attributes, and how multiple garments combine within an outfit. Unlike general object recognition, fashion understanding requires fine-grained labels, including detailed clothing information: category, color, material, and pattern, as well as spatial structures such as clothing landmarks on the human body. To enable this analysis, several large-scale datasets have been developed.

A foundational related work is DeepFashion (Liu et al., 2016), which provides over 800,000 images annotated with categories, attributes, bounding boxes, and landmarks. Its introduction of multi-task learning through FashionNet demonstrated how jointly learning landmarks and attributes improves clothing recognition. DeepFashion2 (Ge et al., 2019) expanded on this foundation with 491,000 images and over 800,000 annotated items, offering segmentation masks, pose estimation, and cross-domain matching pairs. This dataset supports instance-level recognition, retrieval, and fine-grained pose analysis, establishing a comprehensive benchmark for fashion image understanding. Fashionpedia (C. Jia et al., 2020) further unified garment segmentation with attribute labeling by introducing a structured ontology of categories, apparel parts, and detailed visual attributes. Each garment is paired with a mask and multi-attribute labels, advancing research toward an interpretable, fine-grained understanding of clothing appearance.

Beyond single garments, the Polyvore Outfits dataset (Vasileva et al., 2018) extended fashion vision to outfit-level reasoning. It contains over 68,000 user-curated ensembles, enabling tasks such as compatibility prediction and outfit completion. Models trained on this dataset learn embedding spaces that capture both item similarity and cross-type compatibility, supporting queries like “find shoes that match this dress”.

In summary, these datasets collectively shaped modern fashion AI research. DeepFashion and DeepFashion2 established robust foundations for recognition and segmentation, Fashionpedia advanced semantic reasoning with detailed attributes, and Polyvore enabled contextual understanding of outfit compatibility. Together, they define the key benchmarks and methodological backbone for contemporary multimodal fashion analysis.

Vision-Language Models

Fashion vision-language research has rapidly evolved toward models that jointly learn from visual and textual data, aiming to align how images and language describe the same concept. These models bridge perception and semantics, allowing visual understanding to be expressed through natural language and vice versa. The rise of large-scale vision-language pretraining has fundamentally changed multimodal learning, powering tasks such as image captioning, retrieval, and visual question answering.

Among these, CLIP (Contrastive Language-Image Pre-training) (Radford et al., 2021) stands as a landmark contribution. Trained on 400 million image-text pairs, CLIP learns a shared embedding space in which matched images and captions are highly similar. This contrastive objective allows the model to perform zero-shot recognition: without explicit category labels, it can classify new images through natural-language prompts, for instance, identifying “a red dress” without specific training. CLIP’s performance on ImageNet rivaled that of fully supervised models like ResNet50 (He et al., 2016), demonstrating that language supervision at scale can yield general, flexible visual representations.

Building on CLIP, ALIGN (M. Jia et al., 2021) demonstrated the power of scale. Using over one billion noisy web image-text pairs, it showed that massive, weakly supervised data could outperform carefully curated datasets. The model achieved state-of-the-art results on retrieval and zero-shot benchmarks, confirming that scaling contrastive pretraining improves multimodal alignment. In fashion, this paradigm has direct relevance, supporting natural language search, for example, “find a red off-shoulder summer dress” and zero-shot classification of unseen styles. FashionCLIP (Chia et al., 2022), a domain-specific adaptation of CLIP, fine-tunes its embeddings on curated fashion datasets to capture terminology, texture, and stylistic nuance, thereby improving tasks such as fine-grained product classification and semantic outfit retrieval.

Beyond contrastive understanding, generative vision-language models extend multimodal reasoning to produce text from visual input. BLIP (Bootstrapping Language-Image Pre-training) (Li et al., 2022) integrates both understanding and generation into a single architecture. By filtering web data and augmenting it with self-generated captions, BLIP achieves robust cross-modal reasoning and superior image captioning. BLIP-2 (Li et al., 2023) further connects vision encoders to large language models (LLMs) via a Q-Former module, enabling efficient image-conditioned text generation. For fashion applications, this generative ability enables descriptive, context-aware interactions, for example, explaining an outfit's style or suggesting complementary accessories in natural language. Other general-purpose multimodal frameworks, such as FLAVA (Singh et al., 2022), aim to provide a unified representation across image-only, text-only, and image-text tasks. Its fusion architecture integrates recognition, reasoning, and generation into a single model, providing a foundation for comprehensive multimodal agents.

Recently, instruction-tuned multimodal assistants such as LLaVA (Large Language and Vision Assistant) have brought interactivity to vision-language modeling. By connecting a CLIP-like visual encoder with an instruction-following LLM, LLaVA can engage in dialogue grounded in visual content. Text example like "Describe this outfit and suggest shoes for an evening event. This conversational ability points toward fashion AI systems that combine perception, reasoning, and stylistic understanding to deliver personalized recommendations.

In summary, vision-language models form the technological backbone for bridging visual fashion data with natural-language reasoning. Foundational models like CLIP and BLIP provide adaptable feature representations, while emerging multimodal assistants like LLaVA extend these capabilities into interactive, context-aware systems. Building upon these advances, this thesis integrates CLIP-based understanding with BLIP and LLaVA-inspired generative reasoning to develop a multimodal framework for intelligent, interpretable fashion recommendation.

Fashion-Specific Models

Using the generic models and available datasets, various fashion models have been developed to address the unique challenges of fashion analysis and recommendation tasks. These models might leverage domain knowledge, for example, the complementary classes of tops and bottoms, or address tasks that do not fall within the purview of generic vision models, such as fashion model compatibility and trend prediction. In this section, a brief overview of the state of AI fashion models will be presented, broadly classified into: (a) fashion model compatibility models, (b) fashion understanding models, and (c) fashion trend models.

Fashion Model Compatibility Models

A core challenge in computational fashion lies in understanding which garments complement one another. Early studies approached this through metric learning, constructing embedding spaces where compatible items are close and incompatible items are far apart. The Type-Aware Embedding model (Vasileva et al., 2018) introduced a key insight: garments differ by type, such as tops, bottoms, and shoes, and compatibility should be defined across rather than within types. By learning both type-specific and shared compatibility spaces using outfit data, the model jointly captured intra-type similarity and inter-type harmony, achieving state-of-the-art performance in outfit prediction.

Subsequent models expanded this view. BiLSTM-based frameworks (Han et al., 2017) treated an outfit as an ordered sequence: top, bottom, shoes, and accessories, learning implicit style dependencies such as formality or color coherence. Later, attention-based and graph models, such as SCE-Net (Xing et al., 2022), explored higher-order relationships beyond pairs, recognizing that outfit style emerges from the full set of items rather than from isolated matches. The OutfitTransformer (Sarkar et al., 2023) represents the modern standard for set-level outfit modeling. Using a transformer encoder where each item is a token, it jointly predicts outfit compatibility and retrieves complementary items via ranking losses. This design captures multi-item interactions, as an example, color coordination between jacket and shoes, and outperforms graph and embedding models across benchmarks. To evaluate such systems, Aesthetic 100 (Zou et al., 2022) introduced a benchmarking model for aesthetic reasoning, assessing aspects such as color harmony and material balance. Findings revealed that even advanced models struggle with deeper fashion semantics, highlighting the need to embed explicit aesthetic principles into future compatibility models.

Fashion Understanding Models

Beyond outfit reasoning, many applications depend on connecting visual and textual modalities. FashionBERT (Gao et al., 2020) adapts the BERT architecture for cross-modal retrieval between product images and descriptions. By replacing coarse object features with patch-level embeddings and training on image-text pairs with adaptive losses, it learns fine-grained alignment between garment regions and textual details, such as fabric and neckline. This approach markedly improved retrieval accuracy and laid the foundation for multimodal understanding in fashion e-commerce.

Generative multimodal models further extend these capabilities. FashionGPT (Gao et al., 2024) is an instruction-tuning method that trains a vision-language model on curated fashion data, enabling it to describe garments and answer contextual questions such as seasonal suitability. Similarly, FashionReGen (Ding et al., 2024) applies LLMs to generate editorialstyle analyses of fashion shows, combining visual perception with narrative reasoning. These systems exemplify the integration of perception, interpretation, and creative text generation within fashion AI, bridging structured recognition and expressive communication.

Fashion Trend Models

Fashion evolves continuously, making trend forecasting a crucial task. Traditional qualitative analysis is now complemented by data-driven approaches leveraging large-scale visual and temporal data. The KERN model (Ma et al., 2020) introduced a knowledge-enhanced recurrent network that predicts the future popularity of fashion elements by combining time-series learning with domain knowledge such as seasonality or designer influence. This integration improved robustness over purely data-driven LSTMs.

To evaluate temporal generalization, SHIFT15M (Kimura et al., 2023) provides a decade-long dataset of 15 million fashion items organized by year, allowing tests under realistic distribution shifts. It revealed that most models degrade as fashion evolves, underscoring the importance of continual, trend-aware learning.

Additionally, social platforms like Pinterest and Instagram serve as dynamic indicators of emerging trends. By tracking the frequency of visual concepts or search terms, for example, “wide-leg pants”, researchers quantify rising aesthetics and cyclic revivals. Recent methods increasingly combine computer vision, time-series modeling, and LLM-based contextual reasoning to connect visual signals with cultural drivers. From compatibility modeling to multimodal understanding and trend forecasting, fashion AI research has progressed from low-level recognition to high-level reasoning.

Style Classification Models

Recent advances in fashion understanding have focused on leveraging deep learning models to classify clothing styles and attributes from visual data. Early work in fashion recognition primarily relied on convolutional neural networks (CNNs) (O'shea & Nash, 2015) to extract visual features for clothing categorization and attribute prediction. For example, Cheng et al. (2021) provide a comprehensive survey demonstrating how CNN-based architectures can effectively learn fine-grained fashion attributes such as color, pattern, and style, which are essential for downstream tasks like recommendation and retrieval.

Other studies have shifted toward multimodal representation learning, where both visual and textual information are jointly modeled. Chia et al. (2022) introduce a contrastive learning framework tailored for fashion concepts, showing that aligning image and text embeddings significantly improves the model's ability to capture high-level semantic attributes such as style. This approach is closely related to CLIP-based architectures, where models learn generalized representations that can transfer across tasks.

Additionally, large-scale pre-trained models such as CLIP (Radford et al., 2021) and domain-specific variants like FashionCLIP (Chia et al., 2022) have demonstrated strong performance in zero-shot and fine-tuned classification settings. These models enable flexible style classification by encoding both images and text into a shared embedding space, thereby framing attribute prediction as a similarity-matching problem rather than a purely supervised classification task.

In this study, style classification models are used to extract structured attributes that serve as an intermediate representation for retrieval. This design is supported by prior work showing that structured semantic attributes improve interpretability and downstream recommendation performance.

Retrieval Models

Fashion retrieval has evolved from traditional content-based image retrieval (Qazanfari et al., 2023) systems to modern multimodal retrieval frameworks that integrate both visual and textual information. Early retrieval methods relied on handcrafted features or CNN-based embeddings to measure visual similarity between clothing items. However, these approaches often fail to capture semantic relationships such as style compatibility or contextual relevance. The release of large-scale fashion benchmarks, such as DeepFashion2 (Ge et al., 2019) and Fashionpedia (C. Jia et al., 2020), further highlighted that effective fashion understanding requires more than category-level recognition. Instead, retrieval systems must reason over detailed properties such as structure, color, material, and localized attributes. This shift is particularly important in fashion because users do not only seek visually similar items, but also items that satisfy semantic and stylistic constraints.

Recent work addresses these limitations by adopting cross-modal embedding techniques and multimodal representation learning. CLIP (Radford et al., 2021) is a key milestone in this area, learning a joint image-text embedding space using contrastive learning on large-scale datasets. This allows retrieval systems to perform flexible queries using either images or text descriptions. Building upon this, FashionCLIP (Chia et al., 2022) adapts CLIP-like contrastive learning specifically to the fashion domain and shows that domain-specific multimodal representations can improve product retrieval and representation quality. Similarly, FashionBERT (Gao et al., 2020) demonstrates that fashion retrieval requires fine-grained cross-modal matching because fashion-related texts often describe subtle details, such as style, pattern, or material, that are not sufficiently captured by global image representations alone. In addition, FashionIQ (Wu et al., 2021) shows that retrieval performance can benefit from combining visual features with natural-language feedback and attribute information, indicating that fashion retrieval is inherently a multimodal, semantically compositional task rather than a purely visual similarity problem.

In addition to embedding-based retrieval, several studies explore the role of structured attributes in improving retrieval quality. Chia et al. (2022) primarily demonstrate the effectiveness of fashion-specific multimodal representations, whereas Hou et al. (2019) provide stronger evidence that fine-grained semantic attributes improve interpretability and relevance to recommendations in fashion systems. Similarly, multimodal recommendation systems increasingly combine attribute-level matching with embedding similarity to better capture user preferences.

Beyond embedding-based retrieval, prior work in fashion recommendation also suggests that effective item matching depends not only on visual similarity but also on semantic coherence and compatibility. Han et al. (2017) modeled fashion compatibility using bidirectional LSTMs and showed that recommendation quality improves when relationships among outfit items are explicitly learned rather than treated as isolated similarity comparisons. Vasileva et al. (2018) further proposed type-aware embeddings to distinguish similarity within the same item type from compatibility across different item types, showing that a strong fashion representation should capture both interchangeability and complementarity. These findings are highly relevant to retrieval-based recommendation because they suggest that semantically meaningful retrieval should not rely on a single dense embedding, especially when the goal is to recommend related items that are compatible rather than merely visually similar.

In addition to compatibility learning, several studies emphasize the importance of structured semantic attributes for improving recommendation relevance and interpretability. Hou et al. (2019) showed that fashion recommendation benefits from projecting users and items into a fine-grained semantic attribute space, allowing the model to better reflect user preferences and provide interpretable recommendation signals. Compared with approaches based only on global representations, attribute-aware methods are better aligned with how users evaluate fashion items, since consumers often care about explicit details such as collar type, sleeve design, fabric, or pattern. Therefore, prior literature supports the view that structured garment attributes should not be treated merely as auxiliary annotations but as meaningful retrieval signals that complement dense-embedding similarity.

Taken together, the literature suggests that robust fashion retrieval should integrate both symbolic and dense representations. Dense multimodal embeddings are effective for capturing holistic similarity and cross-modal alignment, whereas attribute-based representations provide fine-grained semantic control, interpretability, and stronger alignment with user-preference reasoning. Furthermore, prior work in fashion recommendation indicates that effective item matching depends not only on visual similarity but also on semantic coherence, reinforcing the importance of incorporating structured garment attributes into the retrieval process. Accordingly, hybrid retrieval strategies that combine attribute matching with embedding similarity via score-level or rank-level fusion mechanisms are well supported by prior research. Following this line of reasoning, this work formulates retrieval as a multimodal matching problem that integrates structured attributes with embedding-based similarity, aiming to produce recommendations that are not only visually plausible but also semantically faithful and style-consistent.

Chapter 3: Dataset

Dataset Overview

This section describes the selection criteria, exploratory data analysis (EDA), and preprocessing procedures for all datasets utilized in this study, which focuses on 3 tasks: fine-grained fashion attribute learning, style classification, and attribute-aware retrieval.

The proposed framework is structured into three main stages. First, fashion images are decomposed into multiple fine-grained attributes trained with the Polyvore dataset (Han et al., 2017), which provides a large-scale and diverse collection of fashion items. This dataset enables the model to learn detailed attribute representations, including category, subcategory, color, material, and pattern, forming a structured semantic space for clothing understanding. The attribute taxonomy is further aligned with a predefined label library to ensure consistency and coverage across different garment types. Following an initial error analysis, limitations in attribute granularity and multi-label representation are identified. To address this, the Second-Hand Fashion dataset (Nauman, 2024) is incorporated to enhance the model's ability to capture richer, more realistic multi-label attribute distributions. This dataset provides more information about color, material, and pattern, with a wide variety, which improves robustness and generalization in attribute prediction.

Next, a separate model is developed for style classification. This stage leverages multiple datasets to capture high-level fashion semantics. In this part, we did not use the original Polyvore dataset because it does not provide style class annotations. Therefore, we instead adopted the FashionStyle14K dataset (Takagi et al., 2017), which includes explicit style labels for our task. FashionStyle14K is primarily based on Japanese fashion styles and is adopted as the core dataset. However, due to its limitation in gender diversity, additional datasets are incorporated to ensure coverage of both male and female fashion styles. These include a curated Clothing Style Computer Vision Dataset (FluxBlazeSU2022, 2025), the Fashion Product Images dataset (Param Aggarwal, 2019), and supplementary images collected from Google Image Search. By combining these sources, the model can learn diverse style representations that are not explicitly captured in attribute-level datasets.

Finally, for the recommendation system, the retrieval component is constructed based on the Polyvore dataset (Han et al., 2017). Specifically, the training split is used to build the retrieval gallery, and five different retrieval sets are generated using different random seeds to enhance robustness and reduce sampling bias. The query images are obtained from two sources: (a) randomly collected images from Pinterest to simulate real-world user inputs, and (b) the Fashion Product Images dataset (Param Aggarwal, 2019) to provide structured and cleaned background fashion items. This design ensures a clear separation between query distribution and retrieval candidates, enabling a more realistic evaluation of the recommendation system.

Product-Level Dataset

The Polyvore dataset (Han et al., 2017) was chosen for its rich collection and wide coverage of a variety of fashion items and up-to-date fashion garment images, which make it well-suited for studying multimodal fashion understanding. It contains images of individual clothing items, curated outfit sets, and associated information about the garments, including descriptions and related keywords. These data components enable both visual and text alignment, which is central to this research.

The original dataset consists of item metadata, images, and a category list. The image file comprises 261,057 images at a fixed resolution of 300 x 300 pixels (width x height), each displayed as an isolated clothing item on a clean, clear white background (see Figure 1). This clean background makes it easier for the model to identify and recognize the target garment during training and testing. Each file name corresponds to a unique image ID used to link image files with metadata.

Figure 1

Example images from the Polyvore dataset, which consist of a variety of categories, such as bags, tops, bottoms, shoes, accessories, etc., with a clean background.

The dataset provides metadata in two formats. The first metadata source defines hierarchical category relationships among clothing items. Each record in this file maps a category_id to its corresponding sub_category and main_category (see Table 1). The category_id field is shared between the item-level metadata, enabling cross-referencing between the two sources during preprocessing and label alignment. The second metadata contain item-level information, including title, url_name, description, categories, related keywords, category_id, and semantic_category (see Table 2). These records include fine-grained attributes such as color, material, pattern, gender, and brand, which can be used for category prediction tasks.

Table 1

Category File Information and Example

Schema	Type	Example Value(s)
category_id	int64	3
sub_category	object	dress
main_category	object	all-body

Note. This table represents the column names, data types, and example values from the category.csv file, where category_id links to its related metadata for category reference.

Table 2

Item-level Metadata Information and Example

Schema	Type	Example Value(s)
image_id	object	183179503
url_name	object	christian pellizzari floral jacquard trousers
description	object	Gold and black silk blend floral jacquard trousers from Christian Pellizzari. Color: Metallic. Gender: Female. Pattern: Floral. Material: Viscose/Polyester/Silk/Polyamide.
categories	object	Women's Fashion, Clothing, Pants, Christian Pellizzari pants
title	object	Christian Pellizzari floral jacquard trousers
related	object	Floral pants, Grey pants, Print pants, Patterned pants, Floral-print pants, Metallic pants
category_id	object	28
semantic_category	object	bottoms

Note. This table shows the column name, data types, and example values from item-level metadata, with more detailed information about the image. The category_id is linked to category.csv file.

Exploratory Data Analysis

Before splitting the data into training and evaluation sets, an exploratory data analysis (EDA) was conducted on the original Polyvore dataset to understand its structure and class distributions. The categories.csv data has a hierarchical category structure, with distinct classes for category ID and category, each with 158 and 141 data, respectively, while the main category has only 13 unique classes (see Figure 2). The distribution of category and subcategory is shown in Figure 3, which demonstrates that certain groups, such as tops and shoes, dominate the category column, and in the subcategory column, jeans and purses occur frequently. In addition, it indicates that two columns are right-skewed, so when splitting the data for training and testing, we need to account for this imbalance.

Figure 2

Unique values of category data. This shows the unique values in the dataset for the category we got 158 unique category_id, 141 different sub_category, and 13 unique main_category.

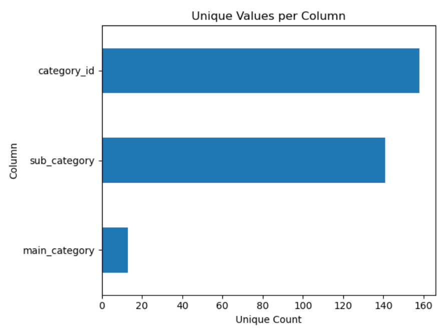
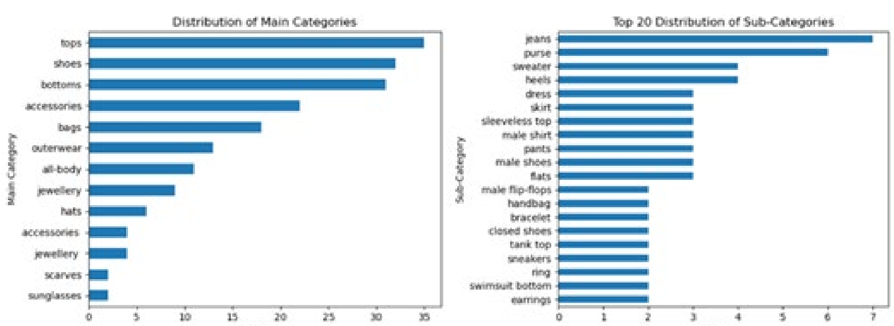


Figure 3

Distribution of category data. The show the distribution of the main category and the top 20 in the subcategory in the category's dataset. Where the top categories have the highest occurrence, and in the subcategory, jeans appear the most.



Data Preprocessing and Splitting

To prepare the dataset for model training and category classes prediction, we first performed data preprocessing and text annotation. First, the metadata was cleaned and standardized by transforming each raw record (see Table 2) into a unified format (see Table 3) consisting of the following labels: category, subcategory, color, material, and pattern, which represent the fundamental descriptive information of a clothing item. To convert unstructured text descriptions into structured attribute labels, we employed a large Language model, LLaMA3 (Grattafiori et al., 2024), a state-of-the-art model suitable for generating precise and structured output. The model was prompted, as shown in Table 5, to extract and categorize the three key attributes: color, material, and pattern from the raw textual metadata. Categories and subcategories are captured from the original metadata and the category.csv file. These preprocessing steps enable a consistent, trainable representation of each clothing item, moreover, to improve training and performance quality, we dropped rows with empty values in the column. For the field category and subcategory, each image contains a single value; on the other hand, for the classes color, material, and pattern, these are multi-value lists with no limit.

Table 3

Formatted Metadata Information and Example

Schema	Type	Example Value(s)
image_id	object	183179503
category	object	bottoms
sub_category	object	pants
color	object	gold,black
material	object	silk blend,viscose,polyester,polyamide
pattern	object	floral jacquard

Note. Formatted metadata schema, data types, and example values for the structured labeling output. For column image_id, each value is unique and represents the image's name. The category and sub_category columns are single-label. In contrast, color, material, and pattern are multiple-label, allowing multiple values per image.

After cleaning the metadata, a total of 35,990 data instances remain, with no missing values across all five attributes, making the dataset more robust for model training. The data was split into training, validation, and test sets with the following proportions: 80%, 10%, and 10%, respectively. We also proposed a label library that includes all unique values from the dataset for the model to train on and predict from, covering the five attributes: category, subcategory, color, material, and pattern (see Table 4).

Table 4

Label Library Example

Class Name	Values
category	accessories, all-body, bags, bottoms, hats….etc
sub_category	backpack, belt, blazer, blouse, bodie….etc
color	black, white, red, blue, yellow…etc
material	925 silver, cotton, viscose, silk, elastane…etc
pattern	abstract print, floral, glitter, fading, blush…etc

Note. This table shows an example of the label library, which includes five classes with all the unique values in the metadata.

Table 5

Prompt Used for Attribute Extraction

Prompt
You are a precise data cleaning and information extraction assistant for fashion product metadata. Extract the following attributes from the text below: – "color": list all color names mentioned (e.g., red, black, gold). If none, return an empty string. – "material": list all materials or fabrics (e.g., silk, cotton, denim, knit). If none, return an empty string. – "pattern": describe any visible or textual pattern (e.g., floral, striped, jacquard, plain). If none, return an empty string. Return ONLY a JSON object with exactly these keys: {{"color":"","material":"","pattern":""}} Text to analyze: {text}

Note. Present the designed prompt used for attribute extraction. The instruction explicitly defines the expected attribute types and output format to ensure consistency and reproducibility. This prompting approach enables automated, streamlined downstream model training and analysis.

Exploratory Data Analysis After Data Splitting

Prior to model training, a comprehensive EDA was conducted after cleaning the dataset into five columns. We examined the distributions and relationships within each column to understand the dataset's composition and potential class imbalance.

The frequency distribution of all the columns (see Figure 4) remains similar, they are long-tailed and right-skewed, with a high dominance of a few major categories. Category distribution shows that tops, bottoms, and shoes dominate, indicating a strong focus on core apparel items. Sub-categories reveal that sweater, skirt, and heels are among the most represented item types. Color distribution highlights black, gold, and white as the most frequent colors, consistent with neutral and versatile fashion tones. Material distribution shows that leather, cotton, and polyester are the dominant materials, reflecting typical fabric compositions in the modern clothing dataset. The pattern distribution indicates that plain, striped, and floral are the most common, spanning both minimalist and decorative styles.

Based on the distribution and correlations of the cleaned dataset, we split the dataset to assess imbalance across the splits. We propose grouped bar charts to investigate proportions across the train, valid, and test splits (see Figure 5). It shows that the dataset splits are well balanced, indicating consistent distribution across subsets; however, the overall category distribution remains long-tailed and right-skewed, showing class imbalance dominated by major categories. Although the dataset exhibits class imbalance, the distribution remains consistent across splits, minimizing the risk of bias introduced by uneven sampling.

Figure 4

Distribution of primary columns of the original dataset. All distributions are long-tailed and right-skewed, which we need to be aware of when splitting the data.

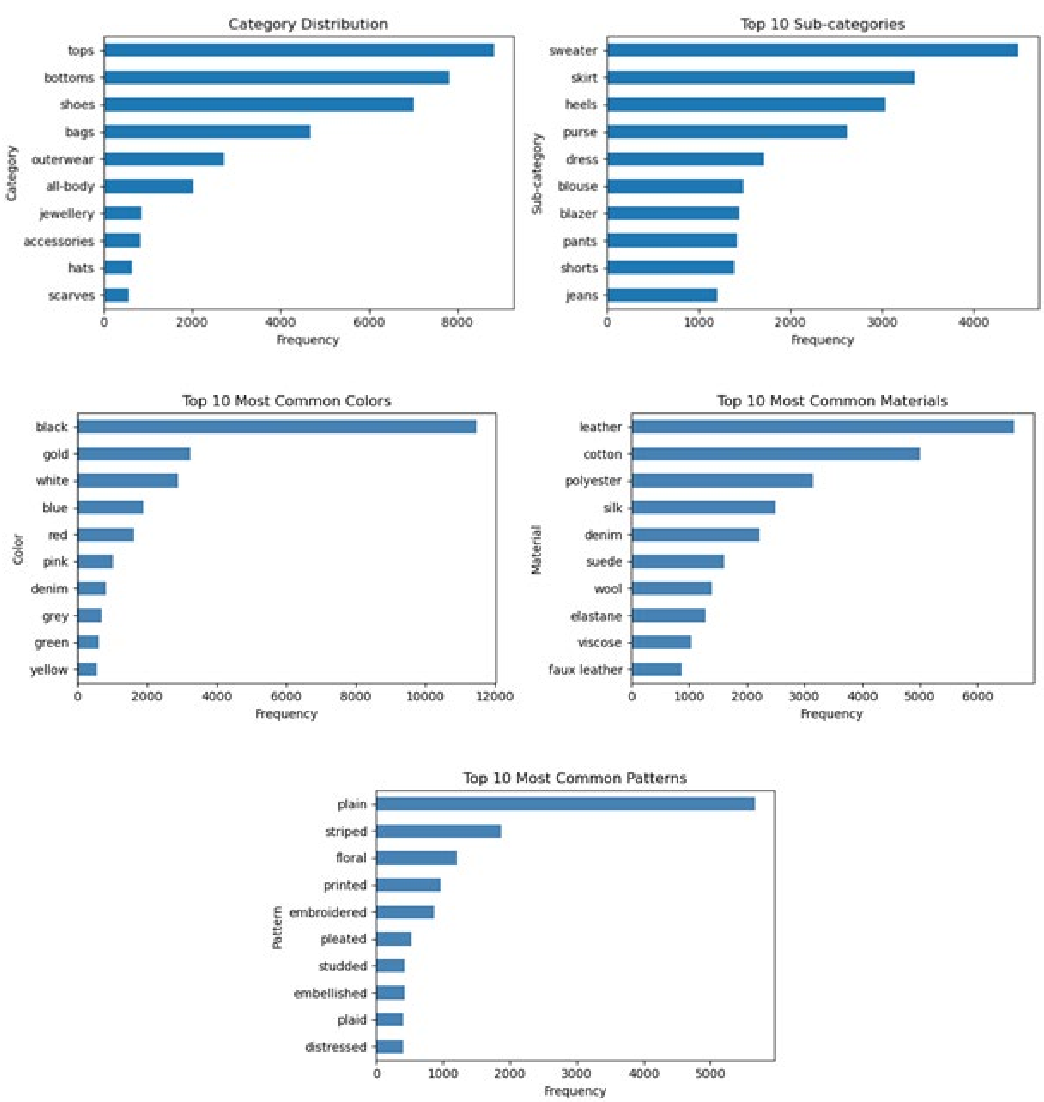
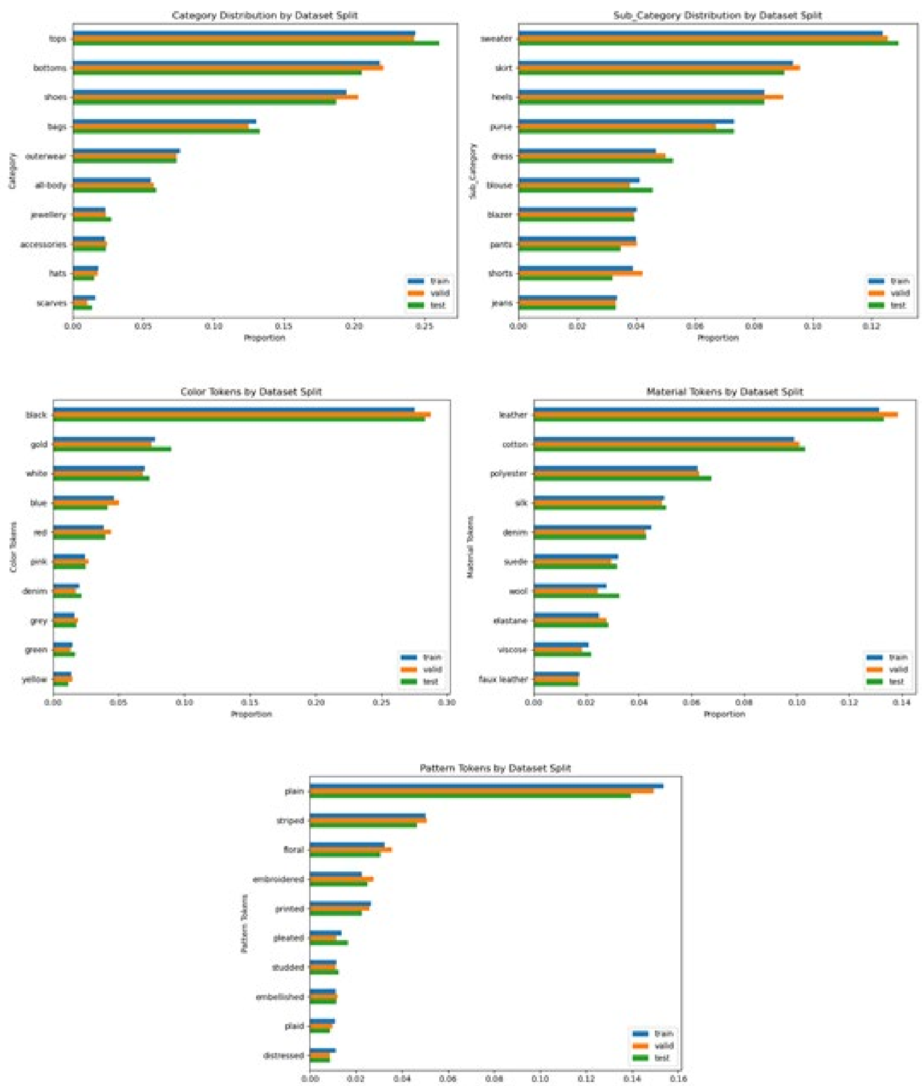


Figure 5

Distribution of primary columns of the split dataset. Although the dataset is highly skewed, we split it in balance, where every split has a similar number of labels for each class.



Attribute-Level Dataset

In this section, we examine Second-Hand Fashion dataset (Nauman, 2024) to understand its distribution and structure. We use the train split, which contains 28,248 samples, and select four columns: image, colors, material, and pattern. Examples of these selected fields are shown in Table 5. The pattern column contains approximately 49% missing values, the material column has around 4% missing values, and the colors column has no missing entries. It includes 25 unique colors, 15 material types, and 936 unique patterns. From Table 6, we can observe that the dataset contains many inconsistent formatting styles. Therefore, we performed several cleaning steps: removing quotation marks in the color column, deleting percentages and numeric values from the material column, and converting both colors and material into a standardized list format, such as [blue, white].

Table 5

Column Data Types and Values

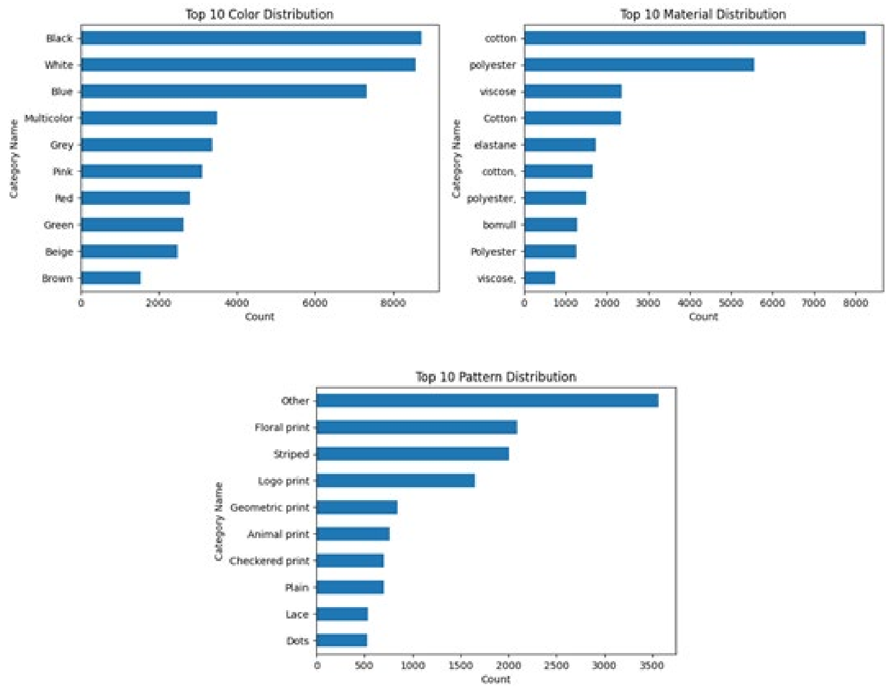
Column	Data Type	Values
colors	object	['Blue', 'White']
pattern	object	Floral print
material	object	33%cotton 67%polyester

Note. This table presents the column data types and values. Where colors can include multiple values, as does the material, moreover, the material contains the percentage of a specific kind of material.

Also, visualized the distribution with the top 10 most frequent classes for clarity, see Figure 6 for the bar chart. For colors, the most common values are basic tones like black and white, followed by blue. For material, the most frequent label is other, followed by floral print and stripe. For patterns, the most common classes are cotton, polyester, and viscose, which align with materials commonly used in clothing.

Figure 6

Top 10 category distribution of multi-label. This reflects the most commonly occurring colors, materials, and patterns in everyday clothing.



Style-Level Dataset

In this study, we focus on binary gender categories: male and female, due to the limitations of publicly available datasets. Nevertheless, we acknowledge and respect the diversity of gender identities; this constraint is solely imposed by the availability and structure of existing datasets. Also, we did not use the original Polyvore dataset because it does not provide style class annotations. Therefore, we instead employed the following datasets, which include explicit style labels for our task.

We adopt FashionStyle14K (Takagi et al., 2017) as the primary dataset for style classification. This dataset comprises 14 Japanese fashion styles: Conservative, Dressy, Ethnic, Fairy, Feminine, Gal, Girlish, Kireime-Casual, Lolita, Mode, Natural, Retro, Rock, and Street. However, it mainly contains female images. Styles such as Girlish, Dressy, Lolita, Fairy, Feminine, and Kireime-Casual are inherently female-oriented. Therefore, no additional male samples are collected for these categories.

To enable both male and female style classification, especially for downstream tasks on the Polyvore dataset Han et al., 2017 (), we further collect male fashion images to complement the missing categories. For male street-style images, data are obtained from Clothing Style Computer Vision Dataset (FluxBlazeSU2022, 2025). Additional samples are collected from the Fashion Product Images dataset (Param Aggarwal, 2019) and web sources.

For web-based data collection, images are retrieved from Google Images, Getty Images, and Pinterest using targeted keywords, such as “Men's rock fashion“, “Men's Gyaru fashion“, “Men's conservative fashion“, “men's retro fashion style“, “Retro Fashion“, “Japanese mode style“, “natural outfit man“, “natural style men outfit“, “men earthy casual outfit“, “Men's rock fashion“, and “Men's Conservative-Chic Fashion.”

Furthermore, three additional styles, Formal, Sports, and Casual, are introduced from the Fashion Product Images dataset (Param Aggarwal, 2019) to enrich the style space. As a result, the final style taxonomy includes 17 categories. Example images for each style are illustrated in Figures 7 and 8. The semantic definitions of each style are summarized as follows:

- Casual: Everyday style focused on comfort and simplicity for day-to-day wear.
- Ethnic: Inspired by traditional or indigenous clothing and cultural motifs, often incorporating bold patterns, handcrafted textiles, and symbolic design elements from specific regions or cultures.

- Formal: A style typically worn for special occasions or professional settings, emphasizing structured tailoring and polished pieces.
- Sports: Athletic-inspired fashion that prioritizes function and comfort.
- Conservative: A traditional and understated style focusing on modesty, neutral colors, and classic silhouettes that avoid bold trends or revealing cuts.
- Dressy: A more elevated version of casual that incorporates dressier elements, for example, blazers, button-downs, or polished accessories.
- Fairy: Whimsical, romantic aesthetic often featuring pastel tones, lace, and delicate details that evoke a magical or fantasy-inspired feel.
- Feminine: Characterized by elegance, softness, and lightness, often conveying a romantic and refined aesthetic associated with mature femininity.
- Gal: Short for "Gyaru". A Japanese fashion subculture featuring bold, trendy outfits, exaggerated makeup, dyed hair, and a playful, rebellious attitude.
- Girlish: Emphasizes youthful, cute, and playful elements, such as pastel colors, bows, ruffles, and motifs that convey innocence or sweetness.
- Kireime-Casual: A Japanese-derived term referring to a neat and refined casual look that combines clean, polished pieces with relaxed items to give an effortless but put-together impression.
- Lolita: A distinct Japanese fashion subculture inspired by Victorian and Rococo aesthetics, emphasizing modest, doll-like silhouettes with puffed skirts, lace, blouses, and decorative headwear.

- Mode: Also called modern or minimalist. A fashion-forward, modern style that emphasizes contemporary cuts, minimalist silhouettes, neutral palettes, and a sleek, trend-aware aesthetic.
- Natural: Prioritizes simplicity and organic materials, often using earthy colors, comfortable fits, and unstructured shapes.
- Retro: A nostalgic aesthetic drawing inspiration from past decades, mostly the 60s, 70s, and 80s, recreating or referencing vintage cuts, patterns, and colors.
- Rock: A rebellious, edgy fashion influenced by rock music culture, usually featuring leather, black-dominated palettes, and bold hardware.
- Street: Urban, youth-driven style rooted in street culture and casual wear, often combining sneakers, hoodies, and graphic tees with an emphasis on individuality and subcultural identity.

Figure 7

FashionStyle14 dataset female-oriented style categories. Representative samples of female-oriented style categories in the FashionStyle14 dataset, covering girlish, dressy, lolita, fairy, feminine, and kireime-casual.

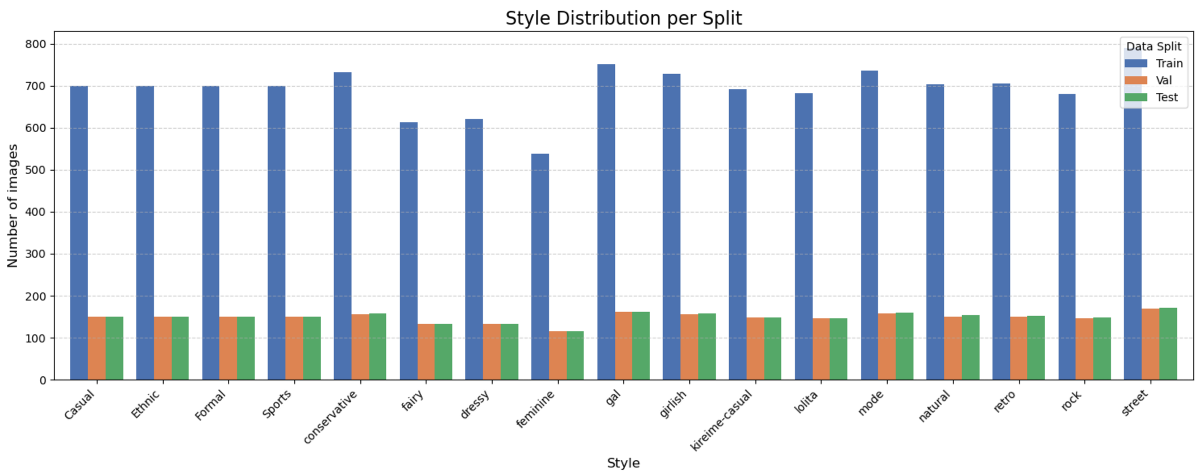
Figure 8

Additional style categories. Additional fashion styles include minimalist, bohemian, vintage, preppy, grunge, athleisure, business-casual, punk, chic, and avant-garde, representing a diverse range of aesthetic preferences beyond the primary categories.

Finally, to mitigate dataset imbalance, we perform data cleaning and sampling strategies to transform the dataset from a highly imbalanced distribution into a more balanced one. The resulting distribution is illustrated in Figure 9.

Figure 9

Distribution of styles across different data splits. By incorporating male images to match the styles, the dataset achieves a nearly balanced distribution.



Recommendation System Dataset

The recommendation system is designed as an image-to-item retrieval task, where a query image is used to retrieve visually and semantically similar fashion items from a predefined gallery. The query images are collected from two sources: (a) Fashion Product Images dataset (Param Aggarwal, 2019), which offers structured and labeled fashion items. (b) Pinterest, which provides diverse and realistic outfit images that simulate real-world user inputs. This combination ensures both diversity and controllability in the input distribution. The query dataset exhibits a deliberately imbalanced distribution across categories, which includes garment types such as sweaters, shoes, pants, shirts, and bags, limited to only two samples per category, whereas more generic or broad categories, including free-items, footwear, apparel, and accessories, contain up to ten samples each. This imbalance is primarily due to the inherent differences in category granularity and semantic coverage. Fine-grained categories are more visually and functionally specific, and a small number of representative queries is sufficient to capture their characteristics. In contrast, broader categories such as apparel and accessories encompass a wider range of styles and visual variations, requiring more queries to adequately capture their diversity. Moreover, this design aligns with real-world recommendation scenarios, where user queries are often skewed toward high-level or ambiguous categories rather than evenly distributed across all item types. As a result, the imbalanced query set allows us to better evaluate the model's robustness under realistic conditions, particularly its ability to generalize across both well-defined and semantically broad categories.

The retrieval gallery is constructed from the Polyvore dataset (Han et al., 2017), specifically using its training split. Polyvore provides curated outfit-level fashion items, making it suitable for downstream recommendation and retrieval tasks. To evaluate the system's robustness, five retrieval sets are constructed with different random seeds. Each seed generates a distinct sampling of the retrieval gallery, allowing us to analyze the stability of model performance under varying data distributions and to reduce potential sampling bias. Importantly, the query set and retrieval gallery are constructed from different data sources, ensuring a clear separation between input and candidate items. This prevents data leakage and better reflects real-world recommendation scenarios.

All images, including both query and retrieval items, are processed by the proposed classification model to extract six fine-grained attributes, forming a unified semantic representation for retrieval. We analyze and report the data distribution of both query and retrieval datasets across different seeds, as shown in Figures 10-15. This includes the distribution of categories and attributes, which provides further insight into the consistency and fairness of the evaluation setup.

Figure 10

Class distribution of the Query dataset.

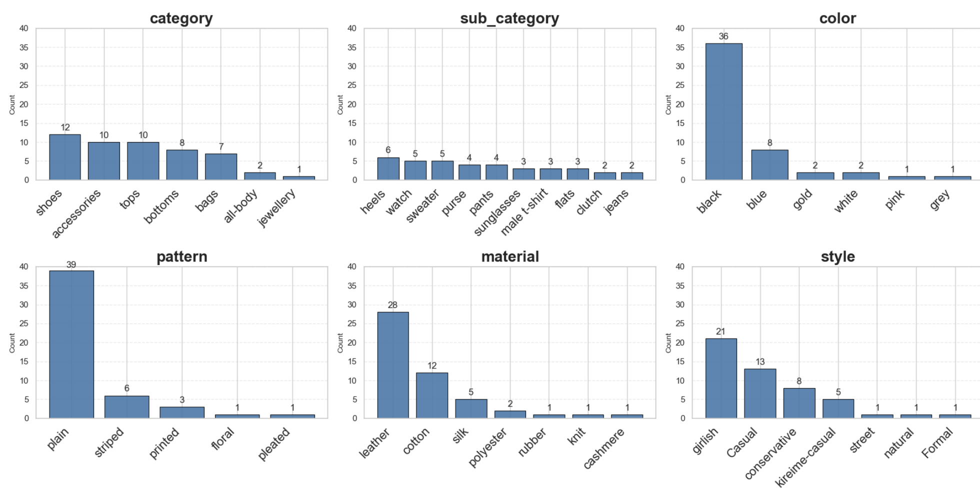


Figure 11

Class distribution of the seed 1 retrieval dataset.

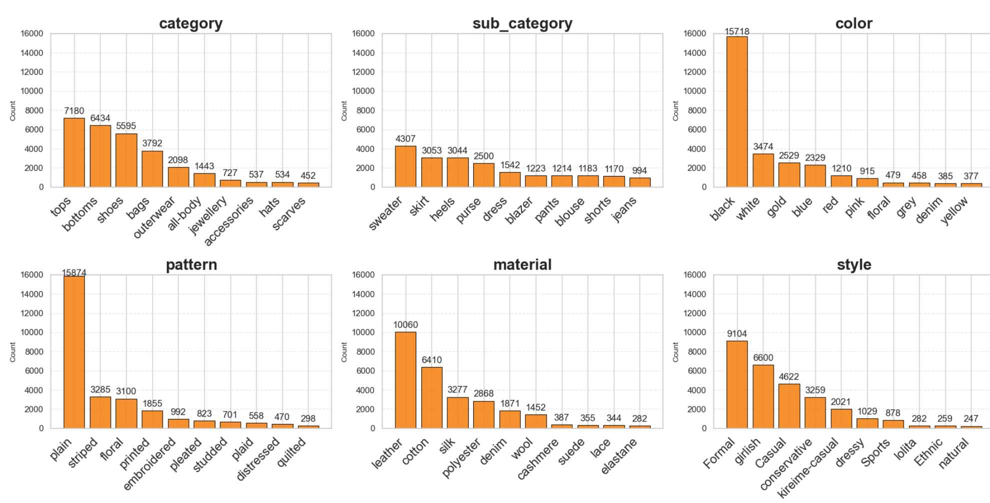


Figure 12

Class distribution of the seed 2 retrieval dataset.

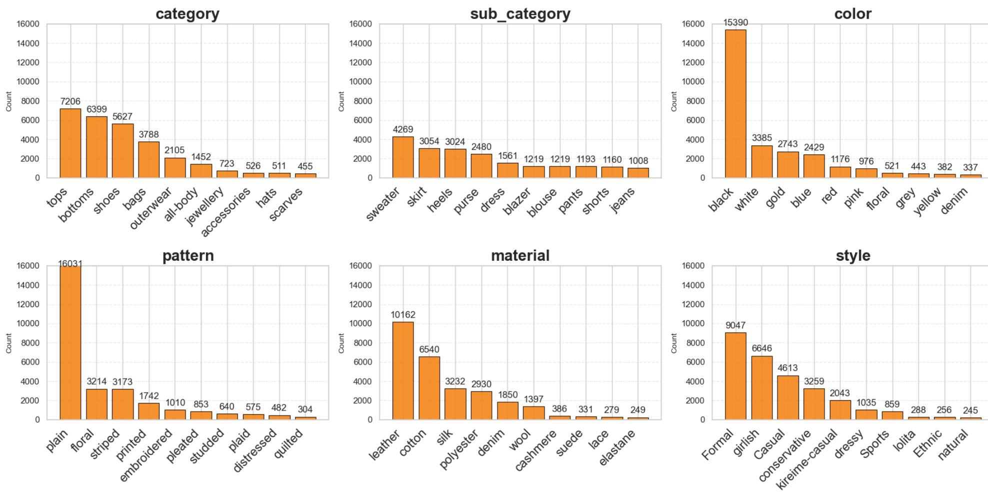


Figure 13

Class distribution of the seed 3 retrieval dataset.

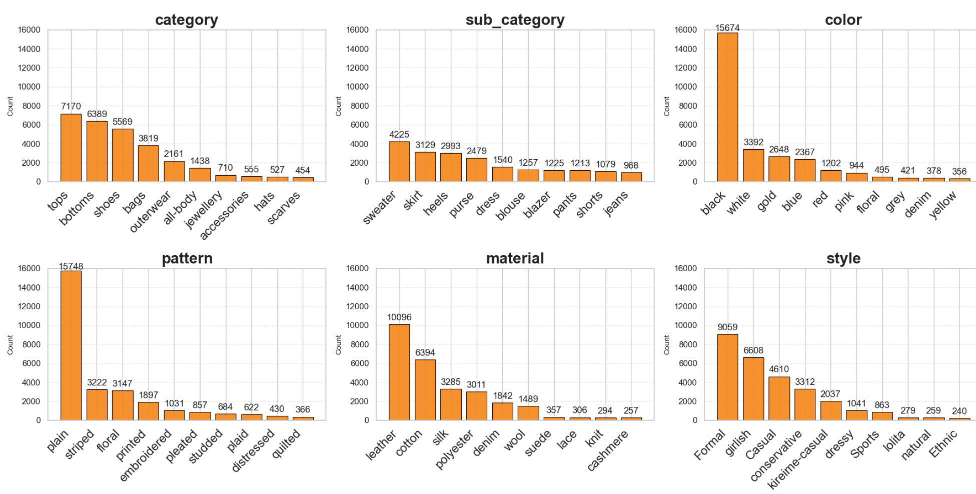


Figure 14

Class distribution of the seed 4 retrieval dataset.

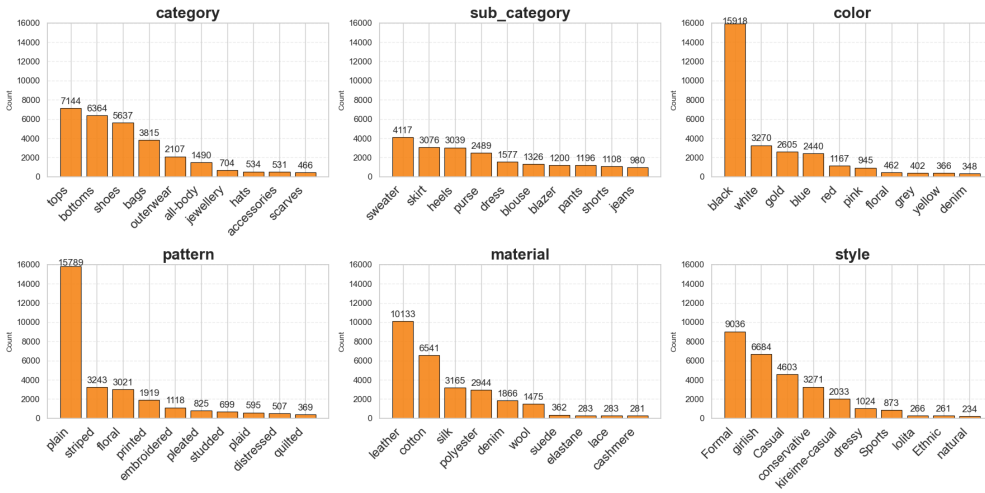
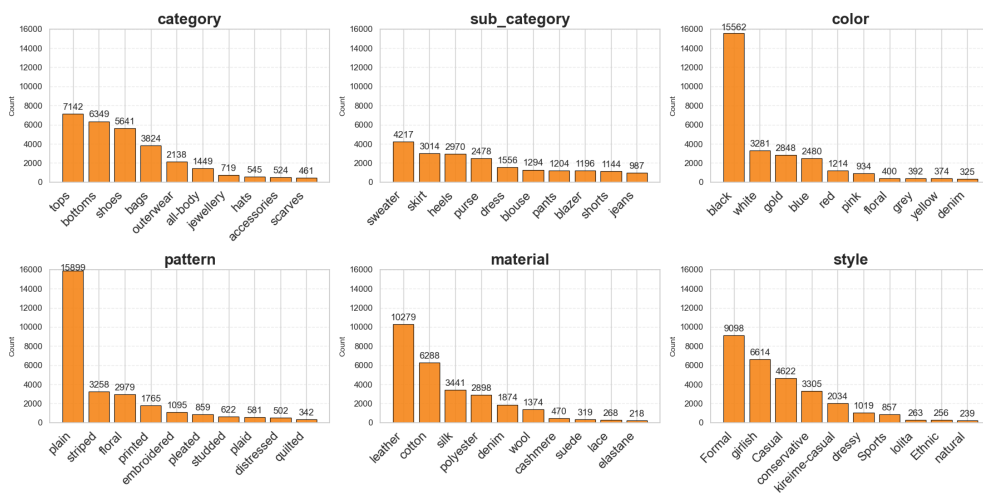


Figure 15

Class distribution of the seed 5 retrieval dataset.



Chapter 4: Evaluation Metrics

To evaluate the performance of the generated prediction model, we use standard classification evaluation metrics, such as Accuracy, Precision, Recall, and F1. Since this study includes both single-label and multi-label classification tasks, different evaluation criteria are applied to each. For single-label classification of attributes such as category and subcategory, we used the standard evaluation metrics mentioned above. However, we still used semantic similarity to assess how well the model predicts, since certain words refer to the same concept but differ slightly in form. For multi-label classification, we may encounter similar words but not the exact same combination of word sets. To address this, we implemented relaxed matching and semantic similarity.

Single-Label Evaluation Metrics

Accuracy

Accuracy measures the overall proportion of correctly classified instances among all predictions, whether positive or negative. A perfect model would have both zero false positives and false negatives. The best accuracy is 1.0 or 100%.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$

Precision

Precision quantifies the proportion of correctly predicted instances among all predicted instances for a class. High precision indicates a low false-positive rate and, therefore, a precision of 1.0 or 100%.

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

Recall

Recall measures the model's ability to correctly classify positives. A good model would have zero false negatives and therefore, its recall would be 1.0 or 100%.

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

F1 Score

The F1 score is the harmonic mean of precision and recall, providing a balanced measure between them. When precision and recall scores increasingly diverge, the quality of the mean declines. A higher score indicates a better-quality classifier.

$$F1\ Score = \frac{2*Precision*Recall}{Precision+Recall} \quad (4)$$

Multi-Label Evaluation Metrics

Unlike single-label classification, where each sample has only one ground truth label, multi-label tasks allow multiple correct labels per instance. Using strict exact match would not represent the best evaluation

Relaxed Match

Relaxed match accuracy measures the accuracy of the intersection of each multi-label, where there is at least one overlapping label between the ground truth and prediction.

$$Relaxed\ Match(gt, pred) = \begin{cases} 1, if |gt \cap pred| \geq 1 \\ 0, otherwise \end{cases} \quad (5)$$

$$Relaxed\ Accuracy(gt, pred) = \frac{1}{N}\sum Relaxed\ Match(gt, pred) \tag{6}$$

For example, ground truth has color $gt = \{red,\ blue\}$, and the prediction has color {red, pink}, they have color red intersect, so in this scenario, the relaxed match score is 1. If the prediction has color $pred = \{pink, white\}$, there is no intersection between the colors, so the relaxed match score would be 0. If the prediction has only color {red}, even if it still intersects with both, the score remains 1. After calculating each row's score, we average the scores to obtain the final relaxed match score.

Semantic Similarity

Semantic similarity assesses the semantic closeness between the ground truth and the predicted labels, rather than relying solely on exact lexical matching. We compute semantics similarity scores, where each label is first converted into a text embedding using the pre-trained text encoder from CLIP (Radford et al., 2021). The cosine similarity between embeddings is computed as:

$$Cosine\ Similarity(E_{pred}, E_{gt}) = \frac{E_{pred} \cdot E_{gt}}{|E_{pred}| \cdot |E_{gt}|} \tag{7}$$

For single-label, the final semantic score is presented after the cosine similarity. For multi-label, after computing all cosine similarities across attributes in the embedding space, we find the maximum score and average it.

$$Semantic\ Similarity = \frac{1}{|num\ of\ gt|}\sum max\ Cosine\ Similarity(E_{pred}, E_{gt}) \tag{8}$$

The pipeline for semantic similarity follows the sequence shown in Figure 16 for single-label and Figure 17 for multi-label. We apply this evaluation to both single-label and multi-label classification to ensure we have many labels that may be semantically similar. When using the accuracy method, it treats "bottom" and "bottoms" as distinct words, even though they are semantically the same and differ only in spelling or form.

Figure 16

Semantic similarity evaluation pipeline for single-label.

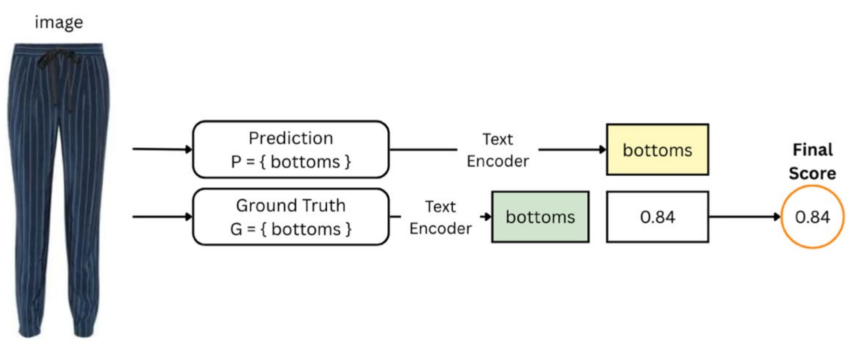
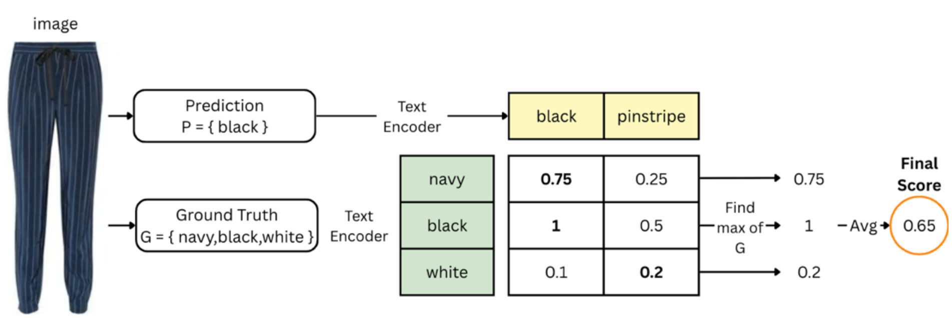


Figure 17

Semantic similarity evaluation pipeline for multi-label.



For instance, ground truth has a color pair $gt = \{red,\ blue\}$, and the model's predicted color labels are $pred = \{maroon, linen, navy\}$, where maroon is a deep and intense red with subtle brown undertones, and linen is a warm-toned brown. Both will be embedded in a vector space and then used to compute pairwise similarities (see Table 7 for an example). Then we find the maximum similarity for each ground-truth row: for red, it's 84%, and for blue, it's 91%. Then we calculate the average, yielding a semantic similarity of 88%.

Table 7

Example of Semantic Similarity

GT / PRED	maroon	linen	navy
red	0.84	0.15	0.12
blue	0.25	0.1	0.91

Note. Example of pairwise cosine similarity between ground-truth colors (rows) and predicted colors (columns); 'red' is closest to 'maroon' and 'blue' is closest to 'navy' in the embedding space."

Attribute Matching Quality Evaluation Metrics

To evaluate the quality of attribute alignment between the query item and the retrieved results, this study adopts three complementary metrics: Attribute Match Rate (AMR), Exact Attribute Match (EAM), and Weighted Attribute Matching (WAM). These metrics are to assess retrieval quality from different perspectives, including partial overlap, exact agreement, and semantically weighted overall matching.

Attribute Match Rate

Attribute Match Rate (AMR) measures the average attribute-level matching quality over the top-k retrieved items. This metric is intended to capture partial correctness of retrieved attributes, especially for attributes that may contain multiple labels.

For a given query q and retrieved item d_i, the attribute matching score for attribute a is defined differently depending on whether the attribute is single-label or multi-label. For single-label attributes, such as category, subcategory, and style, the match score is defined by exact match, where $label_a(q)$ and $label_a(d_i)$ denote the attribute labels of the query and retrieved item, respectively:

$$m_a(q, d_i) = \begin{cases} 1, if\ label_a(q) = label_a(d_i) \\ 0, otherwise \end{cases} \tag{9}$$

For multiple-label attributes, such as color, material, and pattern, the match score is defined using Jaccard similarity:

$$m_a(q, d_i) = \frac{|labels_a(q) \cap labels_a(d_i)|}{|labels_a(q) \cup labels_a(d_i)|} \tag{10}$$

The final AMR@K for attribute a is computed as the average matching score across the top-k retrieved items:

$$AMR_a@K = \frac{1}{K}\sum_{i=1}^{K} m_a(q, d_i) \tag{11}$$

and the dataset-level score is obtained by averaging over all queries.

For example, if the ground-truth color set is {red, white} and the retrieved color set is {red, black, pink}, then:

$$m_{color}(q, d_i) = \frac{|\{red\}|}{|\{red, white, black, pink\}|} = 0.25$$

Exact Attribute Match

While AMR captures partial overlap, Exact Attribute Match (EAM) provides a stricter evaluation by measuring whether the retrieved attribute matches the ground-truth attribute exactly. For both single-label and multi-label attributes, the exact-match score is defined as:

$$e_a(q, d_i) = \begin{cases} 1, & \text{if the attribute values are exactly identical} \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

For single-label attributes, this means the retrieved label must match the query label. For multi-label attributes, this means that the retrieved label set must match the ground-truth label set exactly, with no missing or extra labels. The final EAM@K is computed as the average exact-match score over the top-k retrieved items:

$$EMA@K = \frac{1}{K}\sum_{i=1}^{K} e_a\,(q, d_i) \tag{13}$$

and then averaged across all queries.

For example, the ground-truth color is {red, white}, and the retrieved color set is {red, black, pink} the score is 0. On the other hand, if the retrieved {red, white} score is 1.

Weighted Attribute Score

To assess overall retrieval quality across multiple attributes, we further introduce Weighted Attribute Matching (WAM), which aggregates attribute-level similarity scores into a single overall score. Unlike AMR and EAM, which are evaluated separately for each attribute, WAM reflects the global semantic consistency between the query item and a retrieved item.

The WAM score is computed in four steps: (a) Retrieve top-K candidate items. (b) Compute attribute-level semantic similarity for each attribute. (c) Apply weights. (d) Aggregate weighted scores into a final item-level matching score.

Let $s_a(q, d_i)$ denote the attribute-level semantic similarity for the attribute a between the query q and the retrieved item d_i. Where A is the set of all attributes, w_a is the weight assigned to attribute a, and $s_a(q, d_i)$ is the matching score for attribute a. The weighted attribute matching score is defined as:

$$WAM(q, d_i) = \frac{\sum_{a \in A} w_a \, s_a(q, d_i)}{\sum_{a \in A} w_a} \tag{14}$$

The attribute weights are defined as: $w_{\text{category}} = 4.0$,, $w_{\text{sub\_category}} = 3.0$, $w_{\text{color}} = 3.0$, $w_{\text{style}} = 2.0$, $w_{\text{pattern}} = 1.0$, and $w_{\text{material}} = 1.0$. These weights reflect the relative importance of each attribute in fashion retrieval. Category and subcategory are assigned higher weights because they define the core product type, while color is also emphasized due to its strong influence on user perception and retrieval relevance. In contrast, pattern and material are treated as secondary attributes and thus assigned lower weights. The final WAM@K score can then be computed by averaging the weighted item-level scores across the top-k retrieved results:

$$WAM@K = \frac{1}{K} \sum_{i=1}^{K} WAM(q, d_i) \tag{15}$$

and then averaging over all queries in the dataset.

Retrieval Ranking Quality Evaluation Metrics

The goal here is to measure whether the retrieved results are not only relevant but also correctly ordered. This part adopts two complementary metrics: Normalized Discounted Cumulative Gain (nDCG) and Mean Reciprocal Rank (MRR).

Normalized Discounted Cumulative Gain

We leverage Normalized Discounted Cumulative Gain (nDCG) to evaluate the ranking quality of retrieval results under graded semantic relevance. Unlike binary relevance metrics, nDCG considers not only whether a retrieved item is relevant, but also how relevant it is. This is suitable for our setting because attribute matching is measured by semantic similarity scores in the range [0, 1], rather than exact-match binary labels.

For a given query, let rel_i denote the semantic relevance score of the item ranked at position i. The Discounted Cumulative Gain at rank K is defined as:

$$DCG@K = \sum_{i=1}^{K} \frac{rel_i}{log_2(i+1)} \tag{16}$$

This formulation gives higher weight to highly relevant items appear at earlier ranks, while applying a logarithmic discount to lower-ranked items. To normalize the score, we compute the Ideal DCG (IDCG) by sorting the same top-K relevance scores in descending order:

$$IDCG@K = \sum_{i=1}^{K} \frac{rel_i^*}{log_2(i+1)} \tag{17}$$

Where rel_i^* is the relevance score in the ideal ranking order. The normalized score is then:

$$nDCG@K = \sum_{i=1}^{K} \frac{DCG@K}{IDCG@K} \tag{18}$$

Thus, nDCG@K ranges from 0 to 1, where a higher value indicates that the retrieval system ranks more semantically relevant items closer to the top positions.

For example, assume the top-5 semantic relevance scores of a query are [0.9, 0.8, 0.4, 0.7, 0.2]. Then we calculate DCG@K and IDCG@K:

$$DCG@5 = \frac{0.9}{log_2(2)} + \frac{0.8}{log_2(3)} + \frac{0.4}{log_2(4)} + \frac{0.7}{log_2(5)} + \frac{0.2}{log_2(6)} = 1.983$$

Sort the score in decreasing order: [0.9, 0.8, 0.7, 0.4, 0.2]

$$IDCG@5 = \frac{0.9}{log_2(2)} + \frac{0.8}{log_2(3)} + \frac{0.7}{log_2(4)} + \frac{0.4}{log_2(5)} + \frac{0.2}{log_2(6)} = 2.004$$

$$nDCG@5 = \frac{1.983}{2.004} = 0.989$$

In our implementation, the relevance score of each retrieved item is the semantic similarity score computed between the query attribute and the retrieved attribute. Therefore, nDCG@K measures whether the retrieved results are not only relevant, but also well-ordered according to semantic relevance.

Mean Reciprocal Rank

Mean Reciprocal Rank (MRR) to evaluate how early highly relevant results appear in the ranked retrieval list. MRR emphasizes the position of the most relevant result, making it especially useful for retrieval tasks where the top-ranked result is most important to the user.

In the conventional binary setting, reciprocal rank is defined as the inverse of the rank position of the first relevant item. However, in this work, we use a soft semantic version of MRR based on graded relevance scores. For a retrieved item at rank i with semantic relevance score S_i, we define:

$$MRR@K = \max_{1 \le i \le K} \frac{S_i}{i} \tag{19}$$

This formulation preserves the intuition of reciprocal rank while incorporating semantic relevance. A higher score is obtained when a highly relevant item appears earlier in the ranking list. At the dataset level, the final MRR@K is computed by averaging the per-query MRR values across all queries.

For example, assume the top-5 semantic relevance scores of a query are [0.9, 0.8, 0.4, 0.7, 0.2]. Then calculate and find:

$$MRR@5 = max\left(\frac{0.9}{1}, \frac{0.8}{2}, \frac{0.4}{3}, \frac{0.7}{4}, \frac{0.2}{5}\right) = 0.9$$

Assume another query got score [0.2, 0.3, 0.85, 0.6, 0.1]. Can get:

$$MRR@5 = max(\frac{0.2}{1}, \frac{0.3}{2}, \frac{0.85}{3}, \frac{0.6}{4}, \frac{0.1}{5}) = 0.283$$

We can observe the difference between the two queries: although the second query contains a highly relevant item with a score of 0.85, it only achieves an MRR of 0.283. This indicates that a highly relevant item is ranked lower, and the reciprocal rank formulation significantly penalizes such delayed appearances. Consequently, MRR captures both relevance and ranking order, highlighting the importance of placing highly relevant items at the top of the retrieval list.

Chapter 5: FACTOR: Feature-Aware Contextual Optimization and Refinement

Introduction

This chapter provides a detailed introduction to the research using the Polyvore dataset (Han et al., 2017) to train CLIP (Radford et al., 2021). We build the framework on top of CLIP's multimodal visual language model (VLM) to perform attribute classification for fashion items. Given a clothing garment image, the model extracts five key fashion attributes: category, subcategory, color, material, and pattern. Building on findings from prior research (Hou et al., 2019), which suggest that users tend to express preferences toward specific clothing attributes, we further consider the hierarchical nature of human visual perception in fashion understanding. Specifically, when observing a garment, individuals typically first recognize its overall silhouette or category, followed by its color, and subsequently attend to finer details such as material and pattern. This hierarchical perception motivates the selection of attributes that capture both coarse-grained and fine-grained characteristics of clothing items. We developed the idea of extracting not only the basic category and subcategory, but also extending the attributes to color, material, and pattern. These elements are among the most crucial factors for identifying and keeping up with current fashion trends, which is why we focus on capturing them in our system.

CLIP is a neural network trained on huge image-text pairs to align images and textual descriptions in a shared embedding space (Radford et al., 2021). In our application, first use CLIP in a zero-shot manner as a baseline: given an image, feed CLIP a set of textual prompts describing possible labels and let it choose the closest match. For example, to predict an item's color, we compare the image embedding to text embeddings for prompts like "a photo of a red item", "a photo of a blue item", etc., covering all candidate colors. The label whose prompt is most like the image is selected as CLIP's prediction.

We apply zero-shot method as the baseline, which doesn't require training models on our own data and challenges CLIP's ability to identify fashion attributes from the box. Finally, CLIP's performance will be compared with FashionCLIP, which has the same architecture as CLIP but is specialized in fashion images and their text (Chia et al., 2022). By comparing CLIP and FashionCLIP without additional learning, the extent to which pre-training the model can benefit recognition in the fashion category and for attributes will be determined. For the experimental method, we implement several fine-tuning strategies, including linear probe, prompt fine-tune, hybrid fine-tune (which combines linear probe and prompt fine-tune), and full fine-tune, to identify the best model to proceed with. Finally, we proposed the FACTOR framework with a two-stage linear probe for attribute-aware fine-tuning.

The results show that linear probe has the best accuracy among all attributes and achieves the highest prediction performance among the methods experimented with. This is an interesting finding, because linear probe only adds a new linear head on top of the embedding space after CLIP has encoded the image, which aligns with previous research findings (Huang et al., 2024), that lightweight tuning could perform better than heavy or tuning.

Experimental Framework

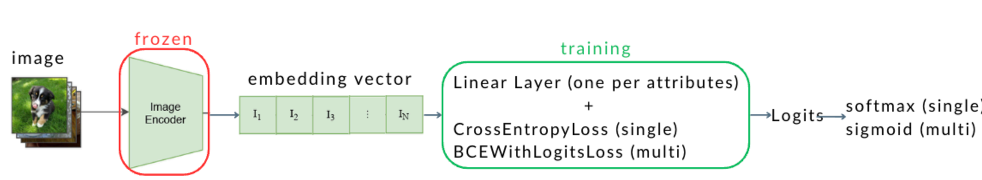
Our goal is to find a method that can provide high accuracy of predicting fashion-specific attributes, covering both single-label (category and subcategory) and multiple-label (color, material, and pattern) tasks. First, evaluate four basic adaptation strategies of fine-tuning CLIP: Linear Probe, Prompt Tuning, Hybrid Tuning, and Full Fine-tuning. Below, we dive deep into the training pipeline, optimization details, and advantages and limitations of each method.

Linear Probe

In this method, the pre-trained image and text encoders of CLIP are frozen only the image classification layer is trained. This technique is an efficient and stable approach with minimal risk of overfitting, in which it doesn't adjust anything significant about the original model but instead adds weights to the embedding space. The workflow is as follows: freeze CLIP image and text encoders, pass each image through them, obtain the embedding space, and, for each attribute, attach a lightweight linear classifier. Then apply loss for both single-label and multi-label, and accordingly, use cross-entropy and binary cross-entropy with logits loss. The complete pipeline is shown in Figure 18.

Figure 18

Linear Probe Pipeline. The CLIP encoders remain frozen and only a lightweight classifier is trained on top of the image embeddings.



Prompt Tuning

Figure 19 shows the whole pipeline of prompt tuning. This technique introduces learnable soft prompts where the text encoder adapts to fashion semantics. In the training process, we frozen both CLIP's image and text encoder, converting each attribute class into an easy prompt template that the vanilla CLIP is been pre-trained for (see Table 8). Then, put the labels into the prompt and pass through the text encoder, Then, the encoded texts and images are aligned in the embedding space, calculate cosine similarity between them, and finally apply cross-entropy and binary cross-entropy loss to the prompt embeddings.

Figure 19

Prompt-tuning pipeline. The text encoder is frozen and trainable prompt embeddings are learned to better align images with attribute-specific text prompts.

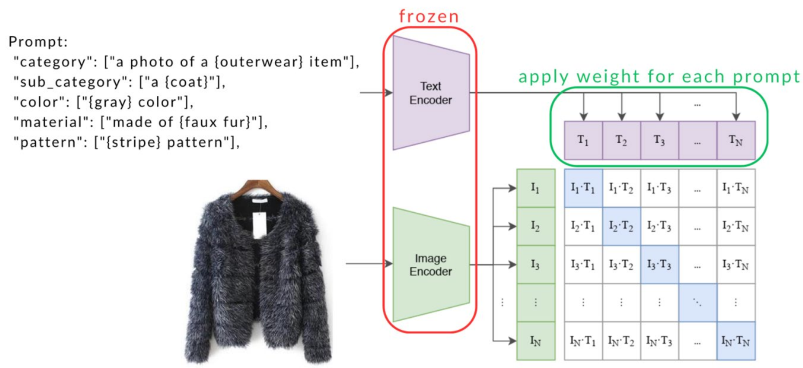


Table 8

Prompt Used in Prompt Tuning Training

Attirbutes	Prompt
category	“a photo of a {label} item", "a fashion photo of {label}"
subcategory	"a {label}", "a {label} garment"
color	"{label} color", "a {label} colored item"
material	"made of {label}", "a {label} material garment"
pattern	"{label} pattern", "a {label} print"

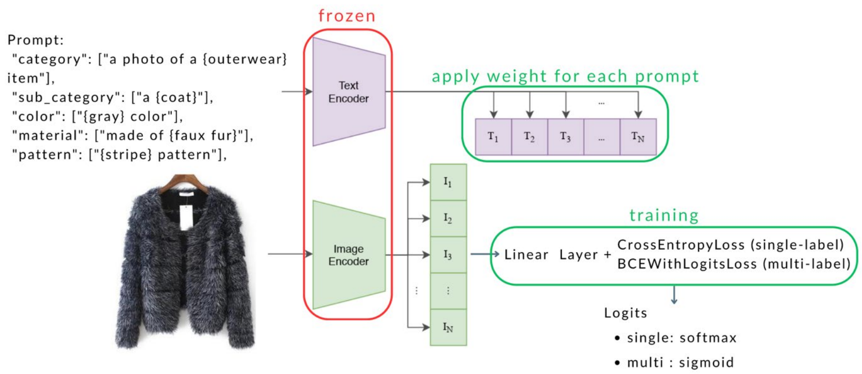
Note. Prompt used in training the model. Showing the short and consistent templates applied across all labels, since CLIP was not trained with longer prompts during pretraining.

Hybrid Tuning

After achieving a high accuracy score with a linear probe and prompt tuning, we combined the prompt and text weights to leverage both methods, adding a linear head to the encoded image and applying weights to each encoded text prompt in the embedding space. The first step of the training process is the same as the previous two techniques: freeze both image and text encoders, attach a linear head to the image encoder, and introduce trainable prompt embeddings after all the applications compute the image and text embedding space, utilizing cosine similarity and leveraging logits for each classification. In addition, enact backpropagation through both linear heads and prompt embeddings. See Figure 20 for the pipeline.

Figure 20

Hybrid-tuning pipeline. Combines prompt-tuning with a trainable linear classifier to further refine attribute prediction.

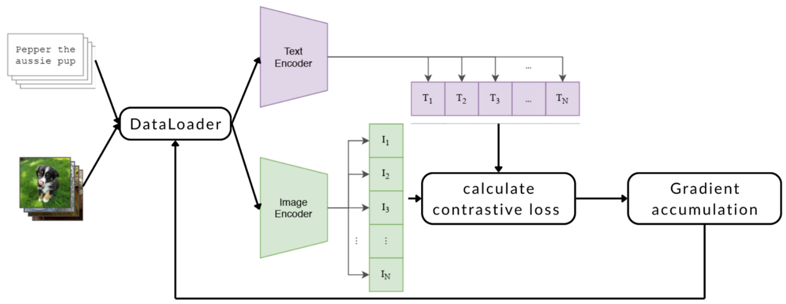


Full Fine-tuning

In this technique, we unfreeze the last six layers of the image encoder to enable the model to achieve deeper domain adaptation to our curated dataset. The training process is to learn the parameters using the log and load, initially pre-trained CLIP, for encoding text and images. Scale the loss by one divided by gradient accumulation steps, then do backpropagation through both encoders, projection layers, and logit scale parameter. After optimization, gradient accumulation is used to handle large batches and increase the effective batch size for many images. Figure 21 shows the pipeline for full fine-tuning.

Figure 21

Full fine-tuning pipeline. Both CLIP encoders are updated end-to-end using a contrastive loss with gradient accumulation.



After comparing all the experiment methods' performance, we found that linear probe, which trains only the linear head of image embedding, got the overall best accuracy. Moreover, for all the attributes, it exceeds the accuracy of the baseline CLIP and FashionCLIP. However, we still need to improve more on the most essential attributes, color, material, and pattern, which are the most critical elements that would affect the whole pipeline for future work. To continuously improve the model, we choose to stick with fine-tuning CLIP with a linear probe and add more example data to better train the model on specific fields like color, material, and pattern, especially when they are multiple-label attributes.

Experimental Results

The results of the experiment demonstrate that linear probe training model has the highest overall accuracy across all attributes. As we can observe from all the evaluation tables below, semantic similarity often yields higher scores for category and subcategory compared to accuracy. This is because accuracy requires the two strings to be identical to count as correct. Semantic similarity, however, evaluates how related the two terms are in meaning. For example, “cotton” and “cottons” are essentially the same word to humans. Still, for a machine, especially when using strict accuracy, they are treated as completely different tokens simply because the second one has an extra “s”. With semantic similarity, the model can interpret their closeness in meaning. Hence, the computed cosine similarity becomes very high. In multi-label setting, relaxed match is still lower than semantic similarity. The reason is similar to accuracy: for a text pair to be counted as 1, every character across the ground truth and the prediction must match exactly. Even if one character differs, the pair is counted as zero. Given such a strict rule, a relaxed match naturally ends up much lower than semantic similarity.

CLIP

CLIP’s zero-shot performance on color, material, and pattern attributes was limited. Relaxed accuracy for these multi-label cases was low, about 2.7% for color and around 2% or less for material and pattern. This reflects that CLIP’s zero-shot predictions often did not match the expected labels, sometimes producing descriptive phrases instead. Despite low exact-match accuracy, the average semantic similarity between predicted and true labels was reasonably high (around 0.7479 for colors), indicating that CLIP often predicted related concepts.

As shown in Table 9, CLIP achieved relatively better performance on coarse-grained fashion classification tasks, such as category prediction, achieving 66.8% accuracy and a high semantic similarity score (0.9484). However, performance dropped substantially for fine-grained subcategory recognition, where the F1 score decreased to 0.1758, indicating difficulty in distinguishing detailed fashion subcategories. Furthermore, Table 10 summarizes the multi-label attribute prediction results. Although relaxed match scores for color, material, and pattern remained low, the semantic similarity scores stayed comparatively high across all attributes. This suggests that CLIP's zero-shot embeddings were still capable of capturing semantically related fashion concepts even when the exact labels were not correctly predicted.

Table 9

Single-Label Performance of Baseline CLIP

Attributes	Accuracy	Precision	Recall	F1	Semantic Similarity
Category	0.6680	0.6974	0.6970	0.6186	0.9484
Subcategory	0.2720	0.2312	0.2076	0.1758	0.8852

Note. CLIP handles category reasonably well but struggles with fine-grained subcategories.

Table 10

Multi-Label Performance of Baseline CLIP

Attributes	Relaxed Match	Semantic Similarity
Color	0.0274	0.7479
Material	0.0203	0.7313
Pattern	0.0135	0.7328

Note. Relaxed match is low, but semantic similarity indicates partial alignment in the embedding space.

FashionCLIP

FashionCLIP, the domain-tuned model, resulted in a substantial boost in zero-shot performance. FashionCLIP achieved 88.9% accuracy on categories and 53.2% across subcategories, indicating that pretraining on fashion data improved recognition of garment types (Chia et al., 2022). However, its performance on color, material, and pattern attributes remained limited, with relaxed accuracy of about 1.8% for color.

As shown in Table 11, FashionCLIP significantly outperformed the baseline CLIP model in single-label classification tasks. The model achieved high semantic similarity scores for both category (0.9726) and subcategory (0.8912) prediction, demonstrating that fashion-specific pretraining improved both coarse-grained and fine-grained garment understanding. Furthermore, Table 12 presents the multi-label prediction results for color, material, and pattern attributes. Although relaxed match accuracy remained low across all attribute types, the semantic similarity scores indicate that FashionCLIP was still able to capture partially related fashion concepts in the embedding space. Compared with baseline CLIP, the model showed improved category understanding but continued to struggle with precise attribute-level zero-shot prediction.

Table 11

Single-Label Performance of Baseline FashionCLIP

Attributes	Accuracy	Precision	Recall	F1	Semantic Similarity
Category	0.8889	0.8528	0.8422	0.8398	0.9726
Subcategory	0.5315	0.33592	0.4168	0.3285	0.8912

Note. This table demonstrates improved category and subcategory prediction compared to vanilla CLIP.

Table 12

Multi-Label Performance of Baseline FashionCLIP

Attributes	Relaxed Match	Semantic Similarity
Color	0.0183	0.6410
Material	0.0119	0.6462
Pattern	0.0147	0.6123

Note. This table shows that multi-label performance remains limited but indicates moderate semantic alignment for color, material, and pattern.

The results suggest that FashionCLIP was not optimized for explicit multi-label attribute assignment, and further fine-tuning is needed to extract specific attribute labels. In summary, CLIP performs well across broad classes but struggles with fine details, while FashionCLIP improves garment-type recognition but still requires adaptation for structured attribute labeling.

Linear Probe Method

See Table 13 and 14 for single-label and multi-label results. This approach proved to be highly effective, delivering a large jump in accuracy on all attributes. By training just a classifier layer on top of frozen CLIP features, we achieved 93.4% accuracy on main category up from 66.8% zero-shot. In fact, this even surpassed FashionCLIP's 88.9%, illustrating that a small amount of supervised learning on our data can adapt CLIP extremely well to our specific label definitions. The model quickly learned, for instance, the exact boundary between similar categories like "outerwear" vs. "tops" which CLIP occasionally confused. The fine-tuned classifier evidently could linearly separate those in CLIP's feature space. For subcategory, linear probing lifted accuracy from 27.2% (CLIP baseline) to 69.6%, a remarkable improvement given the 141-class complexity. Now, most clothing items were assigned the correct sub-category. To put this in context, predicting subcategory is a hard task even for dedicated fashion classifiers due to many classes and subtle differences (e.g., distinguishing "jeans" vs "pants", or "blazer" vs "coat"). An accuracy around 70% indicates the model is correctly identifying many fine-grained types, though there is still considerable room for errors on less common classes. Nonetheless, linear probe nearly matched human-designed FashionCLIP's sub-category performance (53%) and significantly exceeded it, showing the benefit of supervised fine-tuning on the exact task.

Table 13

Single-Label Performance of Linear Probe

Attributes	Accuracy	Precision	Recall	F1	Semantic Similarity
Category	0.9336	0.9121	0.8755	0.8894	0.9890
Subcategory	0.6963	0.3092	0.2837	0.2864	0.9439

Note. This method boosts accuracy for category and provides better semantic alignment for both attributes.

Table 14

Multi-Label Performance of Linear Probe

Attributes	Relaxed Match	Semantic Similarity
Color	0.3957	0.8557
Material	0.3648	0.8649
Pattern	0.2189	0.8401

Note. All attributes achieve semantic similarity scores above 0.84, demonstrating the effectiveness of the proposed method in capturing semantic attribute consistency.

Prompt Tuning Method

See Table 15 and 16 for single-label and multi-label results. Prompt tuning by itself did improve over zero-shot CLIP, but this improvement was much more modest compared to linear probe. On the test set, the prompt-tuned CLIP achieved about 70.0% accuracy on category and 35.5% on subcategory. This is an improvement. For instance, an increase of 3.2% in the category, and an improvement of 8% in the subcategory, compared to CLIP zero-shot, but notably lower than the linear probe results of 93.4% and 69.6%, respectively. The limited improvement suggests that merely tweaking the textual prompts wasn't enough to master the fine-grained details. Likely, CLIP's image embeddings needed some adjustment or dedicated classifier to separate the subtle classes, which prompt tuning cannot provide since it leaves image features untouched. In a nutshell, prompt tuning helped a bit by finding better descriptors for our labels-possibly the learned prompts made CLIP more sensitive to fashion context. For instance, instead of the generic prompt "a photo of a {label} item", the model might learn an internal prompt like "garment {label} style" (implicitly) that resonates better with CLIP's vision features. This yielded small boosts-for example, CLIP might originally have conflated "skirt" and "dress", but a learned prompt can emphasize context words that help distinguish them. However, much confusion persisted due to the same visual features.

Prompt tuning also yielded limited progress on multi-label attributes. Color predictions with learned prompts of the model were still often off the target or overly specific to the pretraining distribution. We observed some qualitative changes: prompt-tuned CLIP sometimes produced outputs that were mixtures of color and style-like "navy white gingham" as a single generated phrase, combining color and pattern in one output, which indicates that the prompt learning tried to capture nuanced descriptors but not in a clean label-by-label way. When we parsed its outputs to match our label set, the effective color accuracy remained low significantly below the linear probe's around 40%.

Table 15

Single-Label Performance of Prompt Tuning

Attributes	Accuracy	Precision	Recall	F1	Semantic Similarity
Category	0.6994	0.6865	0.7145	0.6462	0.9508
Subcategory	0.3545	0.2643	0.3248	0.2267	0.8901

Note. This table shows improved category prediction while subcategory remains challenging.

Table 16

Multi-Label Performance of Prompt Tuning

Attributes	Relaxed Match	Semantic Similarity
Color	0.0061	0.7865
Material	0.0033	0.7994
Pattern	0.0128	0.7816

Note. Prompt tuning achieves moderate semantic alignment despite very low relaxed-match accuracy.

Hybrid Tuning Method

See Table 17 and 18 for single-label and mutli-label results. This approach reached only about 52.9% accuracy on category and 32.1% on sub-category, which is even lower than the prompt-only model. This outcome surprised us, since one would expect adding prompt tuning to either help or at least do no harm if optimized well. A possible explanation is that the hybrid training proved difficult to coordinate with both prompt and classifier, the model might have faced conflicting gradient signals. The prompt-tuning part tries to reshape the text embeddings, while the classifier tries to carve out decision boundaries in the image embedding space. If not carefully balanced, the prompt might alter the scale or distribution of text features, confusing the linear classifier during training. Also, our training scheme for the hybrid used the same data for both objectives, which might have made convergence harder. It’s possible the prompt tuning in the hybrid scenario was redundant, given that the linear head can learn to map image features to labels anyway. The prompt adjustments might add little new information and, in the worst case, could misalign the implicit image-text similarity that CLIP relies on internally.

The hybrid strategy did not yield a positive result in our experiments, it underperformed even the prompt-only model on categories. One symptom we noticed: the hybrid model sometimes made bizarre category mistakes, such as predicting "all-body" or "scarves" as a category for items that were clearly tops or outerwear. These errors hinted that the learned prompts might have biased the model's notion of certain labels. Meanwhile, the linear head might not fully correct for that if the text it's trying to align with is drifting. Our takeaway is that adding more complex mechanisms requires very careful training and tuning.

Table 17

Single-Label Performance of Hybrid Tuning

Attributes	Accuracy	Precision	Recall	F1	Semantic Similarity
Category	0.5288	0.6086	0.6121	0.5184	0.9148
Subcategory	0.3209	0.2554	0.3068	0.2097	0.8864

Note. This table shows moderate gains in semantic similarity but limited improvements in accuracy.

Table 18

Multi-Label Performance of Hybrid Tuning

Attributes	Relaxed Match	Semantic Similarity
Color	0.0067	0.7588
Material	0.0039	0.7582
Pattern	0.0064	0.7081

Note. The semantic similarity is reasonable despite very low relaxed-match accuracy.

Full Fine-Tuning Method

See Table 19 and 20 for single-label and multi-label results. Fully fine-tuning CLIP on our dataset yielded mixed outcomes. On the one hand, we saw continued improvement in some metrics compared to the linear probe, but the gains were not as large as one might expect, and in some respects the model showed signs of overfitting or odd behavior. After carefully training the entire model (with a low learning rate and early stopping), our fully fine-tuned CLIP model achieved around 58.7% accuracy on category and 30.2% on subcategory, which is worse than the linear probe (93.4% and 69.6%). These numbers indicate that something went wrong in the full fine-tuning process for those tasks. We suspect that the model might have overfit on certain classes or lost some generality.

Table 19

Single-Label Performance of Full Fine-Tuning

Attributes	Accuracy	Precision	Recall	F1	Semantic Similarity
Category	0.5866	0.5329	0.6838	0.5196	0.8664
Subcategory	0.3020	0.2618	0.3257	0.1906	0.7386

Note. This method improves recall but still struggles with fine-grained subcategories.

Table 20

Multi-Label Performance of Full Fine-Tuning

Attributes	Relaxed Match	Semantic Similarity
Color	0.0097	0.5831
Material	0.0100	0.5192
Pattern	0.0075	0.4305

Note. The table indicates that where color, material, and pattern remain challenging under both metrics.

For example, we observed that the fully fine-tuned model developed a strange bias: it started predicting the category “scarves” far too often. In a preview of its predictions, we found cases where an image of a pair of pants was misclassified as a “scarf” by the full fine-tune model, an obvious mistake that neither the linear probe nor even the zero-shot CLIP made. This suggests the model’s internal representations shifted in a suboptimal way, perhaps because “scarves” images in the training set had some distinctive feature the model latched onto (like white background and a certain shape), which it then erroneously used for other items. Essentially, by updating all weights, the model might have over-corrected or overemphasized certain patterns it saw during training, at the expense of its previously balanced knowledge.

Error Analysis

After observing the results, we conducted error analysis, focusing primarily on the color attribute, which remained the most challenging after fine-tuning. The color prediction task is difficult because colors can be subtle or images may vary in lighting. We analyzed the errors of all the experiments to identify which specific colors were most frequently misclassified or confused. One clear finding was that unusual or less common color names were most likely to be misclassified. For instance, “ecru,” a light beige or off-white color, was often missed by the model. In many cases, items labeled ecru were predicted to be white or cream by the model. This makes intuitive sense: ecru is a specific term, and to a model, an ecru blouse might just look off-white. CLIP's pretraining likely didn't expose it extensively to the word "ecru," so without fine-tuning, it had virtually no chance of getting that right; even after fine-tuning, distinguishing ecru from white is subtle and requires understanding context or tiny hue differences.

We systematically investigated model behavior by extracting false positive count rates across six configurations: CLIP, FashionCLIP, Hybrid, Full, Prompt, and Linear Probe, and visualizing the ten most frequent misclassified color labels per model (see Figures 22 to 27).

Figure 22

Top 10 false positive count for baseline CLIP.

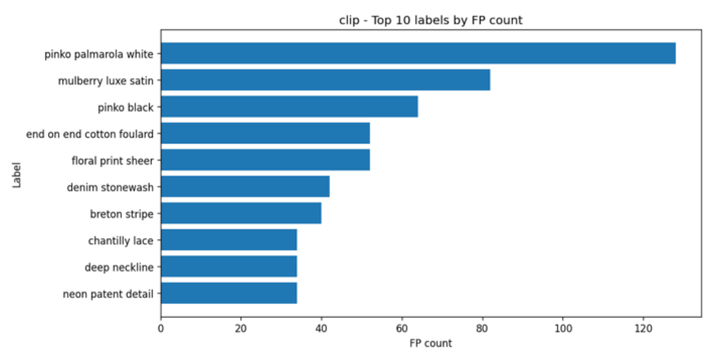


Figure 23

Top 10 false positive count for baseline FashionCLIP.

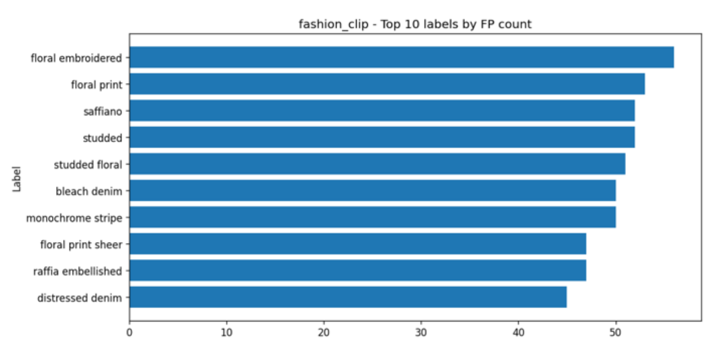


Figure 24

Top 10 false positive count for the linear probe method.

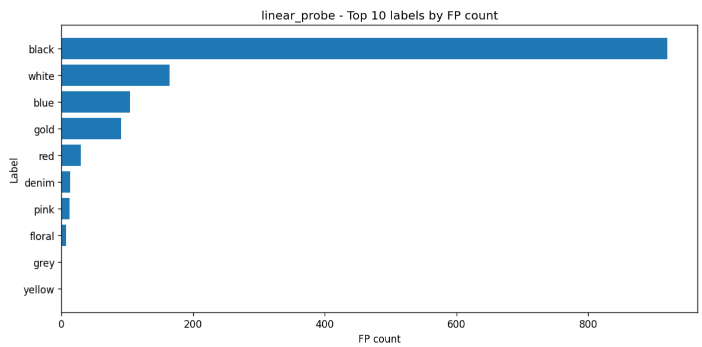


Figure 25

Top 10 false positive count for method prompt tuning.

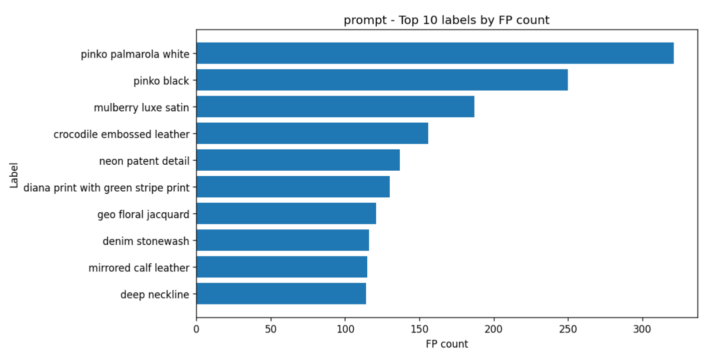


Figure 26

Top 10 false positive count for method hybrid tuning.

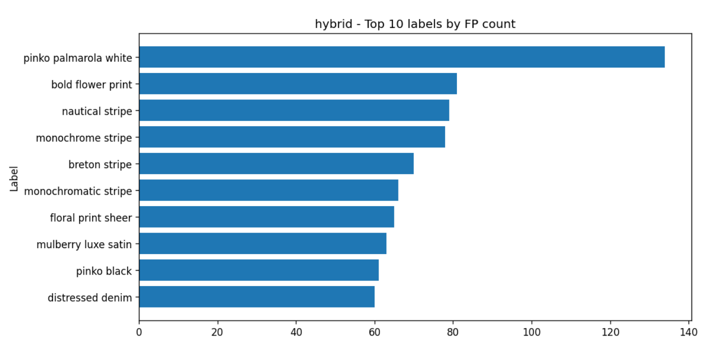
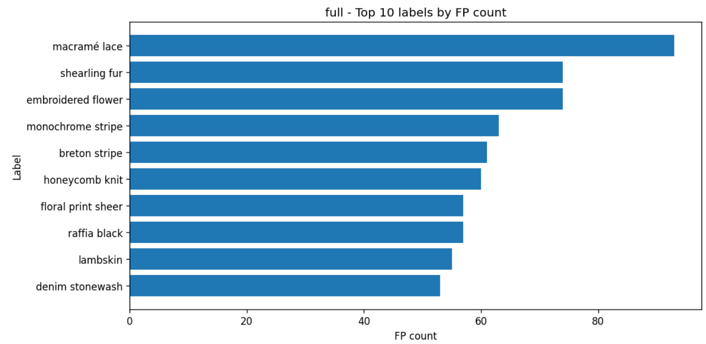


Figure 27

Top 10 false positive counts for full fine-tuning method.



The most over-predicted color label across models was "black" followed by white, blue, and gold, with over 920 false positives in the linear probe setting. This pattern would indicate a systematic color bias toward darker or neutral tones dominating the misclassifications. Such over-predictions are likely due to high intra-class visual similarity collapsing into the single term "black" during feature embedding. On the other hand, models that are more dependent on textual prompts or multimodal fine-tuning tend to misclassify towards material-dependent color descriptors, such as "mulberry luxe satin", "pinko palmarola white", and "denim stonewash". These complex labels combine color with fabric or brand-specific features; such labels indicate models overfit to co-occurring descriptors rather than isolating pure chromatic information.

Understanding these mistake patterns was crucial. It told us that while the linear probe model dramatically improved overall color recognition, it still had blind spots for certain hues and contexts. Many of these errors are also intuitive, which means they are the kind of mistakes a person might make if they have limited experience or if the image is small or blurry. This analysis confirmed that improving the model's sensitivity to these subtle color distinctions would yield a noticeable performance boost.

Proposed FACTOR Framework

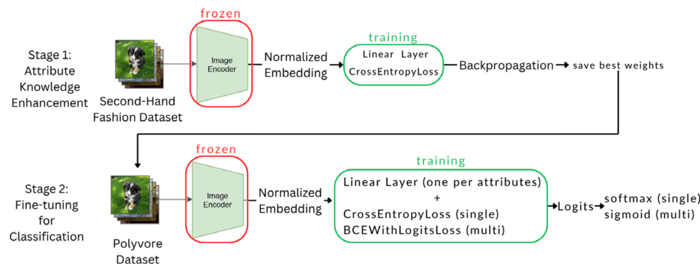
This section presents the attribute-aware fine-tuning process derived from the error analysis results. As the analysis revealed, the diversity of the color attribute had a significant effect on model misclassification. To address this, we refine the model's visual backbone to be attribute-aware on another dataset, then reapply our linear probe training across all attributes using this improved backbone. We extract the top 10 color labels that the linear probe model most frequently misclassifies. Then utilized the Second-Hand Fashion dataset (Nauman, 2024), which includes both image filenames and corresponding color annotations. For detailed dataset characteristics and exploratory data analysis, refer to Chapter 3. From each selected color category, 100 representative images were sampled, yielding a total of 1,000 training samples. These images were subsequently passed through CLIP's encoder to generate embeddings, upon which a linear head was trained. The training process followed the same optimization strategy as described in the experimental design in Chapter 6.

Methodology

To enable fine-grained semantic understanding of fashion items, this study proposes Feature-Aware Contextual Optimization and Refinement (FACTOR), a structured training framework built upon CLIP. The goal of FACTOR is to enhance the visual encoder's sensitivity to attribute-level features and to produce reliable multi-attribute predictions for downstream retrieval tasks. The framework is designed as a two-stage learning process that first improves low-level visual perception and then performs attribute-specific prediction using a controlled probing mechanism (see Figure 28).

Figure 28

FACTOR pipeline. First, fine-tune the encoder leveraging linear probe on attribute-aware dataset, then freeze it and train another linear probe on top of it using Polyvore dataset.



In the first stage, the CLIP visual encoder is fine-tuned to improve its sensitivity to color, which is one of the most dominant visual attributes in human perception of clothing. Specifically, we adopt the pre-trained CLIP ViT-B/32 model and unfreeze the visual encoder while keeping all text-related components fixed. A linear classification head is attached on top of the 512-dimensional visual embedding to perform color classification. Given an input garment image, the model extracts visual features, applies L2 normalization, and passes the embedding through the classification head to produce logits over the predefined color categories. The model is trained with CrossEntropyLoss, and optimization is performed with AdamW.

The training data for this phase is derived from a second-hand fashion dataset, where images are split into training and validation sets using a stratified strategy to ensure balanced color distribution. The model is trained with a batch size of 8 for 150 epochs, using a learning rate of 1e-3 and no weight decay or label smoothing. During training, gradient clipping is applied to stabilize optimization, and early stopping is employed based on validation accuracy to prevent overfitting. The final model checkpoint is selected based on the highest validation accuracy, yielding an attribute-aware visual encoder that is particularly sensitive to color information.

In the second stage, the fine-tuned visual encoder is used as a fixed feature extractor for multi-attribute prediction. Unlike the first phase, the encoder is frozen to preserve the learned representation, and multiple independent linear classification heads are introduced to predict different attributes. Specifically, the model predicts five attributes: category and subcategory as single-label classification tasks, and color, material, and pattern as multi-label classification tasks. Each attribute head operates on the same 512-dimensional visual embedding, enabling efficient and modular prediction without modifying the backbone.

The second phase is trained on the Polyvore dataset, which is split into training, validation, and test sets following an 80/10/10 ratio (28,792 / 3,599 / 3,599 samples). The attribute label space is defined using a structured label taxonomy to ensure consistency across predictions. For optimization, we use the AdamW optimizer with a learning rate of 5e-5, weight decay of 0.02, and a batch size of 16 for 50 epochs. Different loss functions are applied depending on the attribute type: CrossEntropyLoss is used for single-label attributes, while BCEWithLogitsLoss is used for multi-label attributes to account for the non-exclusive nature of properties such as material and pattern.

During inference, different strategies are applied based on the attribute type. For single-label attributes, predictions are obtained using the argmax operation over logits. For multi-label attributes, sigmoid activation is applied followed by a thresholding operation with a threshold of 0.15. If no label exceeds the threshold, a fallback mechanism selects the top-1 prediction to ensure output completeness. This design balances precision and recall while avoiding empty predictions.

To evaluate the effectiveness of the proposed framework, multiple metrics are employed. For single-label attributes, we report accuracy, precision, recall, F1-score, and semantic similarity. For multi-label attributes, evaluation is conducted using relaxed match and semantic similarity metrics, which better capture partial correctness in multi-label scenarios. In addition, different adaptation strategies, including zero-shot inference, linear probing, prompt tuning, hybrid tuning, and full fine-tuning, are explored to analyze their impact on attribute prediction performance. The final model is selected based on overall performance across all attributes.

Overall, FACTOR follows a representation learning followed by attribute-specific probing paradigm, where the first phase enhances visual feature quality and the second phase enables structured and interpretable attribute prediction. This design not only improves fine-grained attribute recognition but also provides a strong foundation for downstream fashion retrieval tasks that require both visual similarity and semantic consistency.

Results

As shown in Figures 29 and 30, the training and validation curves indicate a healthy learning process, with both losses converging after approximately 30 epochs. Similarly, Figures 31 and 32 demonstrate that the overall linear probe training converged stably around epoch 40, suggesting consistent generalization across both datasets.

Figure 29

FACTOR Stage 1 Training and Validation Accuracy Curves.

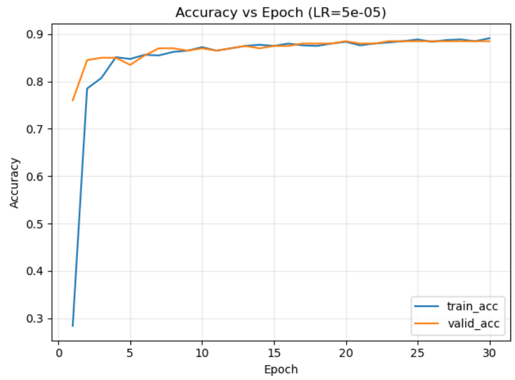


Figure 30

FACTOR Stage 1 Training and Validation Loss Curves.

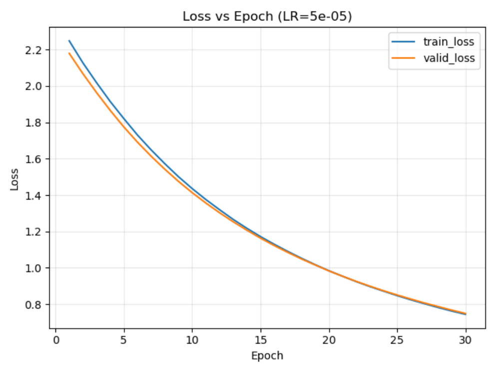


Figure 31

FACTOR Stage 2 Training and Validation Accuracy Curves.

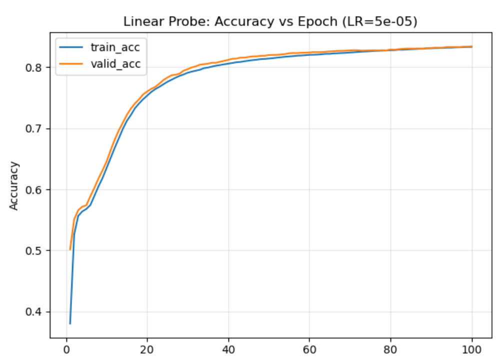
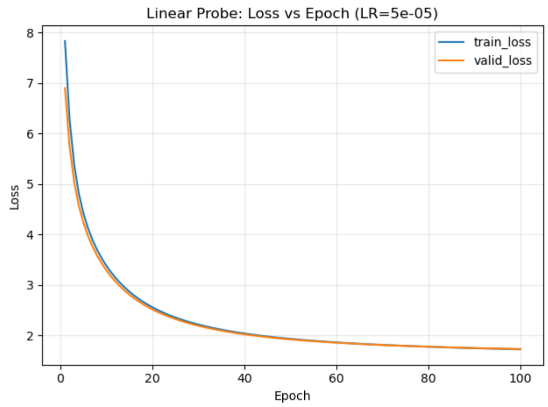


Figure 32

FACTOR Stage 2 Training and Validation Loss Curves.



The final evaluation results, summarized in Tables 21 and 22, show that incorporating attribute-aware fine-tuning led to a modest yet consistent improvement across both single-label and multi-label tasks compared to the original linear probe baseline. This enhancement highlights the model's improved sensitivity to chromatic features and its strengthened capacity for category discrimination.

Table 21

Single-Label Performance of FACTOR

Attributes	Accuracy	Precision	Recall	F1	Semantic Similarity
Category	0.94	0.92	0.89	0.90	0.99
Subcategory	0.72	0.40	0.34	0.35	0.95

Table 22

Multiple-Label Performance of FACTOR

Attributes	Relaxed Match	Semantic Similarity
Color	0.43	0.86
Material	0.40	0.87
Pattern	0.24	0.85

Results Comparison

The results show that semantic similarity scores are consistently higher than relaxed match scores, indicating that the model captures semantic-level information more effectively than exact label overlap. In summary, among the single-stage fine-tuning methods, Linear Probe was the strongest performer for fashion attribute classification. It achieved the highest accuracy on both category and subcategory, with substantial improvements on multi-label attributes, and was simple to train. Prompt tuning was helpful but insufficient on its own, while the hybrid approach underperformed. Full model fine-tuning risked degrading performance, highlighting that constrained fine-tuning can be more effective than extensive modifications.

Baselines Result Comparison

We begin by evaluating zero-shot attribute prediction using CLIP and its domain-adapted variant, FashionCLIP. As summarized in Table 23, CLIP achieves moderate performance for coarse garment category classification (accuracy is 67%), but its ability to discriminate fine-grained subcategories is limited (accuracy is 27%). Performance on structured multi-label attributes is particularly weak, with relaxed match scores below 3% for color, material, and pattern. Despite low exact and relaxed match accuracy, semantic similarity scores remain relatively high across attributes (approximately 73% to 75%). This gap indicates that zero-shot predictions often capture semantically related concepts but fail to align with the discrete and noisy label vocabulary required for structured attribute reasoning.

Table 23

Zero-shot performance of CLIP and FashionCLIP

Method	Attribute Type	Attributes	Accuracy	Precision	Recall	F1	Relaxed Match	Semantic Similarity
CLIP	Single-Label	Category	0.62±0.01	0.66±0.01	0.66±0.01	0.57±0.01	-	0.94±0.00
	Single-Label	Subcategory	0.27±0.00	0.25±0.00	0.31±0.01	0.20±0.00	-	0.88±0.00
	Multi-Label	Color	-	-	-	-	0.00±0.00	0.74±0.00
	Multi-Label	Material	-	-	-	-	0.00±0.00	0.74±0.00
	Multi-Label	Pattern	-	-	-	-	0.00±0.00	0.71±0.00
FashionCLIP	Single-Label	Category	0.88±0.00	0.841±0.01	0.83±0.01	0.83±0.01	-	0.97±0.00
	Single-Label	Subcategory	0.52±0.01	0.370±0.01	0.44±0.02	0.34±0.02	-	0.89±0.00
	Multi-Label	Color	-	-	-	-	0.02±0.00	0.64±0.00
	Multi-Label	Material	-	-	-	-	0.01±0.00	0.65±0.00
	Multi-Label	Pattern	-	-	-	-	0.01±0.00	0.61±0.00

Note. Both models demonstrate reasonable performance for coarse category recognition but struggle with fine-grained subcategories and explicit multi-label attributes. Despite low exact and relaxed match scores, relatively high semantic similarity indicates partial semantic alignment without reliable structured attribute assignment.

FashionCLIP substantially improves zero-shot recognition of garment types, achieving 0.8 accuracy for category and 53% for subcategory. However, this improvement does not translate to structured multi-label prediction. Relaxed match remains below 2% for color, material, and pattern, while semantic similarity scores are consistently lower than those of CLIP. These results suggest that domain-aligned pretraining alone is insufficient for explicit attribute assignment, particularly for attributes defined by subtle visual cues.

Vision Language Models Result Comparison

Among all single-stage adaptation strategies (see Table 24), linear probing consistently delivers the strongest performance. Training lightweight classifiers on frozen CLIP embeddings yields substantial gains across all attributes, achieving 93% accuracy for category and 70% for subcategory. For multi-label attributes, linear probing reaches relaxed match scores of 40% (color), 36% (material), and 22% (pattern), accompanied by strong semantic similarity (84% to 86%). These results indicate that CLIP's pretrained representations already encode substantial attribute-relevant information that can be effectively extracted through constrained decoding.

Table 24

Performance of Vision Language Model Strategies

Method	Attribute Type	Attributes	Accuracy	Precision	Recall	F1	Relaxed Match	Semantic Similarity
Linear Probe	Single-Label	Category	0.93±0.00	0.91±0.01	0.89±0.01	0.90±0.01	-	0.99 ±0.00
	Single-Label	Subcategory	0.72±0.00	0.40±0.02	0.34±0.01	0.35±0.01	-	0.95±0.00
	Multi-Label	Color	-	-	-	-	0.43±0.01	0.86±0.00
	Multi-Label	Material	-	-	-	-	0.38±0.01	0.87±0.00
	Multi-Label	Pattern	-	-	-	-	0.25±0.01	0.85±0.00
Prompt Tuning	Single-Label	Category	0.74±0.01	0.70±0.01	0.73±0.00	0.67±0.01	–	0.96±0.00
	Single-Label	Subcategory	0.34±0.01	0.28±0.01	0.33±0.02	0.23±0.01	–	0.89±0.00
	Multi-Label	Color	–	–	–	–	0.01±0.00	0.79±0.00
	Multi-Label	Material	–	–	–	–	0.00±0.00	0.80±0.00
	Multi-Label	Pattern	–	–	–	–	0.01±0.00	0.78±0.00
Hybrid Tuning	Single-Label	Category	0.75±0.04	0.69±0.02	0.76±0.03	0.69±0.03	–	0.95±0.01
	Single-Label	Subcategory	0.40±0.03	0.28±0.02	0.33±0.02	0.24±0.02	–	0.90±0.00
	Multi-Label	Color	–	–	–	–	0.01±0.00	0.74±0.00
	Multi-Label	Material	–	–	–	–	0.01±0.00	0.77±0.00
	Multi-Label	Pattern	–	–	–	–	0.01±0.00	0.72±0.01
Full Fune-Tuning	Single-Label	Category	0.81±0.04	0.71±0.04	0.79±0.03	0.70±0.04	–	0.94±0.02
	Single-Label	Subcategory	0.35±0.02	0.28±0.01	0.34±0.01	0.23±0.01	–	0.80±0.02
	Multi-Label	Color	–	–	–	–	0.01±0.00	0.62±0.02
	Multi-Label	Material	–	–	–	–	0.01±0.00	0.58±0.02
	Multi-Label	Pattern	–	–	–	–	0.00±0.00	0.45 ±0.02
FACTOR	Single-Label	Category	**0.93±0.00**	**0.91±0.01**	**0.89±0.01**	**0.90±0.01**	–	**0.99 ±0.00**
	Single-Label	Subcategory	**0.72±0.01**	**0.41±0.01**	**0.34±0.01**	**0.35±0.01**	–	**0.95 ±0.00**
	Multi-Label	Color	–	–	–	–	**0.44±0.01**	**0.87±0.00**
	Multi-Label	Material	–	–	–	–	**0.40±0.01**	**0.87±0.00**
	Multi-Label	Pattern	–	–	–	–	**0.25±0.01**	**0.85±0.00**

Note. Linear probing consistently outperforms other single-stage adaptation methods, while prompt tuning, hybrid tuning, and full fine-tuning exhibit reduced stability and degraded performance. The proposed FACTOR framework achieves the strongest results across all single-label and multi-label attributes, demonstrating the effectiveness of constrained, two-stage adaptation over aggressive end-to-end fine-tuning.

Prompt tuning provides only modest improvements over zero-shot CLIP. While category accuracy increases to 70%, subcategory accuracy remains low as 35%, and relaxed match for multi-label attributes stays near zero. Semantic similarity improves moderately, but these gains do not translate into reliable structured prediction. Hybrid tuning and full end-to-end fine-tuning further underperform. Both strategies exhibit degraded accuracy and reduce semantic alignment across attributes. In particular, full fine-tuning shows signs of representation drift, with category and subcategory accuracy dropping to 59% and 30%, respectively, and semantic similarity deteriorating sharply for multi-label attributes. These findings underscore the instability of aggressive adaptation when applied to fine-grained attribute supervision.

Prediction Examples

Figure 33 presents a representative prediction example of the proposed FACTOR framework. Given an input garment image, the model predicts a set of structured attributes by selecting from a predefined label library. As shown in the example, FACTOR can accurately identify all five attributes, including category, subcategory, color, material, and pattern, with predictions that fully match the ground truth. In contrast, baseline and alternative adaptation methods tend to produce partially correct or semantically inconsistent outputs, particularly for fine-grained attributes such as material and pattern. This highlights the advantage of incorporating a label library together with structured attribute prediction, which constrains the output space and improves semantic consistency. Overall, the example demonstrates that FACTOR can generate more precise and interpretable attribute predictions compared to other methods.

Figure 33

Prediction Example of the FACTOR framework. Compared to other methods, FACTOR produces more accurate and semantically consistent predictions that fully align with the ground truth.

\+
Prompt
\+
Label
Library

Model	Category	Subcategory	Color	Material	Pattern
Ground Truth	outwear	blazer	black	leather	plain
CLIP	outerwear	jacket	leather-look	faux patent-leather	patent-leather
FashionCLIP	outerwear	jacket	leather-look	leather-look	asymmetrical-zip belted
Linear Probe	outerwear	blazer	black	leather	plain
Prompt Tuning	outerwear	jacket	leather-look, faux leather	faux patent-leather, lambskin leather	patent-leather, leather panelled, leather look
Hybrid	outerwear	male jacket	leather-look	faux-shearling leather, gucci signature leather, fauxleather	leather panelled
Full Fine Tuning	outerwear	male formal jacket	rich leather, lambskin, leather-look	lambskin leather, calf-leather	layered leather ruffles
FACTOR	**outerwear**	**blazer**	**black**	**leather**	**plain**

\+
Prompt
\+
Label
Library

Model	Category	Subcategory	Color	Material	Pattern
Ground Truth	tops	blouse	coral	polyester	ruffle
CLIP	tops	blouse	babydoll pink	contrast lace insert	tiered ruffled hem with cutouts
FashionCLIP	tops	blouse	coral pink	ruffled fabric	ruffled neck
Linear Probe	tops	blouse	pink	cotton, polyester, silk	plain
Prompt Tuning	tops	blouse	babydoll pink, chiffon, deep neckline	silk-blend chiffon, organza chiffon, floral cotton lace	tiered ruffled hem with cutouts, ruffled peplum hem, scalloped ruffles
Hybrid	tops	blouse	babydoll pink	camisole silk	tiered ruffled hem with cutouts
Full Fine Tuning	tops	blouse	babydoll pink, baby pink flowers, sorbet floral	sheer floral-lace, floral cotton lace, corded floral-lace	floral scalloped v neckline
FACTOR	tops	blouse	pink	silk	plain

Conclusion

This thesis constitutes a deliberate shift from general vision-language models to the task of understanding fashion attributes. What began as research to understand the capability of CLIP's general model to facilitate fashion understanding has evolved over this journey to identify the principles for adapting general models to specialized domains. This work has thus not only helped identify the strengths and weaknesses of the CLIP model and the FashionCLIP but has also helped to outline the principles of adapting models from general to specialized.

Our baseline study has provided a crucial starting point: CLIP has impressive general vision-language abilities, but its zero-shot fashion attributes understanding is not exhaustive. FashionCLIP's fashion pre-training has demonstrated the benefits of alignment, as it outperformed generic CLIP on the zero-shot classification task. However, both CLIP and FashionCLIP shared the same limitation: without task-level fine-tuning, neither could extract the five vital apparel components required for adequate fashion analysis.

This understanding drove our research to evaluate the effectiveness of fine-tuning strategies that manipulate the extent to which the internal representation space of CLIP can be modified. In this array of fine-tuning strategies, linear probe proved to be an essential method. With the base model's weights frozen, a lightweight model demonstrated the profound effectiveness of learning to read CLIP's representations. This yielded high accuracy above 93% at the category-level, about 70% at the subcategory-level, and greatly improved detection of color, material, and pattern, moving from close to random to robust multi-label classification. The effectiveness of the above experiments reaffirms the thesis's crucial point that CLIP already has implicit knowledge about fashion items, which can be harnessed through a carefully written readout layer.

More sophisticated methods, such as prompt tuning and hybrid tuning, revealed the other side of the tradeoff. Although conceptually clean and elegant, prompt tuning alone did not achieve sufficient accuracy for fine-grained multi-label prediction tasks. The hybrid method, which aimed to bring the best of both worlds of prompt tuning and linear probing, was also spotty and showed that the additional flexibility might introduce its own optimization problems. The method requiring the most care in its results when there's limited data available turned out to be full CLIP fine-tuning. CLIP's broad general models can be overwritten relatively easily.

These points of inflection informed the next phase of our approach. Through targeted error analysis, we first pick a color to improve its performance. The approach was to deliberately combine an attribute-aware linear probe. The experiment showed sharp improvements not only within the bounds of color prediction tasks but also boosted the identification of many other attributes as a secondary effect. This experiment illustrates the general truth that fine-tuning will be most valuable when driven by insightful knowledge rather than blanket coverage.

Together, these steps provide a unified story of adapting a foundation model to fashion understanding. The transition from the zero-shot CLIP model to the FashionCLIP pretrained in the fashion domain, to linear probing, and to two-stage adaptation demonstrates how each level of increasing model engagement unearths both opportunities and constraints. Finally, the model can provide richly structured predictions of the form: "Category: Dress; Sub-category: Maxi Dress; Color: Red, White; Material: Cotton; Pattern: Striped". These features are crucial not only for catalog organization and searching but also for the fashion recommendation system. The crucial findings of this thesis lie in:

- Pre-training helps but matters less than task alignment. CLIP makes a good starting point, and FashionCLIP illustrates the value of domain adaptation, but supervised in-domain tuning remains essential for attribute-level accuracy.
- Linear probing can be found to be remarkably effective. Frozen backbone and trained classification head represent the best possible trade-off between leveraging CLIP's strengths and preventing overfitting.
- The application of complex tuning must be handled carefully. The additional tuning parameters can degrade performance when the dataset is small.
- Error analysis provides the engine of progress. By identifying the color bottleneck and developing a two-stage solution, we improved the model’s weakest dimension.
- Enhanced multi-label handling and evaluation. By using relaxed accuracy and semantics via CLIP text encodings, the model achieved a realistic assessment of performance in a particular area of study, where label boundaries might be fuzzy.

To sum up, this part has shown that adapting CLIP to the fine-grained fashion attribute classification task is not a simple learning problem. Instead, it involves a carefully coordinated process of pre-training, efficient adaptation via linear transformations, and targeted fine-tuning of the model’s behavior based on insights from interpretive findings. By pursuing this coordinated approach to adapting the large vision-language model CLIP through pre-training, efficient linear transformation, and targeted fine-tuning of the model’s behavior, this thesis has built a model that not only improves upon FashionCLIP but also provides an effective building block for the development of fashion-aware AI models.

Chapter 6: Comparative Study of Fashion Style Classification Models

Introduction

Beyond low-level garment properties such as category, color, material, and pattern, fashion recommendations also depend on higher-level aesthetic semantics. In practice, users often search for or respond to clothing not only by item type, but also by overall style impressions such as casual, formal, retro, or street. Therefore, to complement the five structured garment attributes introduced in the previous chapter, this chapter incorporates style as an additional semantic dimension for recommendation.

To construct this style attribute, we evaluate several visual classification approaches under a 17-class fashion style recognition setting. Specifically, we compare a conventional supervised baseline, ResNet50 (He et al., 2016), with CLIP-based fashion models, including FashionCLIP (Chia et al., 2022) and OpenFashionCLIP (Cartella et al., 2023). We further examine different adaptation strategies, including frozen-feature classification, partial fine-tuning, prompt-based classification, full fine-tuning, and LoRA-based tuning. This allows us not only to analyze whether CLIP-based representations are useful for style recognition but also to examine how different transfer strategies affect downstream performance. To ensure a comprehensive evaluation, we include accuracy, macro-precision, macro-recall, and macro-F1 score to assess the classification results.

The purpose of this evaluation is not to treat style classification as an isolated end task, but to identify a reliable semantic style encoder that can be integrated into the downstream recommendation framework. As will be shown, partial fine-tuning provides the strongest and most stable performance, enabling style to serve as the sixth attribute in our unified recommendation representation.

Style Classification Framework

The task is formulated as a supervised multi-class image classification problem, given a garment image, the model predicts one style label from 17 predefined fashion style categories. Each sample consists of an image and its associated style annotation, and the objective is to learn a mapping from visual appearance to style semantics. To determine the most suitable style classification model, we evaluate three model families: ResNet50 (He et al., 2016), a convolutional neural network (CNN) (O'shea et al., 2015) baseline; FashionCLIP, a domain-adapted vision-language model; and OpenFashionCLIP, an alternative fashion-oriented CLIP backbone. For the CLIP-based models, we test several adaptation strategies, ranging from frozen-feature classification to parameter-efficient and full-model tuning. Across these settings, the main criterion is not only classification accuracy, but also whether the resulting model provides a sufficiently robust and transferable style signal for downstream recommendation. In this way, the style module extends the semantic representation beyond low-level garment attributes, enabling more user-aligned retrieval.

Baseline Models

To establish a strong benchmark, we first evaluate three baseline models: ResNet50, FashionCLIP, and OpenFashionCLIP. ResNet50 serves as the conventional supervised vision baseline; it is initialized with ImageNet-pretrained weights and replaces the final fully connected layer with a new linear layer that outputs 17 class logits. This model provides a representative CNN benchmark for comparison against CLIP-based approaches. Second is FashionCLIP, which is implemented as a vision encoder with a linear classification head. In the baseline setting, the pretrained backbone is frozen, and only the final linear classifier is trained. The visual feature is extracted from the vision encoder, projected into the CLIP embedding space, normalized, and then passed into the classification layer. Lastly, OpenFashionCLIP follows a similar setup. We leverage the OpenCLIP ViT-B-32 backbone with pretrained weights and attach a linear classifier on top of the normalized image embedding.

CLIP Adaptation Strategies

To systematically evaluate different adaptation strategies for vision language models (VLMs), this study considers four representative approaches: partial fine-tuning, prompt-based classification, full fine-tuning, and LoRA-based adaptation. These methods are designed to explore different trade-offs between adaptation flexibility, computational efficiency, and model stability. The following are details for each strategy:

1. Partial Fine-Tuning: This strategy is intended to preserve general visual knowledge while allowing limited task-specific adaptation. Only the last two transformer blocks of the CLIP visual backbone are updated and jointly trained with newly added classification head, while the remaining layers are kept frozen. This design allows the model to retain most of its pretrained general visual knowledge while enabling limited task-specific adaptation in higher-level feature representations. Compared to full fine-tuning, this approach provides a more controlled optimization process and typically leads to a better trade-off between performance and training stability.

2. Prompt-Based Classification: In prompt-based classification, the task is reformulated as a conventional classification task as an image-text similarity matching problem. For each style category, a text prompt using the template, such as “a photo of a {attribute} fashion outfit”, the model computes similarities between image features and text embeddings, and the predicted class is determined by the highest similarity score. To improve flexibility and adaptability without modifying the backbone, a lightweight trainable adapter is introduced, along with a learnable logit scaling parameter. The main CLIP encoder remains frozen, allowing the model to leverage its pretrained multimodal alignment while adapting minimally to the target task.
3. Full Fine-Tuning: It updates all parameters of the CLIP-based backbone together with the classification head. This strategy provides maximum adaptation capacity; however, it also increases the risk of overfitting and unstable optimization, especially when the dataset size is relatively limited. In practice, full fine-tuning requires careful hyperparameter tuning to prevent degradation of the pretrained representation.
4. LoRA-Based Adaptation: It is applied to FashionCLIP as a parameter-efficient tuning strategy; instead of updating the full pretrained model, it introduces low-rank trainable matrices into selected layers of the model. During training, only these low-rank components are updated, while the original weights remain frozen; this significantly reduces the number of trainable parameters and computational cost, while still enabling effective task-specific adaptation. This method aims to reduce the number of trainable parameters while still enabling task-specific adaptation.

Across all fine-tuning experiments, the optimization process follows a consistent configuration to ensure reproducibility. The random seed is fixed to 42 to guarantee deterministic behavior. The model is trained using the AdamW optimizer with separate learning rates for different components: 1e-4 for the classification head and 1e-5 for the backbone to stabilize updates to pretrained representations. The weight decay is set to 1e-4, and the batch size is fixed at 32. All models are trained for 10 epochs.

For model selection, the best checkpoint is determined by the validation macro F1 score rather than accuracy. This choice is motivated by the class imbalance commonly observed in fashion attribute datasets, where macro F1 provides a more balanced evaluation across all categories. By treating each class equally, this metric ensures that performance improvements are not biased toward dominant classes and more accurately reflect the model's overall effectiveness.

Experimental Results

Overall Performance Comparison

Table 25 summarizes the overall classification performance of all evaluated models, including the baseline models and the different CLIP adaptation strategies. The results are reported using four metrics: Macro F1, Accuracy, Macro Precision, and Macro Recall.

Table 25

Overall Comparison for Fashion Style Classification

	Model	Macro F1	Accuracy	Macro Precision	Macro Recall
BaseLine	ResNet50	0.81 ± 0.01	0.80 ± 0.01	0.81 ± 0.01	0.81 ± 0.01
	OpenFashionCLIP	0.73 ± 0.01	0.74 ± 0.01	0.76 ± 0.00	0.74 ± 0.01
	FashionCLIP	0.73 ± 0.00	0.74 ± 0.00	0.76 ± 0.00	0.73 ± 0.00
OpenFashionCLIP	Partial Fine-Tune	**0.86 ± 0.01**	**0.86 ± 0.01**	**0.86 ± 0.01**	**0.86 ± 0.01**
	Prompt Classify	0.85 ± 0.01	0.84 ± 0.01	0.85 ± 0.01	0.85 ± 0.01
	Full Fine-Tune	0.79 ± 0.02	0.79 ± 0.02	0.80 ± 0.02	0.79 ± 0.02
FashionCLIP	Partial Fine-Tune	**0.86 ± 0.01**	**0.86 ± 0.01**	**0.86 ± 0.01**	**0.86 ± 0.01**
	Prompt Classify	0.85 ± 0.01	0.84 ± 0.01	0.85 ± 0.01	0.85 ± 0.01
	Full Fine-Tune	0.79 ± 0.01	0.79 ± 0.01	0.80 ± 0.01	0.79 ± 0.01
	LoRA	0.74 ± 0.01	0.75 ± 0.01	0.75 ± 0.01	0.75 ± 0.01

Note. Overall comparison of all baseline and adaptation methods for fashion style classification. Partial fine-tuning produced the strongest results for both OpenFashionCLIP and FashionCLIP, while full fine-tuning and LoRA showed comparatively weaker performance.

Among the baseline models, ResNet50 achieved the strongest performance, with a Macro F1 of 81% and an Accuracy of 80%, outperforming both frozen CLIP-based baselines. In comparison, both the frozen OpenFashionCLIP amd FashionCLIP baseline obtained a Macro F1 of 73%. These results indicate that directly attaching a linear classification head to frozen CLIP visual features is insufficient to surpass a strong supervised CNN baseline on this task.

For the adaptation strategies, partial fine-tuning consistently yields the best overall performance across both OpenFashionCLIP and FashionCLIP. Specifically, partial fine-tuning achieves a macro F1 score of 86% for both models, along with balanced improvements across accuracy, precision, and recall. This indicates that selectively updating higher-level layers of the visual backbone enables the model to adapt effectively to the target task while preserving useful pretrained representations. The second-best performance is achieved by prompt-based classification, with a macro F1 score of 85%. This demonstrates that reformulating the task as an image-text similarity problem can effectively leverage CLIP's multimodal alignment capability, even without directly modifying the backbone. However, its performance remains slightly below that of partial fine-tuning, suggesting that prompt-based methods may be limited in their ability to capture fine-grained visual distinctions compared to feature-level adaptation. In contrast, full fine-tuning shows comparatively weaker performance, with a macro F1 score of approximately 79% for both OpenFashionCLIP and FashionCLIP. This indicates that updating the entire pretrained model does not necessarily lead to better generalization and may instead introduce instability or overfitting, especially given the moderate dataset size. Similarly, LoRA-based adaptation achieves a macro F1 score of 74%, which is lower than both partial fine-tuning and prompt-based approaches, and even comparable to the frozen baseline. This suggests that, under the current configuration, parameter-efficient tuning does not provide sufficient capacity for this task.

Overall, three key observations can be drawn from these results. First, frozen CLIP-based models underperform compared to a strong supervised CNN baseline, highlighting the importance of task-specific adaptation. Second, once properly adapted, CLIP-based models can significantly outperform ResNet50, demonstrating the effectiveness of pretrained vision-language representations. Third, among all evaluated strategies, partial fine-tuning provides the most reliable and effective adaptation, achieving the best trade-off between performance and stability, while full fine-tuning and LoRA appear less suitable under the current experimental setting.

Per-Style Classification performance

Across all styles, fine-tuning-based methods consistently outperform both the baseline models and prompt-based approaches, with Partial FT achieving the most stable gains. Categories such as formal, dressy, and lolita reach near-saturation performance (F1 score around 95%-97%), suggesting that these styles have more distinctive visual patterns that are easier to capture through supervised adaptation.

In contrast, styles like feminine, girlish, and conservative show relatively lower performance across all methods, indicating higher intra-class variability and more subtle visual cues that are harder to model. Notably, while prompt-based methods provide moderate improvements over the baseline, they generally underperform compared to fine-tuning, highlighting the limitations of purely text-driven alignment in capturing fine-grained stylistic nuances.

Interestingly, OpenFashionCLIP and FashionCLIP exhibit comparable performance under fine-tuning settings, but FashionCLIP shows slight advantages in several complex styles, such as gal, retro, and natural, suggesting that domain-specific pretraining provides additional benefits when modeling diverse fashion semantics.

Although the prompt-based classification strategy achieves performance comparable to partial fine-tuning, partial fine-tuning consistently yields higher and more stable Macro F1 scores across per-style classification tasks (see Table 26). One key reason is that prompt-based methods only adapt the classifier in the embedding space, whereas partial fine-tuning updates the visual encoder itself, enabling the model to more effectively learn style-related attributes. This allows the model to better capture fine-grained fashion characteristics and style semantics.

From the overall comparison of fashion style classification (Table 25), we observe that partial fine-tuning and prompt-based classification achieve similar accuracy. Therefore, we further analyze their computational cost and latency. We find that, among the baseline models, the CNN-based ResNet50 requires over 2,600 times as many trainable parameters as CLIP-based models (see Table 27). Although it achieves the best classification performance among the baselines, the performance gain, approximately 5 to 8% comes at a disproportionately high computational cost. In contrast, FashionCLIP and OpenFashionCLIP maintain similar numbers of trainable parameters. In terms of inference latency, OpenFashionCLIP achieves the lowest latency, while the CNN-based ResNet50 exhibits higher inference time.

For CLIP adaptation strategies, both partial fine-tuning and prompt-based classification demonstrate comparable computational efficiency and latency (see Table 28). Despite its stronger learning capability, partial fine-tuning introduces only negligible overhead in both training and inference. Prompt-based methods, while efficient, rely heavily on pre-trained representations and may struggle to adapt to domain-specific nuances, such as subtle style variations. Finally, since our downstream tasks require high-quality embeddings for retrieval, adapting the backbone through partial fine-tuning provides a more robust and extensible foundation.

Table 26

Per-Style Classification Performance

	Baseline			OpenFashionCLIP			FashionCLIP			
Style	ResNet50	Open Fashion CLIP	Fashion CLIP	Partial FT*	Prompt Classify	Full FT*	Partial FT*	Prompt Classify	Full FT*	LoRA
casual	0.79±0.01	0.67±0.03	0.64±0.03	**0.85±0.02**	0.83±0.02	0.80±0.03	**0.85±0.02**	0.84±0.02	0.79±0.01	0.71±0.03
ethnic	0.89±0.03	0.82±0.01	0.82±0.02	**0.93±0.02**	0.92±0.01	0.88±0.02	0.93±0.01	0.92±0.01	0.88±0.02	0.88±0.02
formal	0.93±0.01	0.86±0.02	0.86±0.01	**0.95±0.02**	0.94±0.02	0.93±0.02	0.95±0.01	0.94±0.01	0.92±0.02	0.90±0.01
sports	0.88±0.01	0.86±0.02	0.85±0.02	**0.93±0.01**	0.92±0.02	0.89±0.02	0.92±0.02	0.92±0.02	0.88±0.02	0.88±0.02
conserv	0.72±0.03	0.60±0.04	0.61±0.04	0.75±0.02	0.73±0.02	0.65±0.05	**0.76±0.03**	0.74±0.03	0.66±0.03	0.59±0.03
dressy	0.94±0.02	0.94±0.01	0.94±0.01	**0.96±0.01**	**0.96±0.01**	0.94±0.02	**0.96±0.01**	**0.96±0.01**	0.94±0.01	0.94±0.02
fairy	0.92±0.01	0.85±0.02	0.88±0.02	**0.96±0.01**	**0.96±0.01**	0.91±0.03	0.96±0.00	0.95±0.01	0.91±0.03	0.92±0.02
feminine	0.74±0.03	0.47±0.03	0.46±0.03	0.78±0.03	0.77±0.04	0.70±0.06	0.78±0.03	**0.78±0.04**	0.72±0.04	0.64±0.02
gal	0.79±0.02	0.72±0.03	0.73±0.03	0.87±0.04	0.83±0.02	0.79±0.03	**0.88±0.02**	0.84±0.03	0.79±0.02	0.68±0.01
girlish	0.62±0.04	0.57±0.01	0.57±0.02	**0.71±0.03**	0.69±0.03	0.58±0.05	0.69±0.05	0.69±0.04	0.61±0.03	0.48±0.02
kireime	0.66±0.03	0.64±0.04	0.63±0.03	0.73±0.04	0.71±0.03	0.64±0.02	**0.74±0.03**	0.72±0.03	0.64±0.04	0.59±0.05
lolita	0.96±0.01	0.87±0.02	0.87±0.01	**0.97±0.01**	0.95±0.00	0.95±0.00	**0.97±0.01**	0.96±0.01	0.93±0.01	0.94±0.01
mode	0.76±0.03	0.76±0.01	0.75±0.01	0.83±0.01	0.84±0.02	0.74±0.02	**0.84±0.03**	0.83±0.02	0.73±0.04	0.73±0.02
natural	0.76±0.02	0.68±0.04	0.67±0.02	0.81±0.02	0.79±0.02	0.77±0.03	**0.83±0.02**	0.79±0.02	0.73±0.02	0.71±0.02
retro	0.70±0.02	0.59±0.05	0.62±0.04	0.80±0.03	0.79±0.02	0.70±0.02	**0.81±0.01**	0.80±0.03	0.69±0.02	0.60±0.07
rock	0.79±0.01	0.77±0.02	0.75±0.01	0.86±0.01	0.85±0.02	0.78±0.02	**0.86±0.02**	0.84±0.03	0.78±0.02	0.71±0.08
street	0.82±0.03	0.79±0.01	0.79±0.02	**0.90±0.02**	0.87±0.01	0.82±0.02	0.90±0.01	0.86±0.01	0.80±0.02	0.76±0.02

Note. Comparison of per-style classification performance of Macro F1 across models and training strategies, including baseline, partial fine-tuning, prompt tuning, and LoRA.

* FT refers to fine-tuning.

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Table 27

Training Computational Cost Comparison

	Model	Trainable Params	Total Train Time (s)	Avg Epoch Time (s)	Avg Batch Time (s)	Time per Sample (ms)
BaseLine	ResNet50	23542865.0 ± 0.0	2108.77 ± 53.15	178.32 ± 4.32	0.14 ± 0.0	17.9 ± 0.45
	OpenFashionCLIP	8721.0 ± 0.0	2114.88 ± 35.13	174.73 ± 2.91	0.08 ± 0.0	17.95 ± 0.3
	FashionCLIP	8721.0 ± 0.0	2066.78 ± 17.57	170.28 ± 1.59	0.07 ± 0.0	17.54 ± 0.15
OpenFashionCLIP	Partial Fine-Tune	14184465.00 ± 0.00	2114.18 ± 27.95	174.91 ± 2.53	0.08 ± 0.00	17.94 ± 0.24
	Prompt Classify	262657.00 ± 0.00	2094.12 ± 20.15	173.13 ± 1.76	0.08 ± 0.00	17.77 ± 0.17
	Full Fine-Tune	151286034.00 ± 0.00	2263.60 ± 25.86	190.16 ± 2.17	0.13 ± 0.00	19.21 ± 0.22
FashionCLIP	Partial Fine-Tune	14184465.00 ± 0.00	2053.85 ± 17.57	169.55 ± 1.47	0.08 ± 0.00	17.43 ± 0.15
	Prompt Classify	262657.00 ± 0.00	2045.90 ± 22.06	168.65 ± 1.89	0.08 ± 0.00	17.36 ± 0.19
	Full Fine-Tune	151286034.00 ± 0.00	2212.15 ± 32.58	185.26 ± 2.99	0.13 ± 0.00	18.77 ± 0.28
	LoRA	991761.00 ± 0.00	2098.97 ± 30.78	173.89 ± 2.75	0.09 ± 0.00	17.81 ± 0.26

Note. The results show that CNN-based ResNet50 requires significantly more trainable parameters than CLIP-based models, while providing only marginal performance gains. CLIP-based methods, especially under partial fine-tuning and prompt-based strategies, achieve comparable training time and efficiency with substantially reduced parameter overhead. This highlights the advantage of CLIP-based architectures in terms of scalability and computational efficiency.

Table 28

Inference Latency and Efficiency Comparison

	Model	Inference Total Time (s)	Inference Avg Batch Time (s)	Inference Time per Sample (ms)
BaseLine	ResNet50	6.02 ± 0.12	0.08 ± 0.0	2.38 ± 0.05
	OpenFashionCLIP	5.77 ± 0.14	0.07 ± 0.0	2.28 ± 0.06
	FashionCLIP	5.82 ± 0.17	0.07 ± 0.0	2.31 ± 0.07
OpenFashionCLIP	Partial Fine-Tune	5.87 ± 0.28	0.07 ± 0.00	2.32 ± 0.11
	Prompt Classify	5.77 ± 0.20	0.07 ± 0.00	2.28 ± 0.08
	Full Fine-Tune	5.78 ± 0.20	0.07 ± 0.00	2.29 ± 0.08
FashionCLIP	Partial Fine-Tune	6.05 ± 0.11	0.08 ± 0.00	2.40 ± 0.04
	Prompt Classify	5.92 ± 0.17	0.07 ± 0.00	2.34 ± 0.07
	Full Fine-Tune	6.04 ± 0.07	0.08 ± 0.00	2.39 ± 0.03
	LoRA	6.19 ± 0.06	0.08 ± 0.00	2.45 ± 0.03

Note. The results indicate that CLIP-based models achieve comparable or lower latency than the CNN baseline, with OpenFashionCLIP showing the fastest inference time. Among the adaptation strategies, partial fine-tuning and prompt-based classification exhibit similar latency, suggesting that partial fine-tuning does not introduce significant computational overhead despite its improved performance. This demonstrates that the proposed approach maintains efficiency while enhancing representation quality.

Confusion Matrix Analysis

To further examine model behavior, we analyze the confusion matrices of the major CLIP-based adaptation strategies. In general, the stronger models, especially OpenFashionCLIP partial fine-tuning and FashionCLIP partial fine-tuning, show clearer diagonal patterns, indicating better separation across style categories. By contrast, full fine-tuning and LoRA produce more off-diagonal errors, suggesting weaker discrimination among visually similar classes. Across different models, several categories such as dressy, lolita, formal, and street are recognized more consistently, while classes such as conservative, girlish, and kireime-casual remain more difficult. These confusing pairs are likely caused by overlapping visual characteristics and subtle style boundaries. Overall, the confusion matrices support the quantitative results in Figures 34 and 35, showing that partial fine-tuning yields the most stable class separation, whereas full fine-tuning and LoRA lead to more frequent misclassification.

Figure 34

Confusion matrices of OpenFashionCLIP adaptation strategies, including partial fine-tuning, prompt classification, and full fine-tuning.

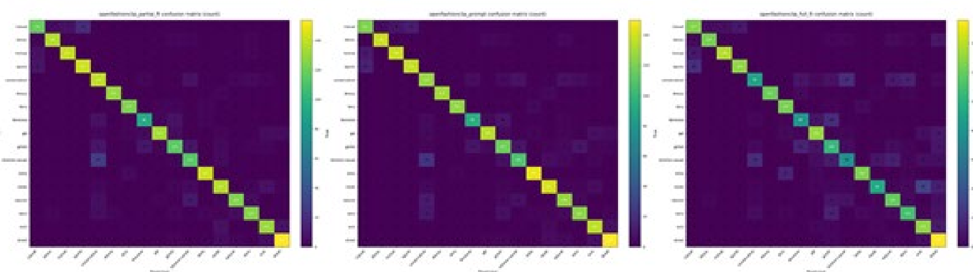
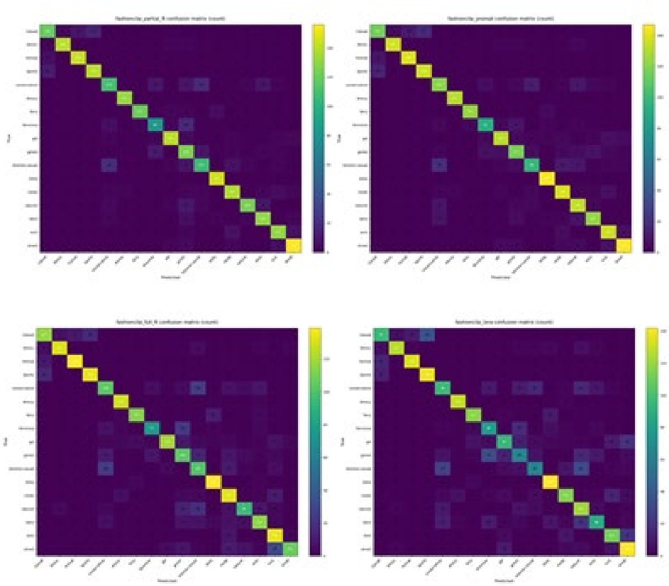


Figure 35

Confusion matrices of FashionCLIP adaptation strategies, including partial fine-tuning, prompt classification, full fine-tuning, and LoRA.



Conclusion

This chapter introduces style modeling as a higher-level semantic component for fine-grained fashion recommendation. While garment attributes such as category, color, material, and pattern capture item-specific properties, style provides an additional representation of overall aesthetic intent. This makes it particularly valuable for recommendation, where semantic compatibility often extends beyond local visual details.

In summary, although FashionCLIP and OpenFashionCLIP are pretrained on large-scale fashion-related datasets, their ability to recognize more specific fashion style attributes remains slightly inferior to a fully supervised model such as ResNet50. However, even limited adaptation significantly improves their performance. Experimental results consistently show that partial fine-tuning is the most effective adaptation strategy for the fashion style classification task. It not only achieves the highest overall performance but also produces cleaner confusion matrices and more balanced per-class results.

Although prompt-based classification achieves comparable overall accuracy, partial fine-tuning yields more stable results across different styles and better captures fine-grained fashion semantics. This is because prompt-based methods primarily operate in the embedding space, while partial fine-tuning updates the visual encoder, enabling more effective learning of style-specific representations. From a computational perspective, partial fine-tuning also offers a favorable trade-off between performance and efficiency. Compared to CNN-based baselines such as ResNet50, which require substantially more trainable parameters for relatively modest performance gains, CLIP-based models remain significantly more lightweight. Moreover, partial fine-tuning introduce only negligible overhead compared to prompt-based methods while providing greater adaptability to domain-specific nuances.

Overall, these findings suggest that a controlled adaptation strategy, rather than full model updating, is more suitable for fashion style modeling. Based on these observations, partial fine-tuning is selected as the core classification strategy and will be used in the subsequent components of this thesis.

Chapter 7: Fine-Grained Attribute-Aware Multimodal Fashion Retrieval Framework

Introduction

This chapter focuses on the thesis's main research question: whether fine-grained semantic garment representations can improve recommendation quality beyond pure visual similarity. To solve this problem, retrieval is formulated as a multimodal recommendation task in which each item can be represented by its image, its predicted six-attribute description, or both. This design allows the study to compare how visual detail, semantic information, and their combinations contribute to recommendation behavior, and whether structured fine-grained attributes can serve as an effective retrieval interface for user-centered fashion recommendation.

In the proposed framework, each fashion item is represented using six attributes: category, subcategory, color, material, pattern, and style. This representation is designed to cover both structural garment identity and fine-grained semantic detail. Categories and subcategories define the basic product type, while color, material, and pattern describe visible item characteristics that often appear in user queries. Style extends the representation to a higher semantic level, capturing broader fashion meaning that cannot be fully described by low-level garment properties alone. Together, these six attributes provide a recommendation-oriented semantic representation for retrieval.

We first evaluate the image-only baselines and the text-only retrieval models. This design allows us to analyze the impact of visual features, attribute-based semantic representations, and retrieval quality of both methods. After establishing these baselines, we further explore native multimodal retrieval and fusion-based methods, in which both image and text information are incorporated via joint embedding models or score and ranking-level fusion mechanisms.

In the fusion setting, score and rank fusion are used to combine the image and text outputs. This allows the system to leverage complementary strengths from visual similarity and semantic consistency, which has been shown to be effective in retrieval systems where different modalities capture different aspects of relevance (Cormack et al., 2009). In the text representations, the experiment is initialized with both structured and sentence-based formats. The structured format directly concatenates attribute values, while the sentence format rewrites the same information into natural language. However, after conducting the initial image-only and text-only experiments, we observe that the two formats produce highly similar retrieval outputs and comparable performance across all evaluation metrics.

Based on this observation, we adopt the structured representation in fusion and native multimodal experiments. This simplifies the system pipeline, reduces preprocessing complexity, shortens the input sequence length, and lowers the computational cost and memory usage of the text encoder. Moreover, this finding aligns with prior work in retrieval and representation learning, where concise and structured semantic inputs are often sufficient for effective similarity matching, especially when the key attributes are explicitly provided (Muennighoff et al., 2023; Reimers & Gurevych, 2019).

Experiment Framework

For the retrieval experiments, a controlled query subset was used to support detailed comparison across multiple retrieval settings and repeated seed-based gallery configurations. This design enabled consistent analysis of image-only, text-only, multimodal, and fusion-based retrieval under the same evaluation protocol. Although the current query size is sufficient for comparative analysis within the present framework, expanding the query pool would be a useful direction for future work to further strengthen robustness and generalizability.

All retrieval experiments are conducted across five seed-based gallery settings under the same evaluation pipeline. This design is adopted to reduce the dependence of the results on a single gallery partition and to provide a more stable estimate of recommendation performance. By repeating the evaluation across five seeds, the study examines whether the observed retrieval trends remain consistent under different balanced gallery configurations. Therefore, the reported results are intended to reflect not only raw effectiveness but also robustness across repeated retrieval settings.

In the first stage, each image is assigned six semantic attributes predicted by FACTOR (see chapter 5), with category and subcategory treated as single-label attributes and color, material, and pattern as multi-label attributes. The sixth attribute, style, is predicted by a separate style classifier (refer to Chapter 6). After this step, each image is represented by a unified attribute profile: category, subcategory, color, material, pattern, and style. In the second stage, the attribute profile is converted into text form when needed. Two text representations are tested: structured and sentence. Structured representation, which directly concatenates the six attributes in a fixed order, for example: tops, sweater, red, cotton, striped, formal. The sentence representation, which rewrites the same information into a natural language description: “A red cotton striped sweater top in a formal style.” These two text forms are then used in the text-only, native multimodal, and fusion settings. In the third stage, the query and gallery embeddings are compared in a shared representation space, and the top-k retrieved items are returned.

The retrieved results are then evaluated from two perspectives. First, we compute ranking-based retrieval metrics such as Precision@k, Recall@k, MRR@k, MAP@k, and nDCG@k. Second, we compute attribute consistency metrics, including attribute matching ratio, exact attribute matching, semantic similarity, and an overall weighted attribute score.

The retrieval models in this study were selected to provide strong, complementary baselines across different embedding paradigms rather than to represent a single model family. BGE-M3 was included because it is designed as a versatile retrieval model with multi-functionality, multilingualism, and multi-granularity support, enabling dense, sparse, and multi-vector retrieval within a single framework. Voyage-3-large was selected as a strong general-purpose embedding model because its developer presents it as a state-of-the-art retrieval model with a favorable accuracy-cost trade-off. SigLIP was chosen as a strong vision-language baseline because it extends CLIP-style image-text pretraining with a sigmoid-based objective and has shown strong performance in visual language retrieval settings. Voyage AI (2025) Multimodal was included to represent recent multimodal embedding systems that jointly encode text and image content in a shared embedding space and are explicitly designed for multimodal retrieval. Together, these models provide a meaningful comparison across text-oriented, vision-language, and multimodal retrieval settings for fashion recommendation. Detailed descriptions of the experiments and their explanations are provided below:

Image-Only Retrieval

In this setting, only the image is used to generate embeddings, and no text or attribute information is included in the retrieval step. Three image-only models are tested:

- FashionCLIP, a CLIP-style model adapted to the fashion domain and designed for tasks such as retrieval, classification, and fashion parsing. This makes it a strong domain-specific baseline for fashion image retrieval.
- OpenFashionCLIP, a fashion-oriented vision-language contrastive model trained on open-source fashion data, was proposed to improve robustness and out-of-domain generalization for fashion retrieval and tagging tasks.
- SigLIP2, a more recent vision language encoder that extends SigLIP and reports stronger image text retrieval and transfer performance than the earlier SigLIP family.

In our implementation, all image embeddings are L2-normalized, and retrieval is performed using cosine similarity via matrix multiplication between the query and gallery embeddings.

Text-Only Retrieval

In this setting, the query image and gallery image are both represented by their attribute-derived text, and recommendation is performed in the text embedding space. This experiment is designed to test whether the six predicted labels already contain enough semantic information for fashion recommendation, even without using the raw image. Two text embedding models are evaluated:

- BGE M3 is a multilingual and multi-functionality embedding model designed to support retrieval with different input granularities.

- Voyage is an instruction-tuned embedding model optimized for retrieval, in which the query and document roles can be specified explicitly during embedding.

To examine how semantic representation should be expressed for retrieval, this study compares two text construction strategies: a structured attribute format, which is originally generated by FACTOR and the style classification model, and a sentence-based format, which applies the originally format in this template: "A {color} {material} {pattern} {subcategory} {category} in a {style} style". The structured format directly concatenates the six predicted attributes, while the sentence-based format presents the same information in a more natural language style. This comparison is important because it helps determine whether retrieval performance depends mainly on linguistic fluency or on the semantic content itself. If structured text performs similarly to sentence-based text comparably, this suggests that the key factor is not language richness but the accuracy and completeness of the underlying semantic representation. This distinction is valuable because structured text is simpler to generate, easier to control, and more computationally efficient.

Native Multimodal Retrieval

The third group is native multimodal retrieval. The native multimodal setting is included to test whether models that jointly encode image and text information in a shared embedding space can provide stronger recommendation retrieval performance than single-modality baselines. In this stage, each query and gallery item is represented by both its image and its semantic text description, allowing the experiment to evaluate multimodal matching directly rather than combining separate rankings after retrieval. This setting is important because fashion recommendation often depends on both visible garment appearance and explicit semantic properties, and native multimodal retrieval offers a direct way to model these two sources of information together. Unlike image-only and text-only settings, native multimodal models directly encode both image and text into a shared representation. This setup is especially suitable for fashion recommendation because the image captures appearance details, while the text provides explicit semantic constraints. The evaluated native multimodal models are:

- Voyage Multimodal, which supports embedding of text, images, or interleaved multimodal inputs into the same retrieval space.
- SigLIP, which is a vision language model trained with sigmoid loss and is widely used for image text alignment and retrieval.
- BGE VL base, which is part of the Visualized BGE family and is designed as a universal multimodal embedding model for hybrid modality retrieval.

In this setting, each query and gallery item is represented by an image and its corresponding attribute text. The model then produces one embedding vector for each multimodal pair, and recommendations are made via nearest-neighbor retrieval in the multimodal embedding space.

Fusion Retrieval

After identifying FashionCLIP and Voyage as the best performing image-only and text-only retrieval models, respectively, we incorporated them into a late-fusion retrieval framework. Instead of using a single model to encode both modalities, this setting combines the scores or rankings from the image and text branches. The fusion setting is designed to examine whether visual similarity and semantic similarity provide complementary signals for recommendation retrieval. In the current experiments, score fusion adopts a balanced setting in which image and text similarity contribute equally, while rank fusion follows the standard reciprocal rank fusion formulation without additional task specific tuning. The purpose of this design is to evaluate the general behavior of late fusion under a simple and controlled setting rather than to optimize the fusion strategy extensively. A more systematic study of fusion weights and rank fusion hyperparameters can be explored in future work. In our experiments, the image branch is FashionCLIP and the text branch is Voyage with the structured text representation. We test two fusion mechanisms:

- Score fusion. Let S_{img} and S_{txt} denote the image similarity matrix and the text similarity matrix. Since the two branches may have different score scales, the system first applies row-wise min-max normalization to both matrices, and then computes the fused score as:

$$S_{fused} = \alpha \hat{S}_{img} + \beta \hat{S}_{txt} \tag{20}$$

  where $\alpha = 0.5$ and $\beta = 0.5$ in our implementation.

- Rank fusion using reciprocal rank fusion. If an item appears at rank r_{img} in the image branch and at rank r_{txt} in the text branch, its fused score is computed as:

$$RRF(d) = \frac{1}{k + r_{img}(d)} + \frac{1}{k + r_{txt}(d)} \tag{21}$$

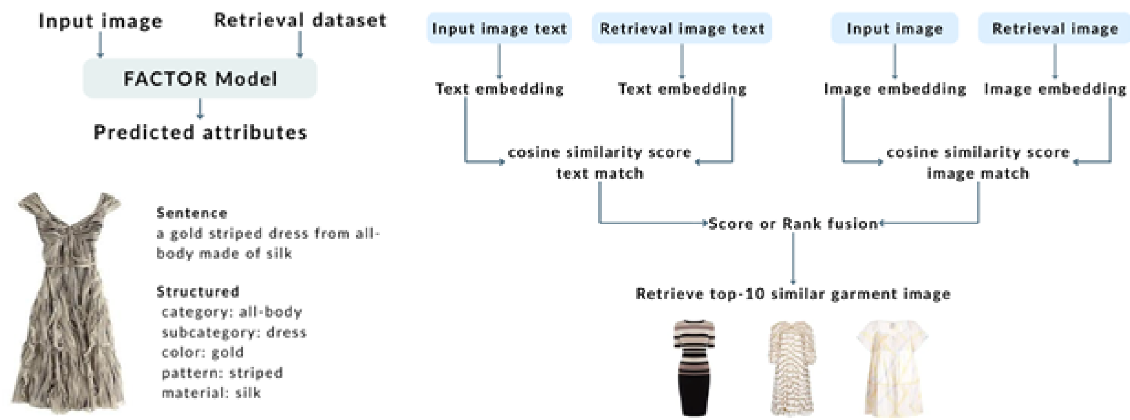
where $k = 60$ is the fusion constant.

This fusion design could improve retrieval quality by preserving the visual strength of image retrieval while also injecting explicit semantic information from the predicted attributes.

Figure 36 presents the proposed recommendation pipeline, from semantic attribute extraction to multimodal late-fusion retrieval. Given an input garment image, the FACTOR model first predicts five fine-grained fashion attributes, namely category, subcategory, color, material, and pattern, as illustrated in the left panel. These outputs are subsequently transformed into textual descriptions, either in sentence form or in a structured format. In the present work, we use the structured representation for retrieval, since it provides a more explicit and stable semantic description of garment properties. As shown in the right panel, retrieval is then performed through two complementary branches. The first branch is an image-based retrieval pathway, where the input image and retrieval images are encoded by FashionCLIP to compute cosine similarity in the visual embedding space. The second branch is a text-based retrieval pathway, where the structured attribute text of the query and the retrieval candidates is embedded by Voyage to measure semantic similarity. Finally, the two similarity signals are integrated using either score fusion or reciprocal rank fusion. This late-fusion strategy is motivated by the observation that visual similarity alone may fail to capture recommendation-relevant semantic attributes, while text similarity alone may overlook appearance-level cues. By combining both modalities, the system balances visual consistency with semantic precision in a fine-grained fashion recommendation.

Figure 36

Overview of the proposed fusion retrieval pipeline. The left panel shows the FACTOR-based attribute extraction process and right panel shows the late-fusion retrieval framework.



Experimental Results

In this section, we evaluate the recommendation framework from three perspectives: overall attribute-matching quality, the retrieval ranking performance, and semantic similarity of each retrieved item. The results demonstrate that attribute-based representations play a critical role in improving recommendation quality and that combining visual and semantic information leads to more balanced performance.

The strong performance of text-only retrieval should not be interpreted simply as text models being universally better than image models. Rather, it indicates that once the upstream a-tribute prediction modules provide sufficiently accurate structured semantics, retrieval can be driven very effectively by semantic matching alone. In other words, the strong text-only results reflect the quality of the six-attribute representation learned in the earlier stages. This finding supports the central design of this thesis, namely that recommendation quality depends heavily on whether the system can first transform garment images into reliable semantic representations.

Another important finding is that compact structured text can perform comparably to sentence-based text in retrieval. This suggests that recommendation quality depends more on whether the semantic attributes are accurate and complete than on whether they are expressed in fluent natural language. For the present framework, this is a useful result because structured attribute text is easier to construct, easier to control, and more suitable for systematic recommendation experiments than longer generated descriptions.

Overall Attribute Matching Performance

We compare all methods using the overall Weighted Attribute Score (WAS), which summarizes attribute consistency across all six attributes. From Table 29, text-only methods significantly outperform image-only methods. Achieves the best performance with a WAS of 99%, in contrast, image-only models such as FashionCLIP and OpenFashionCLIP achieve lower scores around 94 to 95%, indicating that visual similarity alone is insufficient to preserve fine-grained attribute information. Native multimodal and fusion methods do not surpass the text-only models, although multimodal models incorporate both image and text information, their performance is slightly lower than pure text-based approaches. This suggests that when attribute labels are already highly accurate, structured semantic representations alone are sufficient to achieve strong attribute-level consistency.

Table 29

Overall Attribute Matching Performance

	Model	WAS
Image-Only	FashionCLIP	0.95 ± 0.00
	OpenFashionCLIP	0.94 ± 0.00
	SigLIP2	0.94 ± 0.00
Text-Only	BGE-M3 (structure)	**0.99 ± 0.00**
	Voyage (structure)	**0.99 ± 0.00**
	BGE-M3 (sentence)	**0.99 ± 0.00**
	Voyage (sentence)	**0.99 ± 0.00**
Multimodal	Voyage Multimodal	0.98 ± 0.00
	SigLIP	0.97 ± 0.00
Fusion*	Score Fusion	0.98 ± 0.00
	Rank Fusion	0.98 ± 0.00

Note. Text-only models achieve the highest scores, indicating that structured attribute representations are highly effective for preserving semantic consistency in retrieval.

* Fusion is implemented by combining FashionCLIP for image encoding and Voyage for text encoding.

Semantic Similarity Analysis

Text-only methods achieve high semantic similarity across all attributes, reaching values close to 1.00 (see Table 30). This indicates that attribute-based text representations preserve semantic meaning during retrieval. Moreover, both structured and sentence-based formats perform similarly, confirming that the choice of text formulation has minimal impact when the semantic content is identical. In comparison, image-only models show slightly lower performance, especially for color and style attributes. This reflects the limitation of visual embeddings in capturing abstract or high-level semantic concepts. Multimodal and fusion methods improve over image-only baselines but still fall slightly behind text-only approaches in semantic consistency.

Table 30

Semantic Similarity of Top-10 Retrieved Items

	Model	Category	Subcategory	Color	Material	Pattern	Style
Image-Only	FashionCLIP	0.98 ± 0.00	0.95 ± 0.00	0.94 ± 0.00	0.97 ± 0.00	0.95 ± 0.00	0.89 ± 0.00
	OpenFashion CLIP	0.98 ± 0.00	0.94 ± 0.00	0.92 ± 0.00	0.97 ± 0.00	0.94 ± 0.00	0.89 ± 0.00
	SigLIP2	0.98 ± 0.00	0.93 ± 0.00	0.93 ± 0.00	0.96 ± 0.00	0.95 ± 0.00	0.89 ± 0.00
Text-Only	BGE-M3 (structure)	**1.00 ± 0.00**	0.99 ± 0.00	**0.99 ± 0.00**	**0.99 ± 0.00**	0.97 ± 0.00	**0.99 ± 0.00**
	Voyage (structure)	**1.00 ± 0.00**	**1.00 ± 0.00**	**0.99 ± 0.00**	**0.99 ± 0.00**	**0.98 ± 0.00**	0.98 ± 0.00
	BGE-M3 (sentence)	0.99 ± 0.00	0.99 ± 0.00	**0.99 ± 0.00**	**0.99 ± 0.00**	0.97 ± 0.00	**0.99 ± 0.00**
	Voyage (sentence)	0.99 ± 0.00	**1.00 ± 0.00**	**0.99 ± 0.00**	**0.99 ± 0.00**	0.97 ± 0.00	**0.99 ± 0.00**
Multimodal	Voyage Multimodal	**1.00 ± 0.00**	0.98 ± 0.00	0.98 ± 0.01	0.98 ± 0.00	0.97 ± 0.00	0.96 ± 0.00
	SigLIP	0.99 ± 0.00	0.97 ± 0.00	0.93 ± 0.00	0.97 ± 0.00	0.96 ± 0.00	0.96 ± 0.00
Fusion	Score Fusion	**1.00 ± 0.00**	0.99 ± 0.00	0.97 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.94 ± 0.00
	Rank Fusion	0.99 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.95 ± 0.00

Note. Text-based methods consistently achieve near-perfect similarity, while image-only models show limitations in capturing high-level semantic attributes such as style.

Retrieval Ranking Quality

The goal here is to measure whether the retrieved items match the ground-truth attributes of the query. The results (see Tables 31 to 36) show that most methods achieve high ranking performance, especially for text-only and multimodal models, where scores are close to 1.00 across different k values. This indicates that the retrieval task is relatively well solved when accurate attribute information is available.

Attribute-Level Quality

To better understand model behavior, we examine the attribute match rate (AMR), exact attribute match (EAM), and semantic similarity for each attribute. The goal here is to measure whether the retrieved results are not only relevant, but also correctly ordered. The results are shown in Tables 37 to 42.

For category and subcategory attributes, all methods achieve relatively high performance, since these attributes are strongly correlated with visual features. However, text-only methods show superior consistency, particularly on exact-matching metrics. For color, material, and pattern, the gap between image-only and text-based methods becomes more evident. Image-only models struggle with multi-label attributes and subtle variations, while text-only models maintain high performance due to explicit attribute encoding.

Table 31

Retrieval Ranking Performance for Category

	Model	nDCG @1	nDCG @3	nDCG @5	nDCG @10	P @1	P @3	P @5	P @10	MRR @1	MRR @3	MRR @5	MRR @10
Image-Only	FashionCLIP	**1.00 ± 0.00**	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.93 ± 0.02	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.00	0.93 ± 0.02	0.94 ± 0.02	0.94 ± 0.02	0.94 ± 0.02
	OpenFashion CLIP	**1.00 ± 0.00**	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.92 ± 0.01	0.92 ± 0.01	0.93 ± 0.00	0.92 ± 0.00	0.92 ± 0.01	0.92 ± 0.01	0.92 ± 0.01	0.92 ± 0.01
	SigLIP2	**1.00 ± 0.00**	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.95 ± 0.01	0.94 ± 0.01	0.94 ± 0.00	0.93 ± 0.00	0.95 ± 0.01	0.96 ± 0.02	0.96 ± 0.02	0.96 ± 0.02
Text-Only	BGE-M3 (structure)	**1.00 ± 0.00**	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01
	Voyage (structure)	**1.00 ± 0.00**	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01
	BGE-M3 (sentence)	**1.00 ± 0.00**	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01
	Voyage (sentence)	**1.00 ± 0.00**	1.00 ± 0.00	0.99 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01
Multimodal	Voyage Multimodal	**1.00 ± 0.00**	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01
	SigLIP	**1.00 ± 0.00**	0.99 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.96 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01
Fusion*	Score Fusion	**1.00 ± 0.00**	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Rank Fusion	**1.00 ± 0.00**	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01

* Fusion is implemented by combining FashionCLIP for image encoding and Voyage for text encoding.

Table 32

Retrieval Ranking Performance for Subcategory

	Model	nDCG @1	nDCG @3	nDCG @5	nDCG @10	P @1	P @3	P @5	P @10	MRR @1	MRR @3	MRR @5	MRR @10
Image-Only	FashionCLIP	1.00 ± 0.00	0.97 ± 0.01	0.97 ± 0.01	0.96 ± 0.00	0.86 ± 0.03	0.86 ± 0.01	0.85 ± 0.01	0.85 ± 0.01	0.86 ± 0.03	0.86 ± 0.02	0.86 ± 0.02	0.86 ± 0.02
	OpenFashion CLIP	1.00 ± 0.00	0.98 ± 0.00	0.96 ± 0.00	0.95 ± 0.00	0.83 ± 0.02	0.83 ± 0.02	0.84 ± 0.01	0.84 ± 0.01	0.83 ± 0.02	0.84 ± 0.02	0.84 ± 0.02	0.84 ± 0.02
	SigLIP2	1.00 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.95 ± 0.00	0.85 ± 0.01	0.83 ± 0.01	0.83 ± 0.01	0.82 ± 0.01	0.85 ± 0.01	0.86 ± 0.01	0.86 ± 0.01	0.86 ± 0.01
Text-Only	BGE-M3 (structure)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01
	Voyage (structure)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	1.00 ± 0.01	1.00 ± 0.01	1.00 ± 0.01	1.00 ± 0.01
	BGE-M3 (sentence)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01
	Voyage (sentence)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
Multimodal	Voyage Multimodal	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.01	0.97 ± 0.01	0.96 ± 0.01	0.95 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01
	SigLIP	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.94 ± 0.01	0.93 ± 0.02	0.93 ± 0.01	0.91 ± 0.01	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01
Fusion*	Score Fusion	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.00	0.98 ± 0.00	0.95 ± 0.03	0.95 ± 0.01	0.96 ± 0.00	0.96 ± 0.00	0.95 ± 0.03	0.95 ± 0.03	0.95 ± 0.03	0.95 ± 0.03
	Rank Fusion	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.01	0.98 ± 0.01	0.97 ± 0.00	0.96 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01

* Fusion is implemented by combining FashionCLIP for image encoding and Voyage for text encoding.

Table 33

Retrieval Ranking Performance for Color

	Model	nDCG @1	nDCG @3	nDCG @5	nDCG @10	P @1	P @3	P @5	P @10	MRR @1	MRR @3	MRR @5	MRR @10
Image-Only	FashionCLIP	1.00 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.91 ± 0.01	0.90 ± 0.01	0.90 ± 0.01	0.90 ± 0.01	0.91 ± 0.01	0.91 ± 0.01	0.91 ± 0.01	0.91 ± 0.01
	OpenFashion CLIP	1.00 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.89 ± 0.01	0.89 ± 0.00	0.87 ± 0.01	0.87 ± 0.01	0.89 ± 0.01	0.89 ± 0.01	0.89 ± 0.01	0.89 ± 0.01
	SigLIP2	1.00 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.90 ± 0.01	0.89 ± 0.01	0.89 ± 0.01	0.88 ± 0.01	0.90 ± 0.01	0.90 ± 0.01	0.90 ± 0.01	0.90 ± 0.01
Text-Only	BGE-M3 (structure)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	Voyage (structure)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.01	0.99 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	1.00 ± 0.01	1.00 ± 0.01	1.00 ± 0.01	1.00 ± 0.01
	BGE-M3 (sentence)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	Voyage (sentence)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
Multimodal	Voyage Multimodal	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01
	SigLIP	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.00	0.97 ± 0.00	0.90 ± 0.01	0.90 ± 0.01	0.90 ± 0.00	0.89 ± 0.01	0.90 ± 0.01	0.91 ± 0.01	0.91 ± 0.01	0.91 ± 0.01
Fusion*	Score Fusion	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.01	0.97 ± 0.00	0.96 ± 0.01	0.95 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01
	Rank Fusion	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.01	0.97 ± 0.01	0.96 ± 0.00	0.95 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01

* Fusion is implemented by combining FashionCLIP for image encoding and Voyage for text encoding.

Table 34

Retrieval Ranking Performance for Material

	Model	nDCG @1	nDCG @3	nDCG @5	nDCG @10	P @1	P @3	P @5	P @10	MRR @1	MRR @3	MRR @5	MRR @10
Image-Only	FashionCLIP	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01	0.93 ± 0.00	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01
	OpenFashion CLIP	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.00	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01
	SigLIP2	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.95 ± 0.01	0.93 ± 0.01	0.93 ± 0.01	0.92 ± 0.01	0.95 ± 0.01	0.95 ± 0.01	0.95 ± 0.01	0.95 ± 0.01
Text-Only	BGE-M3 (structure)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.96 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01
	Voyage (structure)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01
	BGE-M3 (sentence)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.96 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01
	Voyage (sentence)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.96 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01
Multimodal	Voyage Multimodal	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01	0.96 ± 0.01
	SigLIP	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.01	0.98 ± 0.01	0.95 ± 0.02	0.95 ± 0.01	0.95 ± 0.01	0.94 ± 0.01	0.95 ± 0.02	0.95 ± 0.02	0.95 ± 0.02	0.95 ± 0.02
Fusion*	Score Fusion	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.96 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01
	Rank Fusion	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.97 ± 0.01	0.97 ± 0.01	0.96 ± 0.00	0.96 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01

* Fusion is implemented by combining FashionCLIP for image encoding and Voyage for text encoding.

Table 35

Retrieval Ranking Performance for Pattern

	Model	nDCG @1	nDCG @3	nDCG @5	nDCG @10	P @1	P @3	P @5	P @10	MRR @1	MRR @3	MRR @5	MRR @10
Image-Only	FashionCLIP	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.01	0.97 ± 0.00	0.89 ± 0.02	0.87 ± 0.02	0.85 ± 0.01	0.85 ± 0.01	0.89 ± 0.02	0.90 ± 0.02	0.90 ± 0.02	0.90 ± 0.02
	OpenFashion CLIP	1.00 ± 0.00	0.96 ± 0.00	0.96 ± 0.00	0.95 ± 0.00	0.82 ± 0.01	0.82 ± 0.01	0.82 ± 0.01	0.81 ± 0.01	0.82 ± 0.01	0.84 ± 0.01	0.84 ± 0.01	0.84 ± 0.01
	SigLIP2	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.00	0.97 ± 0.00	0.88 ± 0.01	0.87 ± 0.01	0.86 ± 0.01	0.85 ± 0.01	0.88 ± 0.01	0.88 ± 0.01	0.88 ± 0.01	0.88 ± 0.01
Text-Only	BGE-M3 (structure)	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.96 ± 0.02	0.95 ± 0.01	0.94 ± 0.01	0.92 ± 0.01	0.96 ± 0.02	0.96 ± 0.02	0.96 ± 0.02	0.96 ± 0.02
	Voyage (structure)	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.97 ± 0.01	0.95 ± 0.01	0.94 ± 0.01	0.92 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.01
	BGE-M3 (sentence)	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.00	0.95 ± 0.02	0.94 ± 0.01	0.94 ± 0.01	0.92 ± 0.01	0.95 ± 0.02	0.96 ± 0.02	0.96 ± 0.02	0.96 ± 0.02
	Voyage (sentence)	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.95 ± 0.02	0.93 ± 0.01	0.92 ± 0.01	0.90 ± 0.00	0.95 ± 0.02	0.95 ± 0.02	0.95 ± 0.02	0.95 ± 0.02
Multimodal	Voyage Multimodal	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.93 ± 0.01	0.91 ± 0.01	0.91 ± 0.01	0.91 ± 0.01	0.93 ± 0.01	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.01
	SigLIP	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	0.92 ± 0.03	0.90 ± 0.01	0.90 ± 0.01	0.89 ± 0.01	0.92 ± 0.03	0.93 ± 0.02	0.93 ± 0.02	0.93 ± 0.02
Fusion*	Score Fusion	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.93 ± 0.02	0.92 ± 0.01	0.92 ± 0.00	0.90 ± 0.00	0.93 ± 0.02	0.94 ± 0.02	0.94 ± 0.02	0.94 ± 0.02
	Rank Fusion	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	0.93 ± 0.01	0.92 ± 0.02	0.91 ± 0.01	0.91 ± 0.00	0.93 ± 0.01	0.93 ± 0.01	0.93 ± 0.01	0.93 ± 0.01

* Fusion is implemented by combining FashionCLIP for image encoding and Voyage for text encoding.

Table 36

Retrieval Ranking Performance for Style

	Model	nDCG @1	nDCG @3	nDCG @5	nDCG @10	P @1	P @3	P @5	P @10	MRR @1	MRR @3	MRR @5	MRR @10
Image-Only	FashionCLIP	1.00 ± 0.00	0.94 ± 0.01	0.91 ± 0.01	0.88 ± 0.00	0.62 ± 0.02	0.59 ± 0.02	0.58 ± 0.02	0.56 ± 0.01	0.62 ± 0.02	0.65 ± 0.01	0.65 ± 0.01	0.65 ± 0.01
	OpenFashion CLIP	1.00 ± 0.00	0.93 ± 0.01	0.90 ± 0.01	0.86 ± 0.01	0.60 ± 0.03	0.56 ± 0.01	0.56 ± 0.01	0.55 ± 0.01	0.60 ± 0.03	0.62 ± 0.02	0.62 ± 0.02	0.62 ± 0.02
	SigLIP2	1.00 ± 0.00	0.93 ± 0.01	0.90 ± 0.01	0.86 ± 0.01	0.59 ± 0.01	0.59 ± 0.02	0.58 ± 0.00	0.56 ± 0.00	0.59 ± 0.01	0.62 ± 0.01	0.62 ± 0.01	0.62 ± 0.01
Text-Only	BGE-M3 (structure)	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.02	0.97 ± 0.01	0.96 ± 0.01	0.94 ± 0.01	0.98 ± 0.02	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01
	Voyage (structure)	1.00 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.00	0.94 ± 0.01	0.93 ± 0.01	0.93 ± 0.00	0.92 ± 0.01	0.94 ± 0.01	0.95 ± 0.01	0.95 ± 0.01	0.95 ± 0.01
	BGE-M3 (sentence)	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.01	0.98 ± 0.00	0.97 ± 0.00	0.96 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01
	Voyage (sentence)	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.01	0.97 ± 0.01	0.96 ± 0.01	0.95 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01
Multimodal	Voyage Multimodal	1.00 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.92 ± 0.01	0.87 ± 0.01	0.86 ± 0.01	0.83 ± 0.01	0.92 ± 0.01	0.92 ± 0.01	0.92 ± 0.01	0.92 ± 0.01
	SigLIP	1.00 ± 0.00	0.97 ± 0.00	0.97 ± 0.01	0.95 ± 0.00	0.87 ± 0.01	0.86 ± 0.01	0.85 ± 0.01	0.83 ± 0.01	0.87 ± 0.01	0.89 ± 0.01	0.89 ± 0.01	0.89 ± 0.01
Fusion*	Score Fusion	1.00 ± 0.00	0.96 ± 0.01	0.94 ± 0.00	0.92 ± 0.01	0.80 ± 0.02	0.78 ± 0.02	0.77 ± 0.02	0.76 ± 0.01	0.80 ± 0.02	0.82 ± 0.02	0.82 ± 0.02	0.82 ± 0.02
	Rank Fusion	1.00 ± 0.00	0.97 ± 0.01	0.95 ± 0.00	0.94 ± 0.00	0.85 ± 0.01	0.85 ± 0.02	0.83 ± 0.01	0.81 ± 0.01	0.85 ± 0.01	0.86 ± 0.01	0.86 ± 0.01	0.86 ± 0.01

* Fusion is implemented by combining FashionCLIP for image encoding and Voyage for text encoding.

Table 37

Attribute Matching Performance for Category

	Model	AMR @1	AMR @3	AMR @5	AMR @10	EAM @1	EAM @3	EAM @5	EAM @10	SS @1	SS @3	SS @5	SS @10
Image-Only	FashionCLIP	0.90 ± 0.03	0.91 ± 0.02	0.91 ± 0.01	0.90 ± 0.01	0.90 ± 0.03	0.91 ± 0.02	0.91 ± 0.01	0.90 ± 0.01	0.98 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.98 ± 0.00
	OpenFashion CLIP	0.87 ± 0.02	0.88 ± 0.01	0.89 ± 0.01	0.88 ± 0.00	0.87 ± 0.02	0.88 ± 0.01	0.89 ± 0.01	0.88 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.98 ± 0.00
	SigLIP2	0.93 ± 0.02	0.91 ± 0.01	0.90 ± 0.01	0.88 ± 0.00	0.93 ± 0.02	0.91 ± 0.01	0.90 ± 0.01	0.88 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.98 ± 0.00
Text-Only	BGE-M3 (structure)	0.98 ± 0.01	0.97 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	Voyage (structure)	0.98 ± 0.02	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.02	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	BGE-M3 (sentence)	0.98 ± 0.02	0.97 ± 0.02	0.97 ± 0.01	0.96 ± 0.02	0.98 ± 0.02	0.97 ± 0.02	0.97 ± 0.01	0.96 ± 0.02	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (sentence)	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.02	0.97 ± 0.02	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.02	0.97 ± 0.02	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
Multimodal	Voyage Multimodal	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	SigLIP	0.93 ± 0.02	0.94 ± 0.02	0.94 ± 0.01	0.94 ± 0.02	0.93 ± 0.02	0.94 ± 0.02	0.94 ± 0.01	0.94 ± 0.02	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
Fusion	Score Fusion	0.98 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.01	0.98 ± 0.00	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	Rank Fusion	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00

Note. All methods perform strongly due to the visual nature of category, but text-based and multimodal methods provide more consistent exact matching.

Table 38

Attribute Matching Performance for Subcategory

	Model	AMR @1	AMR @3	AMR @5	AMR @10	EAM @1	EAM @3	EAM @5	EAM @10	SS @1	SS @3	SS @5	SS @10
Image-Only	FashionCLIP	0.75 ± 0.04	0.74 ± 0.01	0.74 ± 0.01	0.73 ± 0.01	0.75 ± 0.04	0.74 ± 0.01	0.74 ± 0.01	0.73 ± 0.01	0.95 ± 0.01	0.95 ± 0.00	0.95 ± 0.00	0.95 ± 0.00
	OpenFashion CLIP	0.70 ± 0.03	0.70 ± 0.03	0.72 ± 0.02	0.71 ± 0.01	0.70 ± 0.03	0.70 ± 0.03	0.72 ± 0.02	0.71 ± 0.01	0.94 ± 0.01	0.94 ± 0.00	0.94 ± 0.00	0.94 ± 0.00
	SigLIP2	0.76 ± 0.03	0.70 ± 0.01	0.69 ± 0.01	0.68 ± 0.01	0.76 ± 0.03	0.70 ± 0.01	0.69 ± 0.01	0.68 ± 0.01	0.95 ± 0.00	0.94 ± 0.00	0.94 ± 0.00	0.93 ± 0.00
Text-Only	BGE-M3 (structure)	0.96 ± 0.03	0.95 ± 0.02	0.95 ± 0.02	0.96 ± 0.02	0.96 ± 0.03	0.95 ± 0.02	0.95 ± 0.02	0.96 ± 0.02	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (structure)	0.98 ± 0.03	0.98 ± 0.02	0.98 ± 0.02	0.97 ± 0.01	0.98 ± 0.03	0.98 ± 0.02	0.98 ± 0.02	0.97 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
	BGE-M3 (sentence)	0.97 ± 0.02	0.95 ± 0.03	0.95 ± 0.02	0.95 ± 0.02	0.97 ± 0.02	0.95 ± 0.03	0.95 ± 0.02	0.95 ± 0.02	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (sentence)	0.98 ± 0.02	0.98 ± 0.02	0.98 ± 0.02	0.97 ± 0.02	0.98 ± 0.02	0.98 ± 0.02	0.98 ± 0.02	0.97 ± 0.02	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00
Multimodal	Voyage Multimodal	0.95 ± 0.02	0.94 ± 0.02	0.92 ± 0.03	0.89 ± 0.02	0.95 ± 0.02	0.94 ± 0.02	0.92 ± 0.03	0.89 ± 0.02	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.01	0.98 ± 0.00
	SigLIP	0.88 ± 0.03	0.86 ± 0.04	0.86 ± 0.03	0.83 ± 0.02	0.88 ± 0.03	0.86 ± 0.04	0.86 ± 0.03	0.83 ± 0.02	0.98 ± 0.01	0.97 ± 0.01	0.97 ± 0.01	0.97 ± 0.00
Fusion	Score Fusion	0.90 ± 0.05	0.90 ± 0.02	0.91 ± 0.01	0.92 ± 0.01	0.90 ± 0.05	0.90 ± 0.02	0.91 ± 0.01	0.92 ± 0.01	0.98 ± 0.01	0.98 ± 0.00	0.98 ± 0.00	0.99 ± 0.00
	Rank Fusion	0.96 ± 0.02	0.95 ± 0.02	0.94 ± 0.01	0.92 ± 0.01	0.96 ± 0.02	0.95 ± 0.02	0.94 ± 0.01	0.92 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.00

Note. Text-only methods significantly outperform image-based approaches, highlighting the importance of semantic representations for fine-grained classification.

Table 39

Attribute Matching Performance for Color

	Model	AMR @1	AMR @3	AMR @5	AMR @10	EAM @1	EAM @3	EAM @5	EAM @10	SS @1	SS @3	SS @5	SS @10
Image-Only	FashionCLIP	0.75 ± 0.04	0.72 ± 0.02	0.72 ± 0.02	0.71 ± 0.02	0.75 ± 0.04	0.72 ± 0.02	0.72 ± 0.02	0.71 ± 0.02	0.94 ± 0.01	0.94 ± 0.00	0.94 ± 0.00	0.94 ± 0.00
	OpenFashion CLIP	0.68 ± 0.01	0.67 ± 0.01	0.64 ± 0.02	0.65 ± 0.01	0.68 ± 0.01	0.67 ± 0.01	0.64 ± 0.02	0.65 ± 0.01	0.93 ± 0.00	0.93 ± 0.00	0.92 ± 0.00	0.92 ± 0.00
	SigLIP2	0.70 ± 0.02	0.68 ± 0.03	0.68 ± 0.02	0.67 ± 0.02	0.70 ± 0.02	0.68 ± 0.03	0.68 ± 0.02	0.67 ± 0.02	0.93 ± 0.00	0.93 ± 0.01	0.93 ± 0.00	0.93 ± 0.00
Text-Only	BGE-M3 (structure)	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.97 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (structure)	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.99 ± 0.01	0.98 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00
	BGE-M3 (sentence)	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.00	1.00 ± 0.00	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.00	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (sentence)	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.01	0.95 ± 0.01	1.00 ± 0.00	0.99 ± 0.01	0.97 ± 0.01	0.95 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
Multimodal	Voyage Multimodal	0.98 ± 0.02	0.97 ± 0.01	0.95 ± 0.02	0.93 ± 0.02	0.98 ± 0.02	0.97 ± 0.01	0.95 ± 0.02	0.93 ± 0.02	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.01
	SigLIP	0.74 ± 0.03	0.72 ± 0.02	0.73 ± 0.01	0.70 ± 0.01	0.74 ± 0.03	0.72 ± 0.02	0.73 ± 0.01	0.70 ± 0.01	0.94 ± 0.01	0.94 ± 0.01	0.94 ± 0.00	0.93 ± 0.00
Fusion	Score Fusion	0.94 ± 0.02	0.92 ± 0.01	0.89 ± 0.02	0.87 ± 0.02	0.94 ± 0.02	0.92 ± 0.01	0.89 ± 0.02	0.87 ± 0.02	0.99 ± 0.01	0.98 ± 0.00	0.98 ± 0.00	0.97 ± 0.00
	Rank Fusion	0.94 ± 0.03	0.91 ± 0.02	0.89 ± 0.01	0.87 ± 0.01	0.94 ± 0.03	0.91 ± 0.02	0.89 ± 0.01	0.87 ± 0.01	0.99 ± 0.01	0.98 ± 0.00	0.97 ± 0.00	0.97 ± 0.00

Note. Image-only models struggle with multi-label color variations, while text-based methods maintain near-perfect consistency.

Table 40

Attribute Matching Performance for Material

	Model	AMR @1	AMR @3	AMR @5	AMR @10	EAM @1	EAM @3	EAM @5	EAM @10	SS @1	SS @3	SS @5	SS @10
Image-Only	FashionCLIP	0.85 ± 0.02	0.85 ± 0.02	0.85 ± 0.02	0.84 ± 0.01	0.85 ± 0.02	0.85 ± 0.02	0.85 ± 0.02	0.84 ± 0.01	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00
	OpenFashion CLIP	0.85 ± 0.02	0.86 ± 0.01	0.86 ± 0.02	0.85 ± 0.01	0.85 ± 0.02	0.86 ± 0.01	0.86 ± 0.02	0.85 ± 0.01	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00
	SigLIP2	0.83 ± 0.03	0.81 ± 0.02	0.83 ± 0.01	0.81 ± 0.01	0.83 ± 0.03	0.81 ± 0.02	0.83 ± 0.01	0.81 ± 0.01	0.97 ± 0.00	0.96 ± 0.00	0.97 ± 0.00	0.96 ± 0.00
Text-Only	BGE-M3 (structure)	0.93 ± 0.03	0.93 ± 0.02	0.93 ± 0.02	0.92 ± 0.02	0.93 ± 0.03	0.93 ± 0.02	0.93 ± 0.02	0.92 ± 0.02	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (structure)	0.96 ± 0.03	0.95 ± 0.02	0.94 ± 0.02	0.93 ± 0.02	0.96 ± 0.03	0.95 ± 0.02	0.94 ± 0.02	0.93 ± 0.02	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	BGE-M3 (sentence)	0.92 ± 0.03	0.93 ± 0.02	0.93 ± 0.02	0.92 ± 0.02	0.92 ± 0.03	0.93 ± 0.02	0.93 ± 0.02	0.92 ± 0.02	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (sentence)	0.94 ± 0.03	0.93 ± 0.02	0.94 ± 0.02	0.92 ± 0.02	0.94 ± 0.03	0.93 ± 0.02	0.94 ± 0.02	0.92 ± 0.02	0.99 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
Multimodal	Voyage Multimodal	0.92 ± 0.02	0.91 ± 0.03	0.91 ± 0.02	0.91 ± 0.02	0.92 ± 0.02	0.91 ± 0.03	0.91 ± 0.02	0.91 ± 0.02	0.99 ± 0.00	0.98 ± 0.01	0.98 ± 0.00	0.98 ± 0.00
	SigLIP	0.89 ± 0.04	0.89 ± 0.03	0.88 ± 0.02	0.86 ± 0.02	0.89 ± 0.04	0.89 ± 0.03	0.88 ± 0.02	0.86 ± 0.02	0.98 ± 0.01	0.98 ± 0.01	0.98 ± 0.01	0.97 ± 0.00
Fusion	Score Fusion	0.94 ± 0.02	0.92 ± 0.02	0.92 ± 0.02	0.91 ± 0.02	0.94 ± 0.02	0.92 ± 0.02	0.92 ± 0.02	0.91 ± 0.02	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.00
	Rank Fusion	0.92 ± 0.03	0.91 ± 0.02	0.90 ± 0.01	0.90 ± 0.02	0.92 ± 0.03	0.91 ± 0.02	0.90 ± 0.01	0.90 ± 0.02	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00

Note. Material remains challenging for image-based models, whereas text-based representations effectively capture this attribute through explicit encoding.

Table 41

Attribute Matching Performance for Pattern

	Model	AMR @1	AMR @3	AMR @5	AMR @10	EAM @1	EAM @3	EAM @5	EAM @10	SS @1	SS @3	SS @5	SS @10
Image-Only	FashionCLIP	0.83 ± 0.02	0.79 ± 0.04	0.77 ± 0.02	0.77 ± 0.02	0.83 ± 0.02	0.79 ± 0.04	0.77 ± 0.02	0.77 ± 0.02	0.96 ± 0.01	0.95 ± 0.01	0.95 ± 0.01	0.95 ± 0.00
	OpenFashion CLIP	0.72 ± 0.02	0.72 ± 0.01	0.72 ± 0.01	0.70 ± 0.01	0.72 ± 0.02	0.72 ± 0.01	0.72 ± 0.01	0.70 ± 0.01	0.94 ± 0.01	0.94 ± 0.00	0.94 ± 0.00	0.94 ± 0.00
	SigLIP2	0.80 ± 0.02	0.78 ± 0.01	0.78 ± 0.01	0.77 ± 0.01	0.80 ± 0.02	0.78 ± 0.01	0.78 ± 0.01	0.77 ± 0.01	0.96 ± 0.00	0.96 ± 0.00	0.95 ± 0.00	0.95 ± 0.00
Text-Only	BGE-M3 (structure)	0.93 ± 0.03	0.92 ± 0.02	0.91 ± 0.01	0.87 ± 0.01	0.93 ± 0.03	0.92 ± 0.02	0.91 ± 0.01	0.87 ± 0.01	0.99 ± 0.01	0.98 ± 0.00	0.98 ± 0.00	0.97 ± 0.00
	Voyage (structure)	0.96 ± 0.01	0.92 ± 0.02	0.90 ± 0.01	0.88 ± 0.01	0.96 ± 0.01	0.92 ± 0.02	0.90 ± 0.01	0.88 ± 0.01	0.99 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.97 ± 0.00
	BGE-M3 (sentence)	0.92 ± 0.04	0.91 ± 0.01	0.90 ± 0.01	0.87 ± 0.01	0.92 ± 0.04	0.91 ± 0.01	0.90 ± 0.01	0.87 ± 0.01	0.98 ± 0.01	0.98 ± 0.00	0.98 ± 0.00	0.97 ± 0.00
	Voyage (sentence)	0.92 ± 0.04	0.89 ± 0.01	0.88 ± 0.01	0.85 ± 0.01	0.92 ± 0.04	0.89 ± 0.01	0.88 ± 0.01	0.85 ± 0.01	0.98 ± 0.01	0.98 ± 0.00	0.98 ± 0.00	0.97 ± 0.00
Multimodal	Voyage Multimodal	0.89 ± 0.01	0.87 ± 0.02	0.86 ± 0.01	0.85 ± 0.01	0.89 ± 0.01	0.87 ± 0.02	0.86 ± 0.01	0.85 ± 0.01	0.98 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00
	SigLIP	0.88 ± 0.05	0.85 ± 0.02	0.84 ± 0.01	0.82 ± 0.01	0.88 ± 0.05	0.85 ± 0.02	0.84 ± 0.01	0.82 ± 0.01	0.97 ± 0.01	0.97 ± 0.00	0.96 ± 0.00	0.96 ± 0.00
Fusion	Score Fusion	0.89 ± 0.02	0.87 ± 0.01	0.87 ± 0.01	0.85 ± 0.00	0.89 ± 0.02	0.87 ± 0.01	0.87 ± 0.01	0.85 ± 0.00	0.98 ± 0.01	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00
	Rank Fusion	0.89 ± 0.02	0.87 ± 0.03	0.87 ± 0.01	0.86 ± 0.00	0.89 ± 0.02	0.87 ± 0.03	0.87 ± 0.01	0.86 ± 0.00	0.98 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00

Note. Text-based methods outperform others, demonstrating robustness in capturing repetitive or subtle visual patterns.

Table 42

Attribute Matching Performance for Style

	Model	AMR @1	AMR @3	AMR @5	AMR @10	EAM @1	EAM @3	EAM @5	EAM @10	SS @1	SS @3	SS @5	SS @10
Image-Only	FashionCLIP	0.46 ± 0.02	0.42 ± 0.02	0.40 ± 0.02	0.37 ± 0.01	0.46 ± 0.02	0.42 ± 0.02	0.40 ± 0.02	0.37 ± 0.01	0.91 ± 0.01	0.90 ± 0.00	0.90 ± 0.00	0.89 ± 0.00
	OpenFashion CLIP	0.41 ± 0.04	0.36 ± 0.02	0.36 ± 0.01	0.35 ± 0.01	0.41 ± 0.04	0.36 ± 0.02	0.36 ± 0.01	0.35 ± 0.01	0.90 ± 0.00	0.89 ± 0.00	0.89 ± 0.00	0.89 ± 0.00
	SigLIP2	0.43 ± 0.01	0.42 ± 0.02	0.41 ± 0.01	0.38 ± 0.01	0.43 ± 0.01	0.42 ± 0.02	0.41 ± 0.01	0.38 ± 0.01	0.90 ± 0.00	0.90 ± 0.00	0.90 ± 0.00	0.89 ± 0.00
Text-Only	BGE-M3 (structure)	0.97 ± 0.02	0.95 ± 0.01	0.94 ± 0.01	0.92 ± 0.01	0.97 ± 0.02	0.95 ± 0.01	0.94 ± 0.01	0.92 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (structure)	0.91 ± 0.02	0.89 ± 0.01	0.89 ± 0.01	0.88 ± 0.01	0.91 ± 0.02	0.89 ± 0.01	0.89 ± 0.01	0.88 ± 0.01	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.98 ± 0.00
	BGE-M3 (sentence)	0.99 ± 0.01	0.97 ± 0.01	0.95 ± 0.01	0.93 ± 0.01	0.99 ± 0.01	0.97 ± 0.01	0.95 ± 0.01	0.93 ± 0.01	1.00 ± 0.00	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
	Voyage (sentence)	0.97 ± 0.02	0.95 ± 0.02	0.94 ± 0.01	0.93 ± 0.01	0.97 ± 0.02	0.95 ± 0.02	0.94 ± 0.01	0.93 ± 0.01	1.00 ± 0.00	0.99 ± 0.00	0.99 ± 0.00	0.99 ± 0.00
Multimodal	Voyage Multimodal	0.86 ± 0.01	0.80 ± 0.01	0.78 ± 0.01	0.74 ± 0.01	0.86 ± 0.01	0.80 ± 0.01	0.78 ± 0.01	0.74 ± 0.01	0.98 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.96 ± 0.00
	SigLIP	0.82 ± 0.02	0.79 ± 0.02	0.78 ± 0.02	0.76 ± 0.01	0.82 ± 0.02	0.79 ± 0.02	0.78 ± 0.02	0.76 ± 0.01	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.96 ± 0.00
Fusion	Score Fusion	0.69 ± 0.02	0.67 ± 0.02	0.66 ± 0.02	0.65 ± 0.02	0.69 ± 0.02	0.67 ± 0.02	0.66 ± 0.02	0.65 ± 0.02	0.95 ± 0.00	0.94 ± 0.00	0.94 ± 0.00	0.94 ± 0.00
	Rank Fusion	0.77 ± 0.01	0.77 ± 0.02	0.76 ± 0.01	0.72 ± 0.01	0.77 ± 0.01	0.77 ± 0.02	0.76 ± 0.01	0.72 ± 0.01	0.96 ± 0.00	0.96 ± 0.00	0.96 ± 0.00	0.95 ± 0.00

Note. Style is the most challenging attribute for image-based models, while text-based methods achieve significantly higher consistency due to explicit semantic representation.

The most noticeable difference appears in the style attribute. Image-only models show significantly lower AMR and EAM scores, indicating difficulty in capturing abstract stylistic concepts. In contrast, text-only methods achieve near-perfect performance, highlighting the importance of semantic representations for high-level attributes.

Multimodal and fusion methods generally provide a trade-off between visual and semantic performance. While they do not always achieve the highest attribute-matching scores, they maintain strong performance across attribute types, making them more balanced in practical recommendation scenarios.

Best Retrieval Performance Example

Figures 37 to 40 are some examples of the query image and the retrieval results. Each example shows the query image with its predicted attributes, followed by retrieved items from the best model of different methods. The results demonstrate that the image-only approach tends to capture visual similarity more accurately but may fail to preserve style semantics, whereas text-only methods better reflect semantic attributes but sometimes lack visual consistency. This also indicates that all retrieval methods rely heavily on the FACTOR model. Therefore, we introduce fusion and multimodal approaches to reduce the strong dependency on either visual or textual information in image-only and text-only methods. In contrast, both fusion and multimodal approaches produce retrieval results that are more closely aligned with the query's visual appearance and underlying style attributes.

Figure 37

Query Image of a Sweater and the Retrieval Outputs

Figure 38

Query Image of a Blouse and the Retrieval Outputs

Figure 39

Query Image of a Heel and the Retrieval Outputs

Figure 40

Query Image of a Pants and the Retrieval Outputs

Conclusion

In this chapter, we proposed an attribute-guided multimodal retrieval framework for fashion recommendation and conducted a comprehensive evaluation across image-only, text-only, multimodal, and fusion-based methods.

The experimental results reveal several key findings. First, structured attribute representations alone are highly effective for recommendation, as demonstrated by the strong performance of text-only models across all evaluation metrics. This indicates that when accurate attribute labels are available, semantic similarity becomes more important than pure visual similarity. Second, image-only models are limited in their ability to capture fine-grained and abstract attributes, particularly for multi-label attributes such as color, pattern, and style. This highlights the gap between visual similarity and semantic consistency in fashion recommendation. Third, multimodal and fusion-based approaches provide a more balanced trade-off between visual and semantic information. Although they do not always achieve the highest attribute-matching scores, they improve robustness and provide realistic recommendation behavior by combining complementary signals from both modalities. Finally, the results demonstrate that traditional ranking metrics alone are insufficient for evaluating recommendation systems in this domain. Attribute-level evaluation, including AMR, EAM, and semantic similarity, is essential for capturing whether the retrieved items truly satisfy user intent.

Overall, the findings of this thesis suggest that the next step for fashion recommendation is not simply to build larger similarity models, but to build better semantic representations. By treating recommendation as a fine-grained attribute-aware retrieval problem, this work provides a more interpretable, controllable, and user-aligned direction for multimodal fashion recommendation. Overall, this study shows that fashion recommendation can be effectively formulated as an attribute-aware retrieval problem, where structured semantic information plays a central role in bridging the gap between visual appearance and user expectations.

Chapter 8: Discussion and Limitations

Discussion

The core contribution of this study is the transition from visual perception to semantic understanding. Conventional e-commerce recommendation systems primarily rely on similarity to prior user interactions, like purchase or browsing history. However, from a user-centric viewpoint, the actual need is not to retrieve similar items, but to obtain recommendations that can form a coherent outfit and align with specific contexts or occasions. Therefore, transforming garment images into fine-grained semantic attributes serves as a critical intermediate representation that bridges the gap between user intent and system output. This positioning is consistent with earlier research on outfit compatibility, in which sequence-based or type-aware embedding methods have shown that effective recommendations require modeling cross-category relationships rather than simple item-level similarity.

In attribute recognition, the proposed FACTOR framework achieves strong performance across both single-label and multi-label settings. Specifically, it attains high accuracy in category and subcategory prediction, 93.1% and 72.1%, respectively, while achieving semantic similarity scores in the range of 85% to 87% for color, material, and pattern. The comparative performance across different training strategies provides actionable insights into how vision language models should be adapted for small to medium scale fashion datasets. Linear probing and attribute-aware adaptation significantly outperform the zero-shot baseline, and the attribute-aware variant further yields consistent improvements in multi-label semantic similarity. In contrast, full fine-tuning leads to substantial performance degradation in multi-label attribute prediction. This phenomenon can be attributed to limited data scale, long-tailed class distributions, and the risk of disrupting the pretrained representation space through excessive parameter updates. This observation aligns with established findings in CLIP-based models, where parameter-efficient adaptation tends to be more robust than full end-to-end fine-tuning, especially under constrained data conditions.

For style classification, we compare a supervised CNN baseline with adapted vision-language models. ResNet50 achieves an F1 score of approximately 81.1%, while OpenFashionCLIP with partial fine-tuning reaches 86.1%. This demonstrates that for high-level semantic concepts such as style, appropriately adapted vision language representations can surpass purely visual supervised models. At the same time, both full fine-tuning and certain parameter-efficient configurations underperform in this task, reinforcing the conclusion that partial fine-tuning provides the most effective trade-off. This finding is consistent with domain-specific contrastive pretraining approaches, such as FashionCLIP, which emphasize aligning multimodal representations with domain-specific semantics (Chia et al., 2022).

The retrieval experiments highlight two key observations. First, the weighted attribute score measures overall consistency across six attributes, with text-based embedding methods achieving near-saturation performance of around 99%, whereas image-based models remain at around 94% to 95%. This indicates that structured attributes themselves form highly discriminative retrieval keys that support semantically aligned recommendations. Second, the ranking-based metrics nDCG and MRR show consistently high scores across different numbers of retrieval items, with most methods approaching 1.00. This indicates that the retrieval task becomes relatively well resolved when accurate, structured attribute information is available, suggesting that attribute-level representations provide strong discriminative power for ranking relevant items. To assess the quality of the retrieved results, we use AMR, EAM, and semantic similarity. The results demonstrate that across different numbers of retrieval items, more than 90% of the retrieved items preserve correct attribute alignment with the query. This confirms that the proposed approach can maintain semantic consistency not only at the overall ranking level but also within fine-grained attribute dimensions.

Limitations

This study has several limitations from both methodological and data perspectives. From a methodological perspective, the attribute annotation process relies on LLaMA3 to extract attributes from product descriptions. While this approach improves scalability and consistency, it could introduce noise, including parsing errors, contamination from brand-related terms, and inconsistencies in semantic granularity. Such noise may also interact with dataset distributions, potentially amplifying biases toward specific item categories or styles. Additionally, the attribute-aware adaptation is based on a limited subset of 1,000 sampled images from selected confusing color categories. As a result, the improvement is localized to specific color groups rather than the entire color space.

From a data point of view, Polyvore dataset exhibits a long-tailed distribution, where dominant categories disproportionately influence the learning process, leading to underrepresentation of rare attributes. For style classification, the availability of public datasets remains limited. Existing datasets often focus on specific demographics, with some styles predominantly represented by female samples. This restricts diversity in gender and cultural representation and may introduce biases in both classification and recommendation outcomes.

Chapter 9: Conclusion

This study proposes a unified framework that leverages fine-grained garment attributes as interpretable semantic representations, bridging image understanding and attribute-aware recommendation. The frameworks integrate image-to-attribute prediction, structured semantic representation, and attribute-guided retrieval, and provide reproducible empirical evidence across attribute recognition, style classification, and recommendation evaluation.

In attribute recognition, FACTOR framework achieves state-of-the-art performance, with semantic similarity reaching 99% and 85% for category and subcategory, respectively, and 87%, 87%, and 85%, respectively, for color, material, and pattern. Comparative analysis across adaptation strategies further demonstrates that lightweight adaptation outperforms full fine-tuning, providing a robust guideline for adapting vision-language models in domain-specific settings. In style understanding, partially fine-tuned vision-language representations outperform strong CNN baselines, confirming their effectiveness in modeling abstract semantic concepts.

In recommendation, the results show that when high-quality structured attributes are available, semantic text-based retrieval can achieve near-saturated attribute consistency. Moreover, this study highlights the limitations of traditional ranking metrics and emphasizes the need for attribute-level evaluation to accurately reflect recommendation quality. From a practical standpoint, this work provides a pathway to transition e-commerce recommendation systems from item-level similarity to semantic consistency.

Chapter 10: Future Work

Future work can extend this framework toward full outfit recommendation, enabling the system to recommend complete outfit combinations rather than retrieval of similar individual items. This direction is particularly relevant for e-commerce scenarios, where users seek not only products but also styling guidance.

From a sustainability perspective, the system can be extended to incorporate personal wardrobe data, enabling it to recommend daily outfits using existing garments. With this functionality, we could promote garment reuse, reduce unnecessary purchases, and help users rediscover underutilized clothing items. Additionally, future research can expand the semantic space beyond style classification by introducing occasion-based categories, such as wedding, party, family reunion, etc. This would allow users to directly specify contextual requirements through natural-language prompts, enabling more personalized, context-aware recommendations.

Furthermore, we could strengthen the retrieval framework in several directions. First, the current retrieval evaluation is conducted on a controlled query subset, future studies can expand the query pool to test robustness across larger, more diverse recommendation scenarios. Second, the present fusion experiments adopt a balanced-score setting and the standard reciprocal-rank fusion formulation, without extensive hyperparameter tuning. A more systematic investigation of fusion weights, rank fusion constants, and adaptive modality balancing may provide additional insight into how visual and semantic signals should be combined for fashion recommendation. Another promising direction is to incorporate large language models into the recommendation framework. LLM-based methods may further improve conversational query understanding, explanation generation, occasion-aware reasoning, and more flexible personalization. However, they are not included in the current experimental comparison because they introduce substantially higher computational cost, larger model size, and more complex inference requirements. Since the main goal of this thesis is to evaluate the effect of fine-grained semantic representation on retrieval quality in a controlled setting, the current study focuses on compact and directly comparable retrieval pipelines.

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